

BANOQABIL GUJARKHAN



PROJECT FINAL REPORT

Project Title:

Windows Security Assessment and Monitoring using Wazuh SIEM

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Submission date:	14-08-2026

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Abstract

The Windows Security Assessment and Monitoring using wazuh siem project focuses on implementing a centralized security monitoring solution for a Windows 11 Pro endpoint in a controlled virtual laboratory environment. The project uses an Ubuntu Server as the Wazuh server infrastructure and a Windows 11 Pro system with the Wazuh Agent installed for endpoint monitoring.

The system was configured to collect and monitor security-relevant information from the Windows endpoint, including Windows Event Logs,

authentication activities, file changes, and other endpoint security events. The project also included an assessment of Windows security controls such as Microsoft Defender, firewall configuration, password policies and installed applications. The Wazuh Agent was registered with the Wazuh Manager and communication between the endpoint and server was verified through the Wazuh environment.

Controlled security scenarios were used to demonstrate monitoring capabilities, including failed login attempts and File Integrity Monitoring events. The collected information was analyzed through the Wazuh Dashboard to provide centralized visibility into endpoint activity.

The project demonstrates how Wazuh SIEM can support security monitoring, event analysis, and endpoint visibility in a small-scale SOC laboratory environment. Although the implementation is limited to a controlled environment and a small number of endpoints, it provides a practical foundation that can be extended to multiple endpoints, customized detection rules, additional log sources, threat intelligence, and automated incident-response capabilities.

1 INTRODUCTION

1.1 Background:

Modern organizations depend heavily on Windows-based computers and servers. These systems continuously generate security logs and events. Examples include successful and failed login attempts, changes to system settings, application errors, security policy events, and modifications to important files. Manually checking these logs is difficult because of the large amount of information generated. A security analyst may miss an important event if logs are not collected and analyzed centrally.

A SIEM solution helps solve this problem by collecting security information from endpoints, analyzing events, applying detection rules, and presenting relevant information through a centralized dashboard.

Wazuh is an open-source security monitoring platform that provides capabilities such as security event monitoring, log analysis, File Integrity Monitoring (FIM), vulnerability detection, and alert generation. Because it can be deployed in a laboratory environment without requiring expensive commercial infrastructure, it is suitable for this project.

1.2 Problem Statement:

In a normal Windows environment, an administrator or security analyst may need to manually inspect Event Viewer to identify suspicious login attempts, file changes, or other security-related activities. This approach becomes inefficient when the number of events increases. There is therefore a need for a centralized monitoring solution that can collect Windows security events and provide a clear view of endpoint security activity.

This project addresses this problem by implementing Wazuh SIEM to monitor a Windows endpoint and demonstrate how security events can be collected and analyzed from a centralized platform.

1.3 Aim of Project:

The main aim of this project is to design and implement a Wazuh-based security monitoring environment for assessing and monitoring a Windows 11 Pro endpoint and detecting relevant security events through centralized log collection and analysis.

1.4 Objectives:

The main objectives of the project are:

- To set up a virtual cybersecurity laboratory using VMware.
- To install and configure Ubuntu Server as the Wazuh Manager.
- To install and configure the Wazuh Agent on a Windows 11 Pro endpoint.
- To establish communication between the Windows endpoint and Wazuh Manager.
- To collect and monitor Windows security events.
- To monitor successful and failed authentication activities.
- To implement and test File Integrity Monitoring.
- To monitor relevant Windows firewall/security events.
- To generate controlled security events in the laboratory environment.
- To document the implementation, testing, findings, and recommendations.

1.5 Scope:

The project will focus on monitoring a **Windows 11 Pro virtual machine** using Wazuh.

The scope includes:

- Ubuntu Server deployment
- VMware installation
- Kali linux installation
- Wazuh Manager installation
- Wazuh Dashboard configuration
- Windows Wazuh Agent installation
- Agent enrollment and communication
- Windows Event Log monitoring
- Authentication monitoring
- File Integrity Monitoring
- Windows firewall/security event monitoring
- Controlled security testing
- Documentation of findings

The project will be implemented in a controlled virtual environment using VMware. The testing will be performed only on the project's own virtual machines.

1.6 Significance:

The project provides practical experience in implementing a SIEM-based monitoring environment and demonstrates the complete workflow from endpoint security assessment and event collection to detection, investigation, and analysis. It also provides a controlled environment for understanding basic SOC monitoring activities and evaluating how endpoint security events can be investigated using Wazuh.

2 Technologies and System Requirements

2.1 Virtualization:

Virtualization is a technology that allows multiple virtual computers, known as virtual machines (VMs), to run on a single physical computer. Each virtual machine operates as an independent computing environment with its own operating system, virtual CPU, memory, storage, and network interface.

Virtualization was used in this project to create a controlled and isolated cybersecurity laboratory. Instead of installing different operating systems directly on the physical computer, separate virtual machines were created for the Windows endpoint, Ubuntu Server, and Kali Linux. This approach allowed the project environment to be developed and tested without directly affecting the host operating system.

2.2 VMware Workstation:

VMware Workstation Pro is a desktop virtualization platform that allows users to create, configure, and run multiple virtual machines on a physical computer. It provides virtual hardware resources such as processors, memory, storage, and network adapters to each virtual machine.

VMware Workstation Pro was used as the virtualization platform for this project. It provided the environment required to create and manage the Windows, Ubuntu Server, and Kali Linux virtual machines used throughout the security assessment and monitoring project.

VMware also provided virtual networking capabilities that allowed the virtual machines to communicate with each other. The project environment used a NAT-based virtual network configuration, allowing the virtual machines to access required network resources while remaining within the controlled laboratory environment.

The VMware Workstation version used during the project was **VMware Workstation Pro 17.6.4**

2.3 Windows 11Pro:

Windows 11 Pro was used as the target endpoint in this project. The Windows virtual machine represents an endpoint that requires security assessment and continuous monitoring.

The Windows system was assessed for several security-related areas, including Windows Update status, firewall configuration, Microsoft Defender status, user accounts, password policy, running services, installed software, listening network ports, and Windows Event Logs.

A Wazuh Agent was installed on the Windows endpoint to collect relevant security and system information and forward it to the Wazuh Manager running on the Ubuntu Server. This allowed Windows activities to be investigated centrally through the Wazuh Dashboard.

Controlled security activities were also performed on the Windows endpoint, including user-account creation and deletion, firewall state changes, file-integrity testing, and failed authentication attempts. These activities were used to evaluate the monitoring and detection capabilities of the implemented environment.

2.4 Ubuntu Server:

Ubuntu Server was used as the server operating system for the Wazuh deployment. It provided the Linux-based environment on which the Wazuh server components were installed and operated.

Ubuntu Server was selected because it is a widely used Linux server operating system and provides a suitable environment for deploying server-side security monitoring components. In this project, Ubuntu Server acted as the central server machine receiving and processing security data from the Windows endpoint.

The Ubuntu Server virtual machine was configured within VMware and connected to the project's virtual network. Wazuh server components were then installed on this machine and verified through the Wazuh Dashboard.

2.5 Wazuh SIEM:

Wazuh is an open-source security platform that provides capabilities for security monitoring, threat detection, endpoint monitoring, log analysis, file integrity monitoring, vulnerability detection, and security event investigation.

Wazuh was selected as the central security monitoring solution for this project. The main purpose of using Wazuh was to collect security-related information from the Windows endpoint, process the collected events using its detection rules, and provide a centralized interface for investigating events and alerts.

The Wazuh platform was integrated with the Windows endpoint through the Wazuh Agent. Events collected from the Windows system were sent to the Wazuh server environment and made available through the Wazuh Dashboard for monitoring and investigation.

2.6 Wazuh Manager:

Wazuh Manager is the central processing component responsible for receiving data from Wazuh Agents and analyzing the collected events using Wazuh rules and decoders. When an incoming event matches a relevant rule, Wazuh can generate a security alert with information such as the rule ID, rule level, description, timestamp, and affected agent.

In this project, the Wazuh Manager was deployed on the Ubuntu Server virtual machine. It received security-related data from the Windows Wazuh Agent and processed the collected events for monitoring and investigation.

2.7 Wazuh Agent:

The **Wazuh Agent** is a lightweight component installed on a monitored endpoint. Its purpose is to collect security and system information from the endpoint and send the collected data to the Wazuh Manager.

In this project, the Wazuh Agent was installed on the Windows 11 Pro virtual machine. It enabled the Windows endpoint to communicate with the Wazuh Manager and allowed relevant Windows event information to be collected for centralized monitoring.

2.8 Wazuh Indexer:

The **Wazuh Indexer** is responsible for storing and indexing security data so that it can be efficiently searched and analyzed. It allows collected security information to be organized and retrieved by the Wazuh Dashboard.

In the implemented Wazuh architecture, the Indexer works with the Manager and Dashboard to provide centralized storage and investigation of security data.

2.9 Wazuh Dashboard:

The **Wazuh Dashboard** is the web-based interface used to visualize, monitor, search, and investigate security information collected by the Wazuh platform.

The Dashboard provided a centralized view of the monitored Windows endpoint and allowed security events to be investigated without manually reviewing every event directly on the endpoint.

2.10 Kali Linux:

Kali Linux is a Linux distribution designed for penetration testing, security assessment, network analysis, and digital forensics. It includes a large collection of tools used by security professionals to assess systems and networks.

Kali Linux was included in the virtual cybersecurity laboratory as the security assessment environment. Its primary role in this project was to provide a separate environment from which network-level assessment activities could be performed against the Windows endpoint.

2.11 Virtual Network/NAT:

Network Address Translation (NAT) was used for the virtual machine networking environment. NAT allows virtual machines to communicate with external network resources through the host system while maintaining their own virtual network addresses.

2.12 Windows PowerShell:

Windows PowerShell is a command-line shell and scripting environment used for system administration, configuration and security-related tasks in Windows environments. It provides commands that allow administrators and security analysts to inspect and manage Windows system components efficiently.

PowerShell was selected because it provides direct access to Windows administrative and security functions through commands, making it suitable for performing repeatable security checks and controlled testing without requiring additional third-party software.

2.13 System Requirements:

2.13.1 Hardware Requirements:

- 64-bit processor with virtualization support
- Minimum 16 GB RAM recommended
- Sufficient storage for virtual machines
- Stable network connection
- VMware Workstation

2.13.2 Software Requirements:

Host System

- Windows 11 Pro
- VMware Workstation

Server VM

- Ubuntu Server 24 LTS
- Wazuh Manager
- Wazuh Dashboard

Client VM

- Windows 11 Pro
- Wazuh Agent

3 System Environment and Setup

3.1 System Environment:

Firstly , the system settings are checked wether they meet requirements or not.

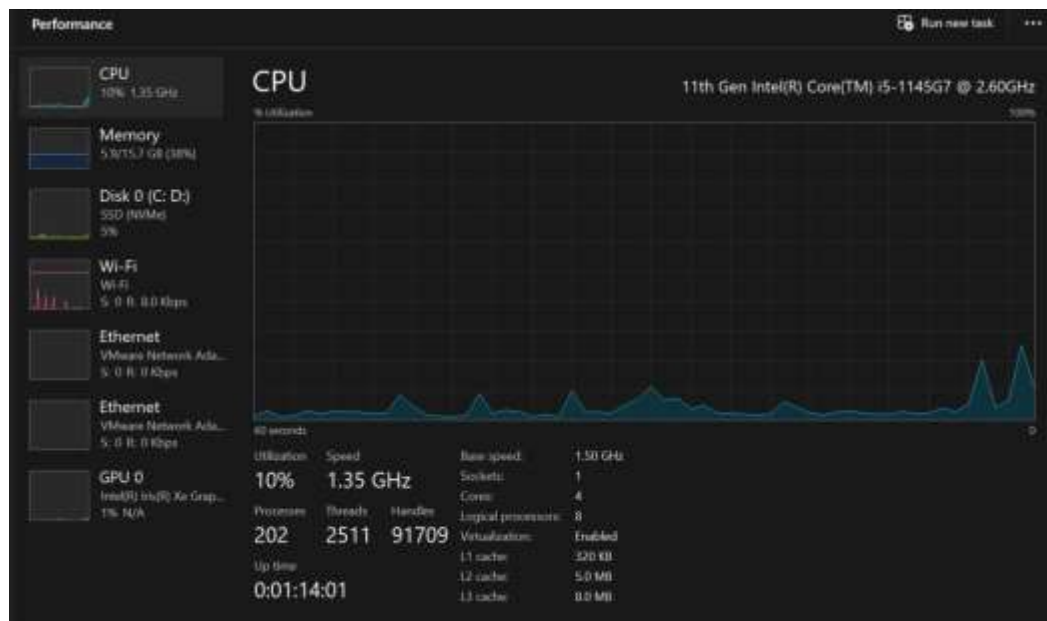


Figure 3.1 System CPU info

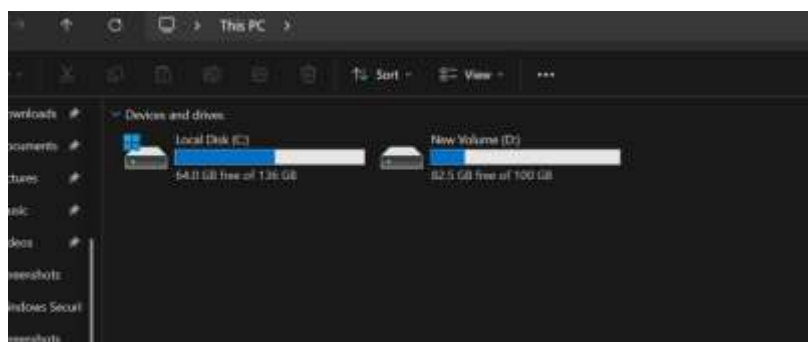


Figure 3.2 RAM info

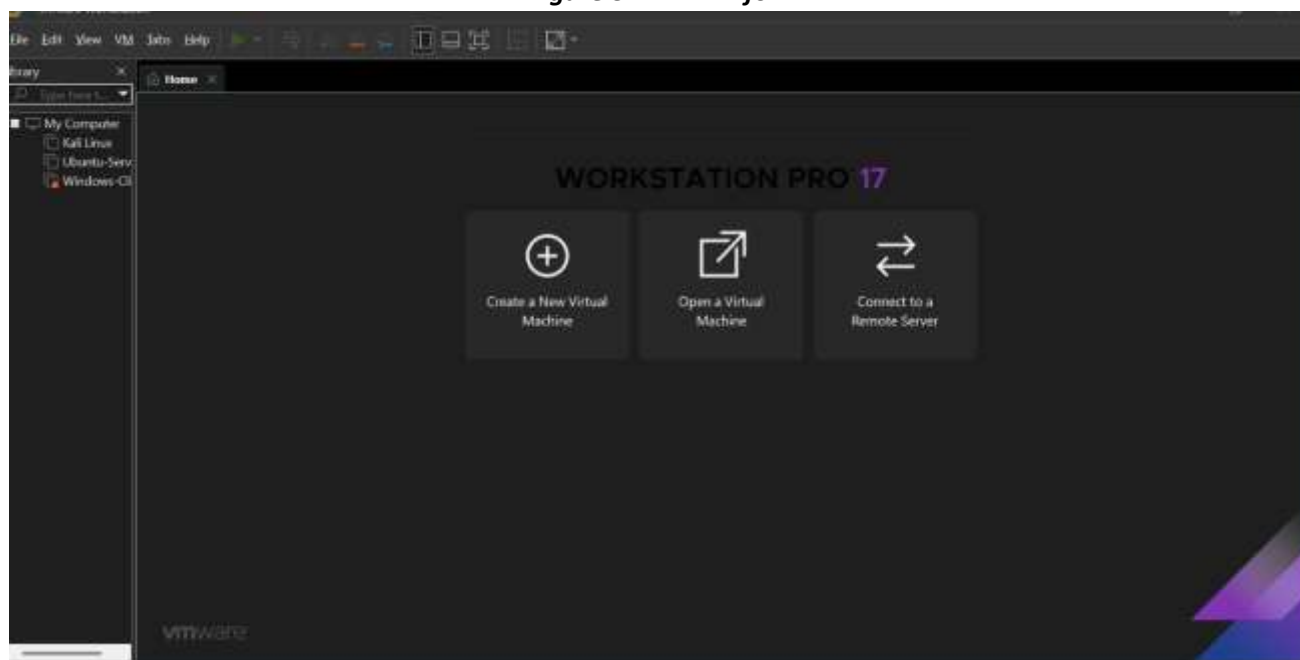


Figure 3.3 Final Architecture

3.2 VMware Workstation Installation:

VMware workstation 17 Pro was downloaded from the official website to create a virtual lab environment. VMware Workstation Pro was installed on the host Windows system to provide the virtualization platform required for the project.

One thing that before installing VMware workstation, its mandatory to make virtualization enabled as for above Figure 3.1 it shows virtualization feature is enabled.

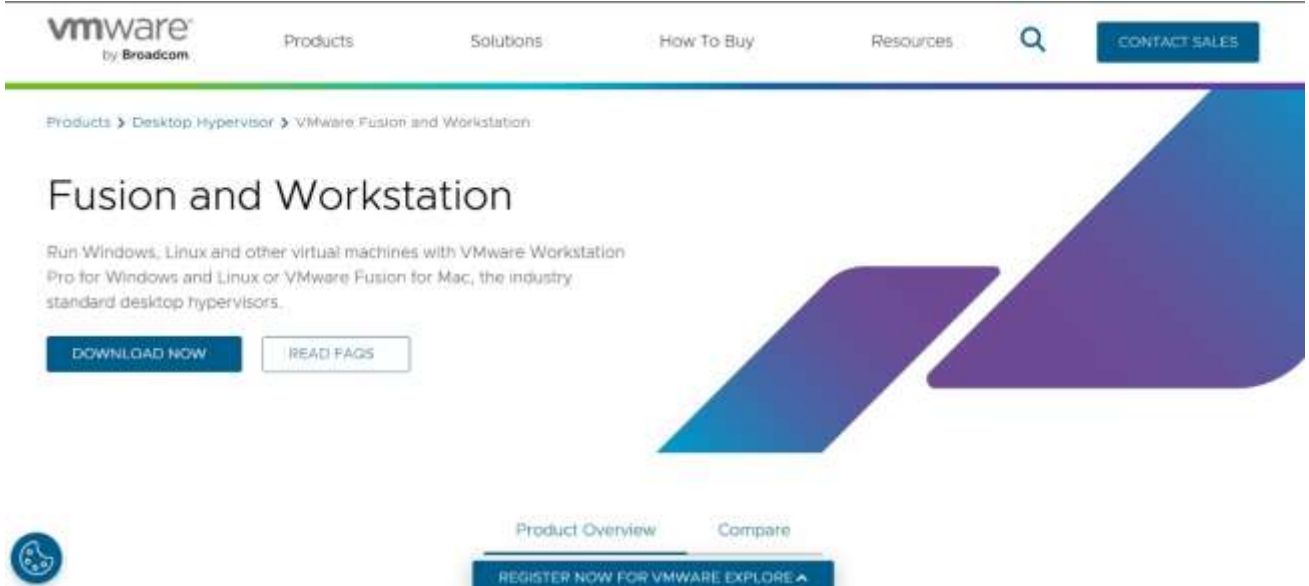
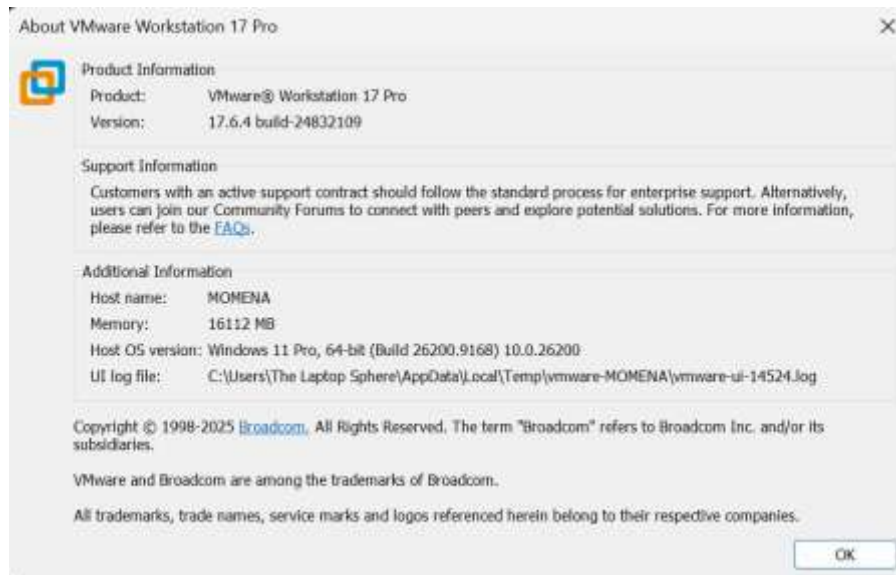


Figure 3.4 VMware Installation

Then after all configurations, a complete setup is ready for the project.



Figure

3.5 Environment after Installation

3.3 Kali Linux :

3.3.1 Kali Linux Created:

Kali Linux is a Debian based linux distribution designed for cybersecurity ,penetration testing and security testing. It provides pre-installed tools and Nmap is one of them which will be used. It was already installed in my system but complete installation process is included in B Appendix .

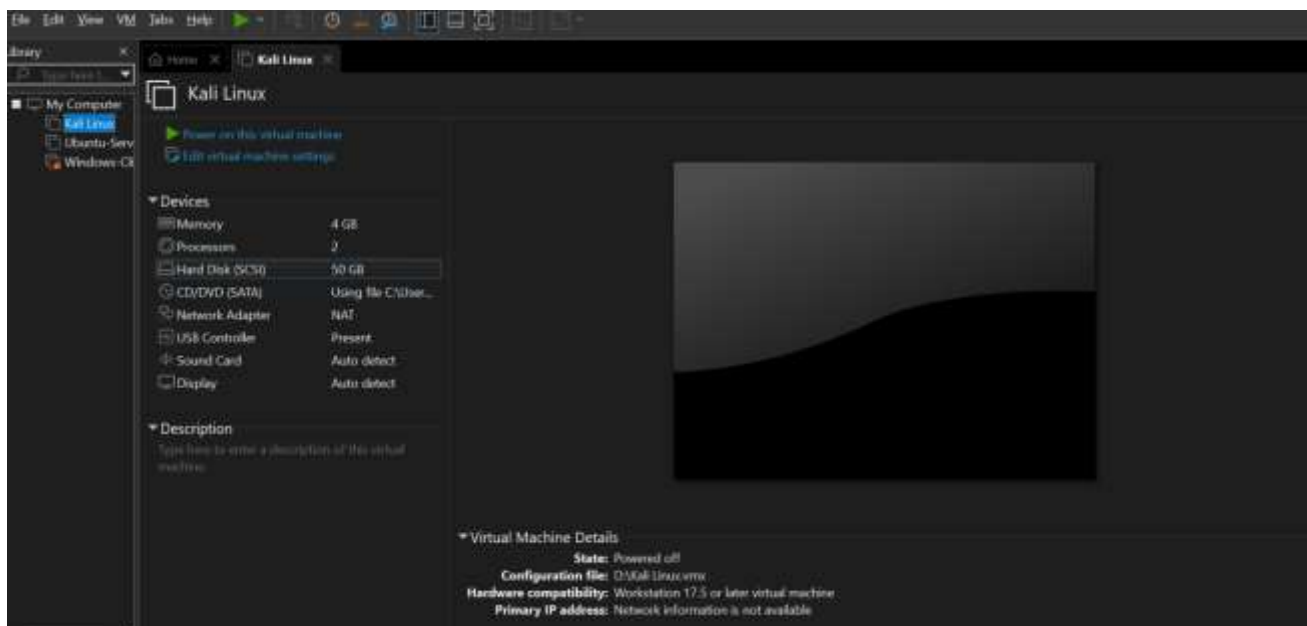


Figure 3.6 VM created

3.3.2 Virtual Hardware Configuration:

The Ubuntu Server virtual machine was configured with virtual processor, memory, storage, and network resources suitable for running. The virtual disk was configured with **50 GB** of storage, while the remaining resources were allocated according to the available host system resources.

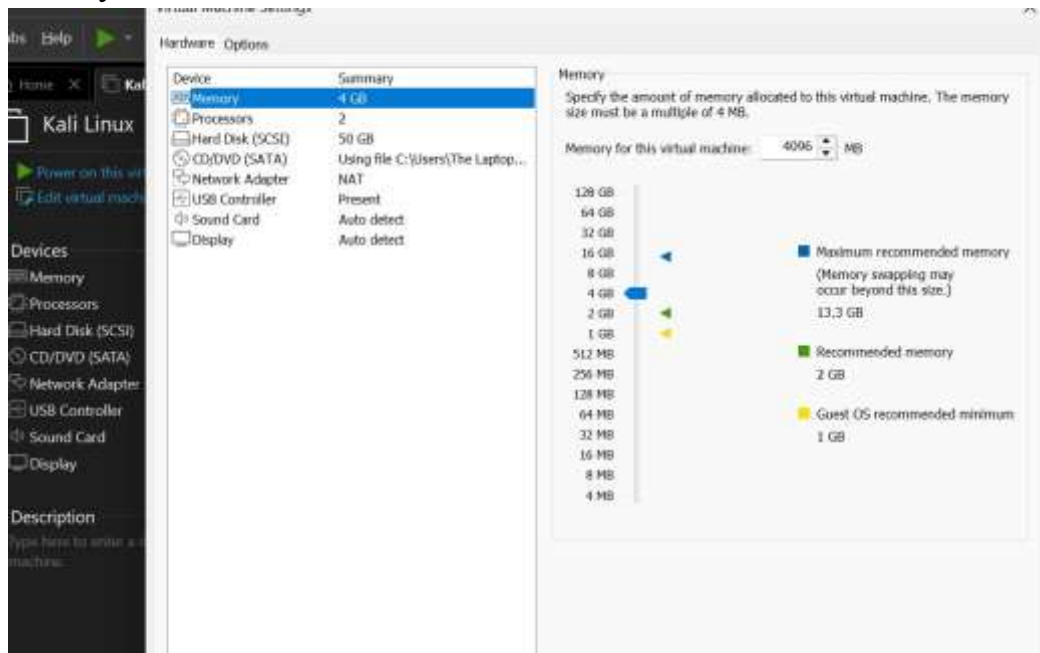


Figure 3.7 Hardware settings

3.4 Windows 11 pro VM:

3.4.1 Windows VM Creation:

A Windows 11 Pro virtual machine was created in VMware Workstation to serve as the monitored endpoint for the project. During the VM creation process, the required virtual hardware resources were assigned and the Windows installation media was attached to the virtual machine. The network adapter was configured to allow communication with the other virtual machines in the project environment.

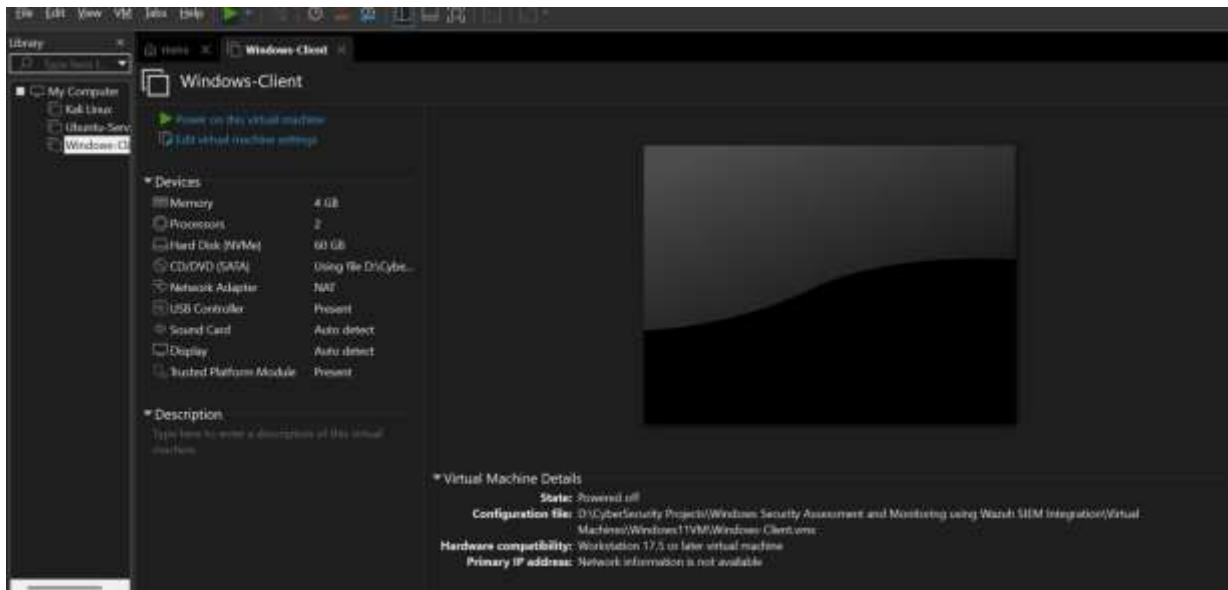


Figure 3.8 VM created

3.4.2 Virtual Hardware Configuration:

The Windows 11 Pro virtual machine was configured with virtual CPU, memory, storage, and network resources appropriate for the security monitoring laboratory. The configuration was selected to provide sufficient resources for Windows and the Wazuh Agent while maintaining adequate resources for the other virtual machines running on the host system.

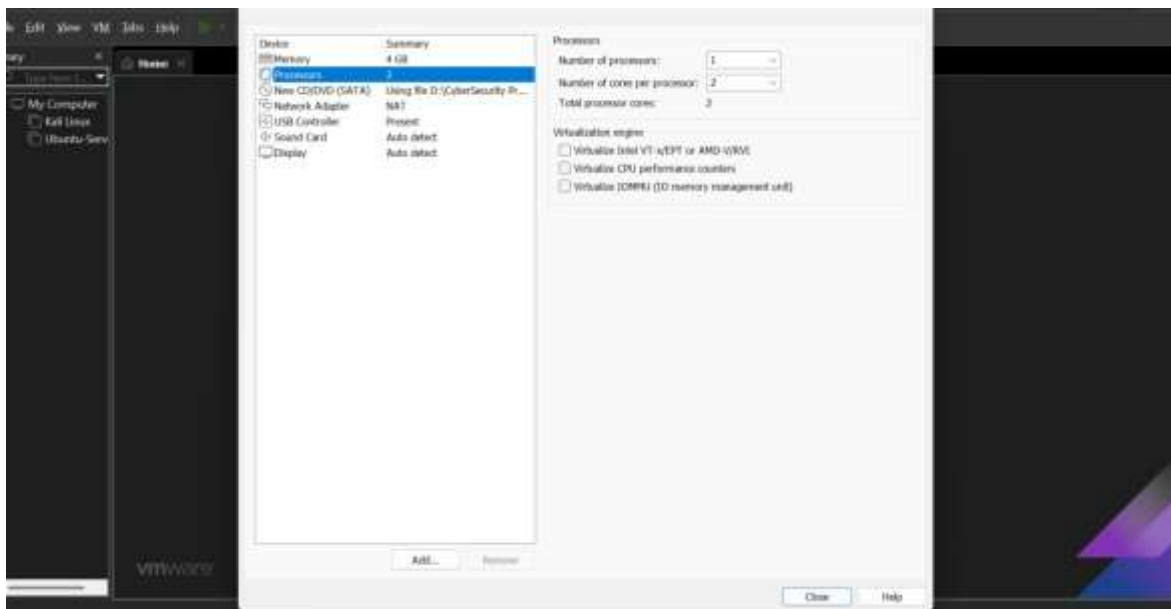


Figure 3.9 Hardware Configuration

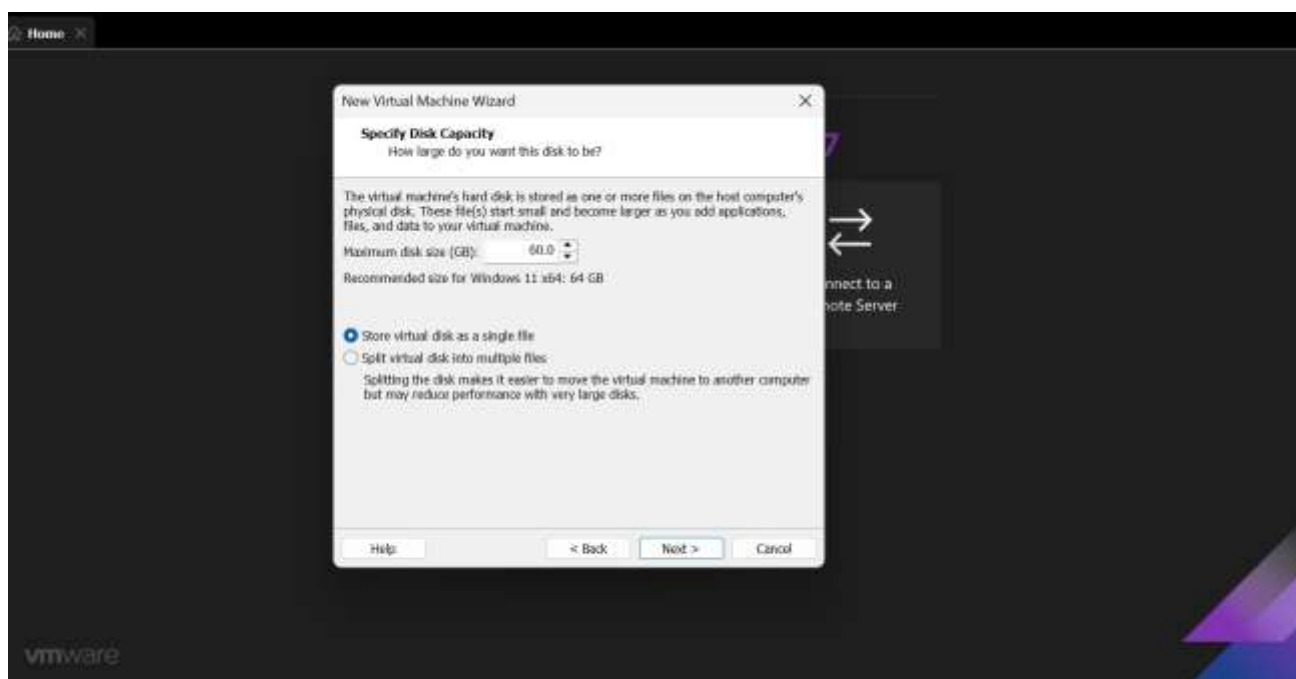


Figure 3.10 Disk capacity

3.4.3 Windows 11 pro Installation and Verification:

Windows 11 Pro was installed on the newly created virtual machine. After installation, the operating system was configured and prepared as the target endpoint for the security assessment. The system was subsequently used for Windows security checks, controlled security-event generation, and Wazuh Agent integration. The installed operating system was verified from the Windows system information page to confirm that the required Windows edition was running correctly inside the virtual environment.



Figure 3.11 Window VM installation



Figure 3.12 Windows Vm verified

Configuration	Windows 11 Pro VM
Operating System	Windows 11 Pro
Virtualization Platform	VMware Workstation Pro 17.6.4
Role	Monitored Windows Endpoint
Configuration	Windows 11 Pro VM
Network Mode	NAT
CPU	2
RAM	4GB

3.5 Ubuntu Server VM:

3.5.1 Ubuntu Server VM creation:

A separate Ubuntu Server virtual machine was created in VMware Workstation to host the Wazuh server environment. The Ubuntu Server VM was kept separate from the Windows endpoint so that the security monitoring infrastructure could operate independently from the monitored system.

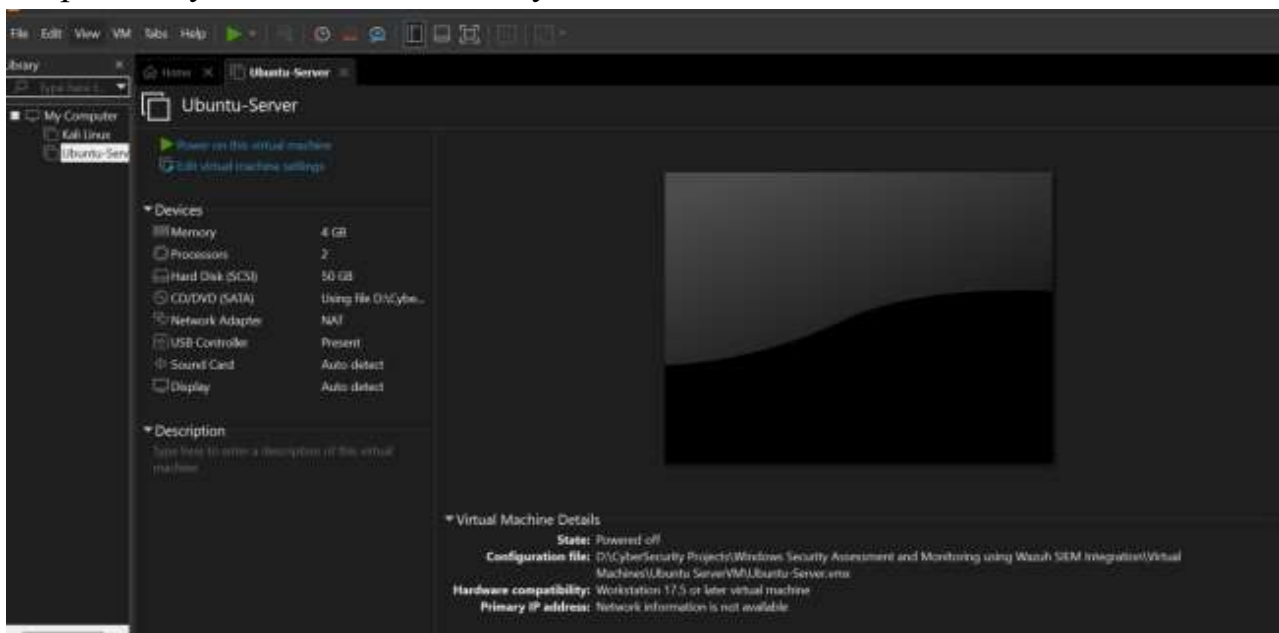


Figure 3.13 ubuntu VM created

3.5.2 Virtual Hardware Configuration:

The Ubuntu Server virtual machine was configured with virtual processor, memory, storage, and network resources suitable for running the Wazuh server environment. The virtual disk was configured with **50 GB** of storage, while the remaining resources were allocated according to the available host system resources.

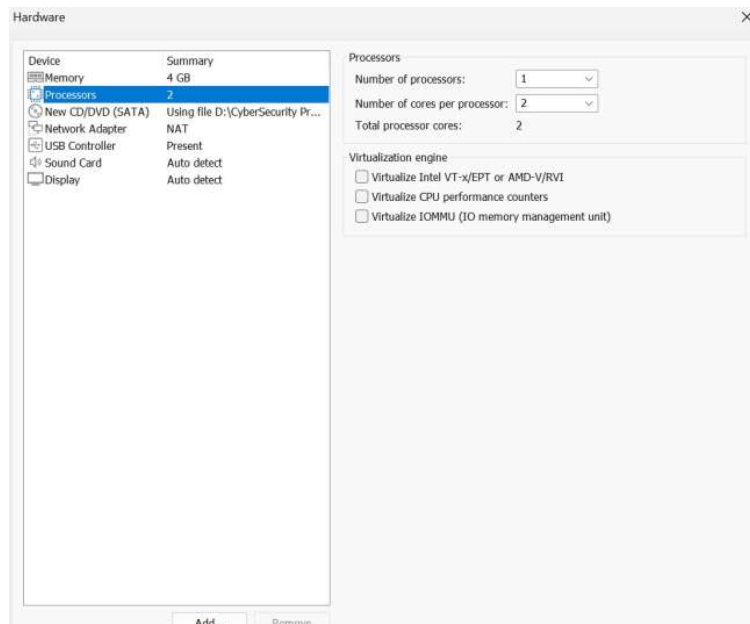


Figure 3.14 Hardware Configuration

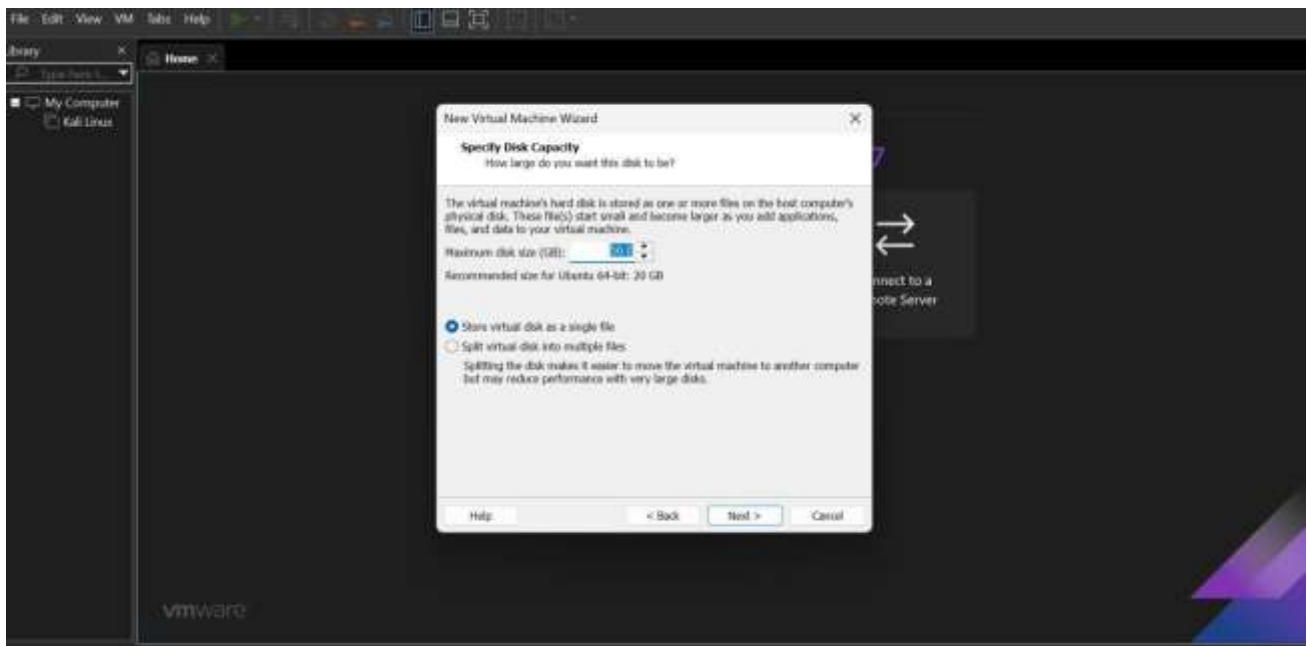


Figure 3.15 Disk Capacity

3.5.3 Ubuntu Server Installation:

Ubuntu Server 24.04 LTS was installed on the virtual machine. During installation, the network configuration was set to DHCP, and the virtual disk was configured for the server operating system. After installation was completed, the system was booted from the virtual disk and the Ubuntu Server login environment was verified.

The Ubuntu Server VM was subsequently used as the central server environment for deploying Wazuh and providing the backend infrastructure required for endpoint monitoring.

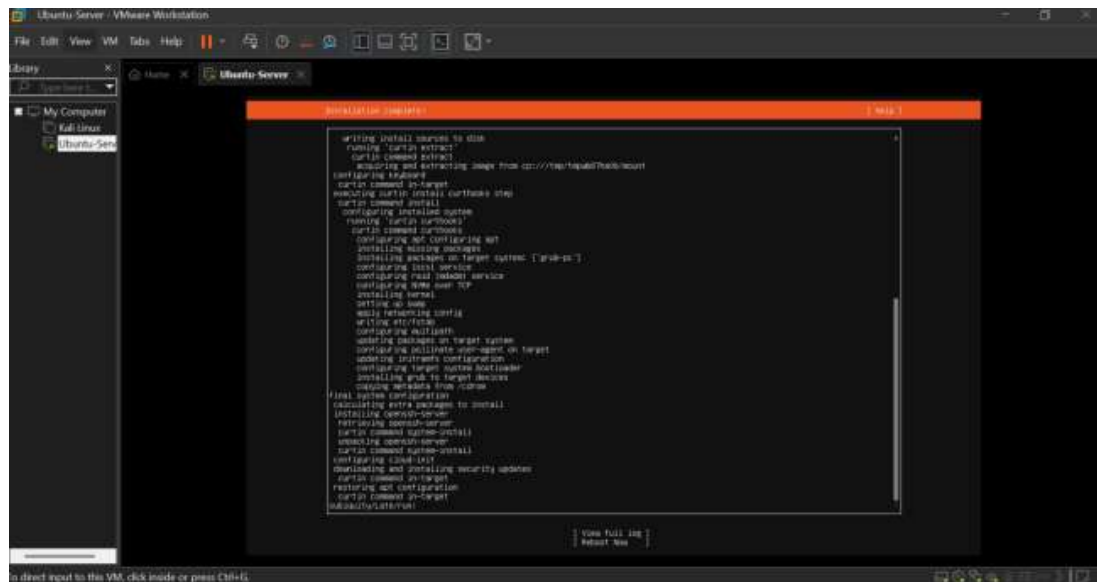


Figure 3.16 Ubuntu Installation

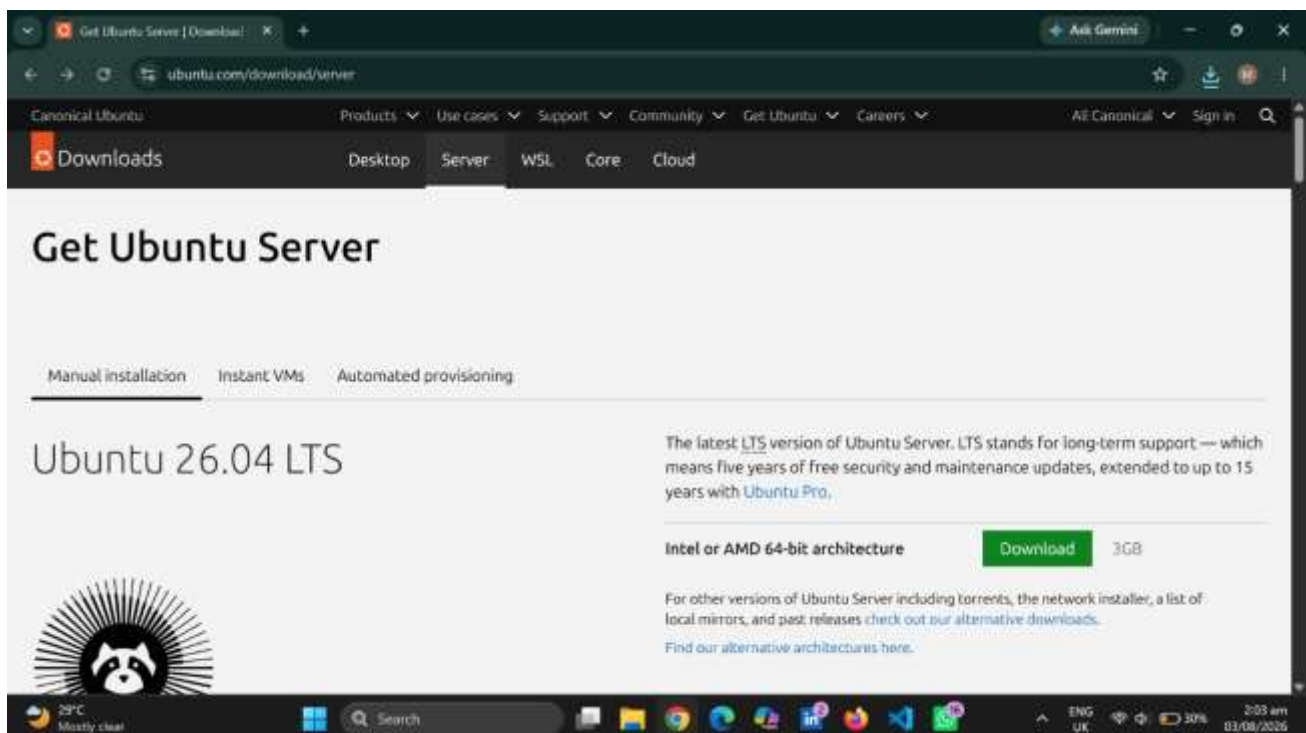


Figure 3.17 Ubuntu version

3.5.4 Ubuntu Server Network Configuration:

The Ubuntu Server VM was connected to the VMware NAT network to allow communication with the Windows endpoint and the required Wazuh components. The server's IP address was later used when configuring and verifying communication between the Wazuh Agent and the Wazuh server environment.

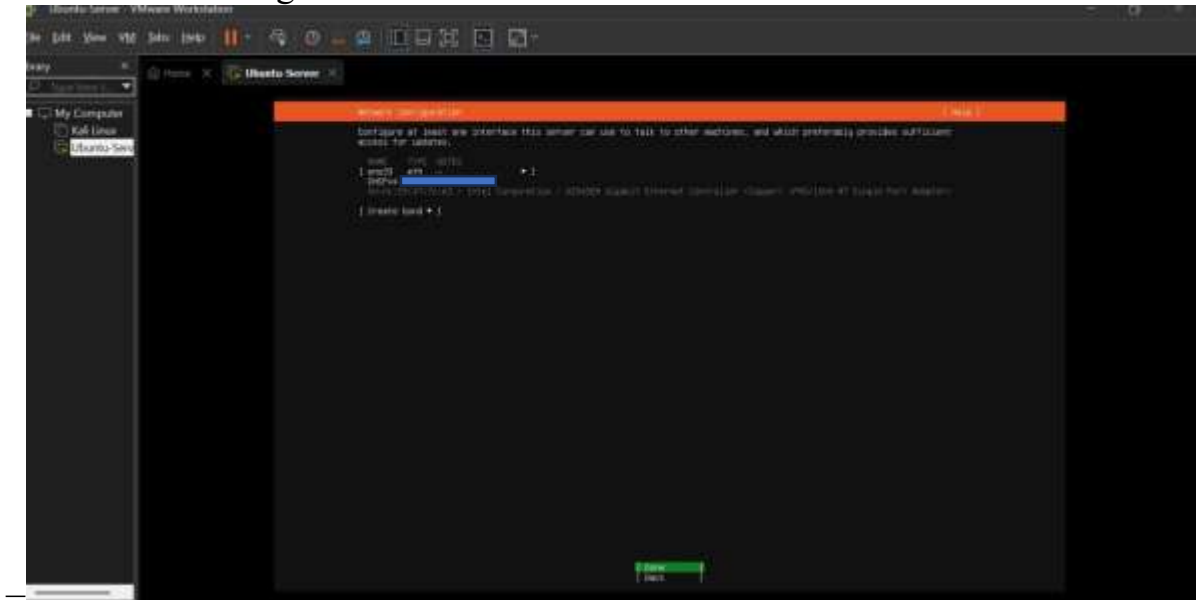


Figure 3.18 Network Configuration

Configuration	Ubuntu Server VM
Operating System	Ubuntu Server 24.04 LTS
Role	Wazuh Server
Virtualization	VMware Workstation Pro 17.6.4
Virtual Disk	50 GB
Network	NAT
CPU	2
RAM	4GB

3.6 Virtual Machine Resource Allocation:

The resources were distributed among virtual machines according to available host hardware while ensuring sufficient resources are available for working.

VM	OS	Purpose	CPU	RAM	Storage	Network
Windows	Windows 11 pro	Monitored Endpoint	2	4GB	60GB	NAT
Ubuntu	Ubuntu Server 24.04 LTS	Wazuh Server	2	4GB	50GB	NAT
Kali	Kali Linux	Security Assessment	2	4GB	50GB	NAT

3.7 Network Configuration:

The virtual machines were configured using NAT networking. This provided network connectivity for virtual machines.

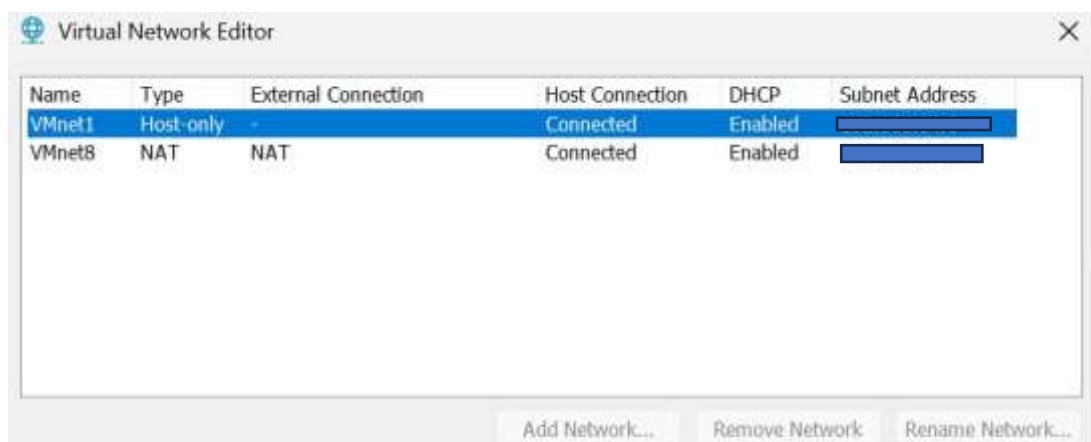


Figure 3.19 VMware Network configuration

3.8 IP Address Configuration:

The IP addresses of all VMs are find that are useful for further working.

System	Role	IP Address
Windows Pro	Monitor Endpoint	192.168.44.131
Ubuntu Server	Wazuh Server	192.168.44.130

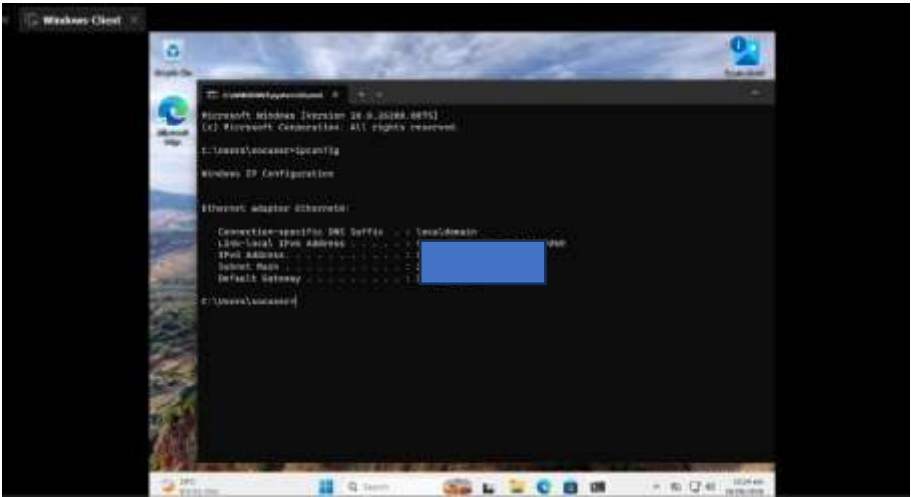


Figure 3.20 Window IP Address



Figure 3.21 Ubuntu IP Address

```

zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:4a:0e:ae brd ff:ff:ff:ff:ff:ff
    inet 10.10.10.10/24 scope global dynamic noprefixroute eth0
        valid_lft 1783sec preferred_lft 1783sec
    inet6 fe80::20c:29ff:fe4a:eae/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(kali@kali)-[~]
$

```

Figure 3.22 Kali IP address

3.9 Connectivity Verification:

Connectivity between the virtual machines was verified before beginning the monitoring and security-testing activities. This step ensured that the Windows endpoint could communicate with the Wazuh server and that the virtual laboratory was functioning correctly.

```

C:\Users\socuser>ping 10.10.10.10

Pinging 10.10.10.10 [10.10.10.10] with 32 bytes of data:
Reply from 10.10.10.10: bytes=32 time=1ms TTL=64
Reply from 10.10.10.10: bytes=32 time=1ms TTL=64
Reply from 10.10.10.10: bytes=32 time=1ms TTL=64
Reply from 10.10.10.10: bytes=32 time=1ms TTL=64

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\socuser>

```

Figure 3.23 connectivity Verified

3.10 Final Lab Architecture:

The final laboratory architecture consists of three virtual machines operating within VMware Workstation. Kali Linux provides the security assessment environment, Windows 11 Pro represents the monitored endpoint, and Ubuntu Server hosts the Wazuh

server environment. The Wazuh Agent installed on Windows collects relevant endpoint information and communicates with the Wazuh server. The Wazuh server processes and stores the collected security information, which is then accessed through the Wazuh Dashboard for monitoring and investigation.

4 Wazuh Server and Agent Implementation

4.1 Ubuntu Server Preparation:

4.1.1 Server Login and Environment Preparation:

the server was started and accessed using the configured administrative user account. The server environment was verified before beginning the Wazuh installation to ensure that the operating system and network configuration were functioning correctly.

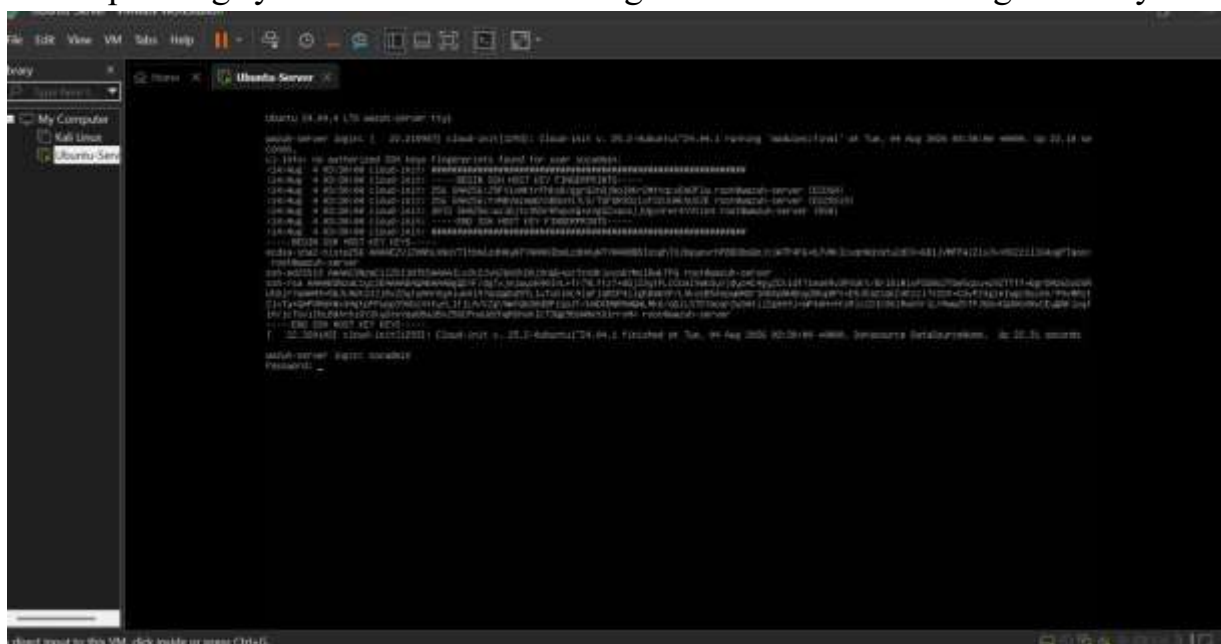


Figure 4.1 Ubnutu server after Installation

4.1.2 System Update:

Before deploying Wazuh, the Ubuntu package repository information was updated and available system packages were upgraded. This ensured that the server had the latest available package information and security updates required before beginning the Wazuh deployment.

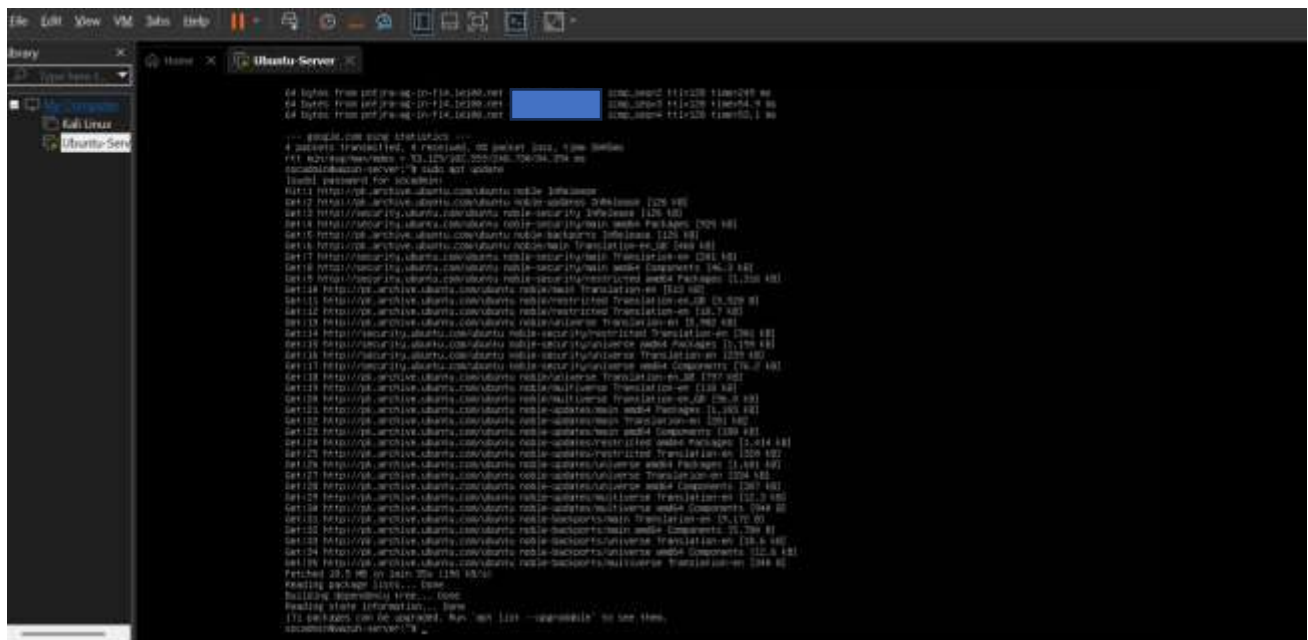


Figure 4.2 System Update

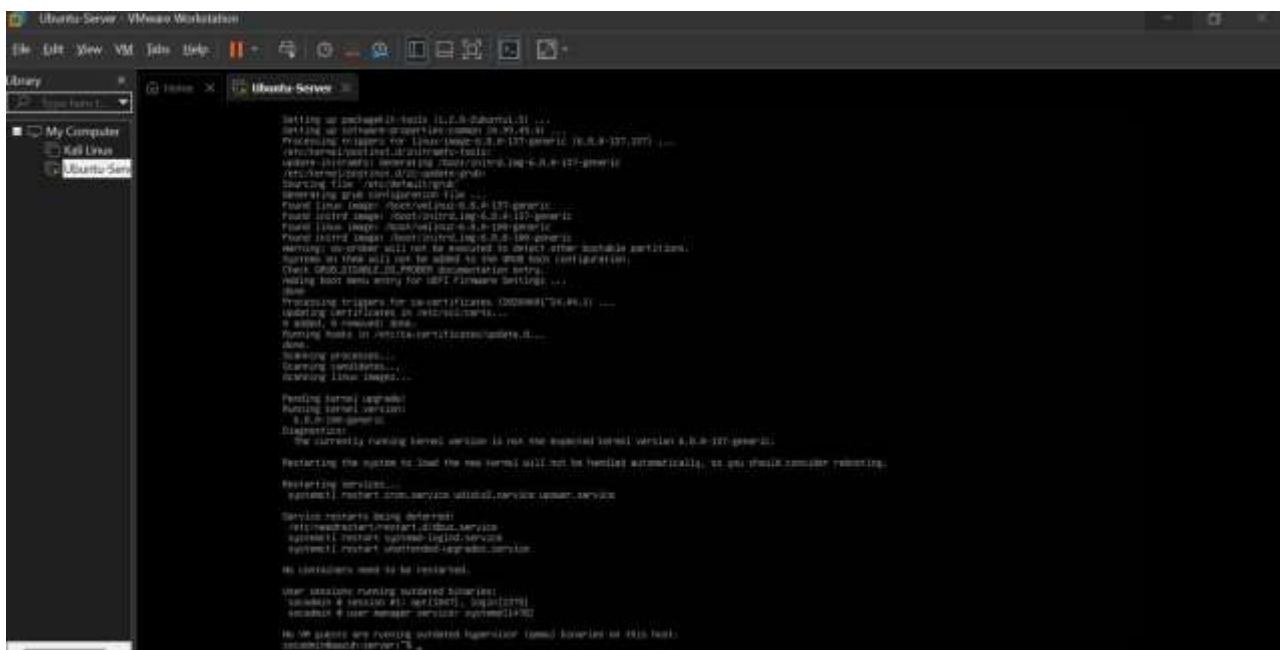


Figure 4.3 System upgraded

4.2 Wazuh Server Installation:

4.2.1 Wazuh Deployment:

The Wazuh server environment was deployed on Ubuntu Server using the official Wazuh installation method. The installation process configured the components required for centralized endpoint security monitoring. This deployment provided the backend infrastructure required to receive, process, index, and visualize security information collected from monitored endpoints.

The screenshot shows a terminal window titled "Ubuntu Server" running on a host named "My Computer". The terminal output displays the following information:

```

ubuntu@ubuntu:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION="24.04.4 LTS (Noble Numbat)"
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://ubuntu.com"
SUPPORT_URL="https://ubuntu.com/support"

```

Below the terminal output, there are several security notices and links:

- A notice about the Ubuntu 24.04.4 LTS security status, mentioning the release date (April 25, 2024) and the security team's commitment to providing timely updates.
- A link to the Ubuntu security page: <https://ubuntu.com/security>.
- A notice about the Ubuntu 24.04.4 LTS security status, mentioning the release date (April 25, 2024) and the security team's commitment to providing timely updates.
- A link to the Ubuntu security page: <https://ubuntu.com/security>.
- A notice about the Ubuntu 24.04.4 LTS security status, mentioning the release date (April 25, 2024) and the security team's commitment to providing timely updates.
- A link to the Ubuntu security page: <https://ubuntu.com/security>.

Figure 4.4 Wazuh Installation Command

4.3 Wazuh Server Components:

4.3.1 Wazuh Indexer:

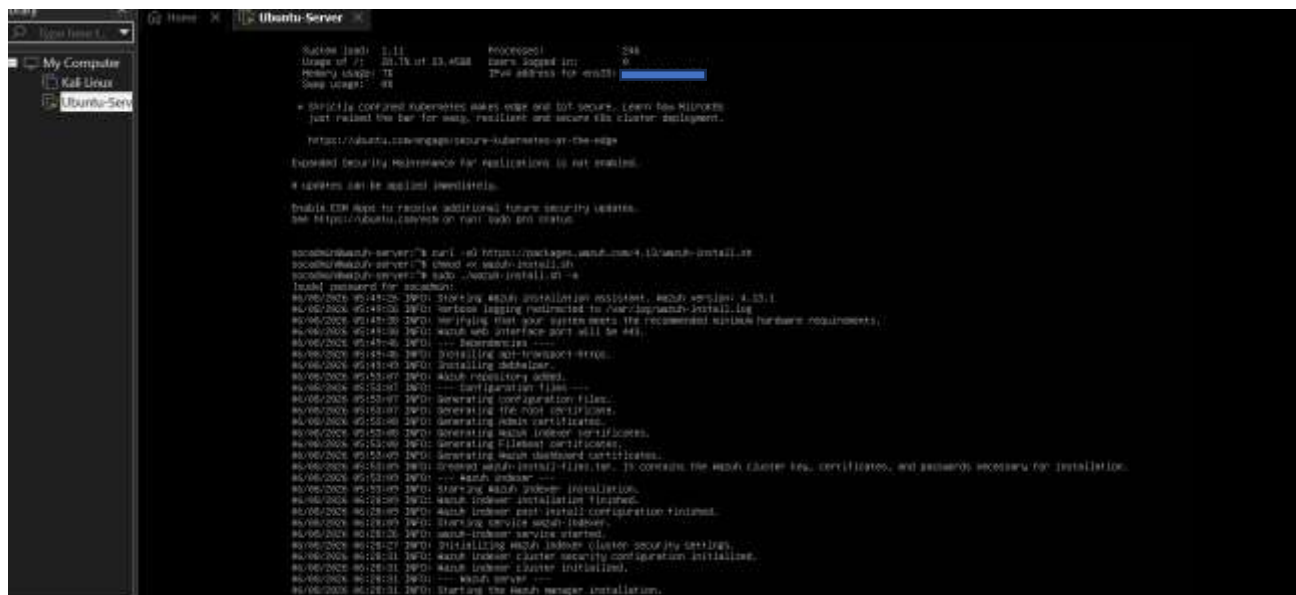
The **Wazuh Indexer** stores and indexes the security data processed by the Wazuh environment. Indexing allows collected events and alerts to be efficiently searched and retrieved during security monitoring and investigation.

```
succeeded@wecuh-server:~$ curl -sD https://packages.wecuh.com/4.13/wecuh-install.sh
succeeded@wecuh-server:~$ chmod +x wecuh-install.sh
succeeded@wecuh-server:~$ sudo ./wecuh-install.sh -y
[auth] Password for root@wecuh:
06/08/2026 05:49:26 INFO: Starting Wecuh installation assistant. Wecuh version: 4.13.1
06/08/2026 05:49:26 INFO: Verbose logging redirected to /var/log/wecuh-install.log
06/08/2026 05:49:28 INFO: Verifying that your system meets the recommended minimum hardware requirements.
06/08/2026 05:49:28 INFO: Wecuh web interface port will be 443.
06/08/2026 05:49:46 INFO: -- Dependencies --
06/08/2026 05:49:46 INFO: Installing apt-transport-https.
06/08/2026 05:49:49 INFO: Installing jq-helper.
06/08/2026 05:50:07 INFO: Wecuh repository added.
06/08/2026 05:50:07 INFO: --- Configuration files ---
06/08/2026 05:50:07 INFO: Generating configuration file.
06/08/2026 05:50:07 INFO: Generating the root certificate.
06/08/2026 05:50:08 INFO: Generating nodejs certificates.
06/08/2026 05:50:08 INFO: Generating wecuh indexer certificates.
06/08/2026 05:50:08 INFO: Generating Filebeat certificates.
06/08/2026 05:50:08 INFO: Generating wecuh dashboard certificates.
06/08/2026 05:50:09 INFO: Created wecuh-install-files.tar. It contains the Wecuh cluster key, certificates, and passwords necessary for installation.
06/08/2026 05:50:09 INFO: -- Wecuh Indexer --
06/08/2026 05:50:09 INFO: Starting Wecuh indexer installation.
```

Figure 4.5 Indexer Installation

4.3.2 Wazuh Manager:

The **Wazuh Manager** is the central component responsible for receiving and analyzing security information from Wazuh Agents. It applies detection rules to collected events and generates security alerts when configured conditions are met.



```
Wazuh
My Computer
Kali Linux
Ubuntu-Server

System load: 1.11
Usage of /: 20.1% of 33.4GB
Memory usage: 7%
Swap usage: 0%

Processes: 248
Users logged in: 0
IPv6 address for eth0:

* Strictly configured Kubernetes makes edge and IoT secure. Learn how Microsoft
just raised the bar for edge, resilient and secure K8s cluster deployment.

https://ubuntu.com/blog/kubernetes-at-the-edge

Expanded security references for applications is not enabled.
* updates can be applied immediately.

Enable ESM apps to receive additional future security updates.
See https://ubuntu.com/esm for more details.

root@ubuntu-server:~# curl -sO https://packages.wazuh.com/4.3/wazuh-install.sh
root@ubuntu-server:~# chmod +x wazuh-install.sh
root@ubuntu-server:~# ./wazuh-install.sh
Install package for wazuh-agent
Wazuh/2025-05-14:25:25: INFO: Starting Wazuh installation assistant, Wazuh version: 4.3.1
Wazuh/2025-05-14:25:25: INFO: System logging requirements for /var/log/wazuh-agent.log
Wazuh/2025-05-14:25:25: INFO: Verifying that your system meets the recommended minimum hardware requirements.
Wazuh/2025-05-14:25:25: INFO: Wazuh web interface port will be 443.
Wazuh/2025-05-14:25:25: INFO: --- Dependencies ---
Wazuh/2025-05-14:25:25: INFO: Installing dependencies.
Wazuh/2025-05-14:25:25: INFO: Installing database.
Wazuh/2025-05-14:25:25: INFO: Wazuh repository setup.
Wazuh/2025-05-14:25:25: INFO: --- Configuration files ---
Wazuh/2025-05-14:25:25: INFO: Generating configuration files.
Wazuh/2025-05-14:25:25: INFO: Generating the Wazuh certificates.
Wazuh/2025-05-14:25:25: INFO: Generating Wazuh certificates.
Wazuh/2025-05-14:25:25: INFO: Generating Wazuh internal certificates.
Wazuh/2025-05-14:25:25: INFO: Generating Wazuh dashboard certificates.
Wazuh/2025-05-14:25:25: INFO: Wazuh cluster key, certificates, and passwords necessary for installation.
Wazuh/2025-05-14:25:25: INFO: --- Wazuh daemon ---
Wazuh/2025-05-14:25:25: INFO: Starting Wazuh daemon installation.
Wazuh/2025-05-14:25:25: INFO: Wazuh daemon installation finished.
Wazuh/2025-05-14:25:25: INFO: Wazuh daemon post-install configuration finished.
Wazuh/2025-05-14:25:25: INFO: Stopping service wazuh-daemon.
Wazuh/2025-05-14:25:25: INFO: Wazuh daemon service started.
Wazuh/2025-05-14:25:25: INFO: Initializing Wazuh internal cluster security settings.
Wazuh/2025-05-14:25:25: INFO: Wazuh internal cluster security configuration initialized.
Wazuh/2025-05-14:25:25: INFO: Wazuh internal cluster initialized.
Wazuh/2025-05-14:25:25: INFO: --- Wazuh server ---
Wazuh/2025-05-14:25:25: INFO: Starting the Wazuh manager installation.
```

Figure 4.6 Manager installation

4.3.3 Wazuh Dashboard:

The **Wazuh Dashboard** provides the graphical interface used by security analysts to monitor agents, investigate events, review alerts, perform threat hunting, and analyze security information collected from monitored endpoints.

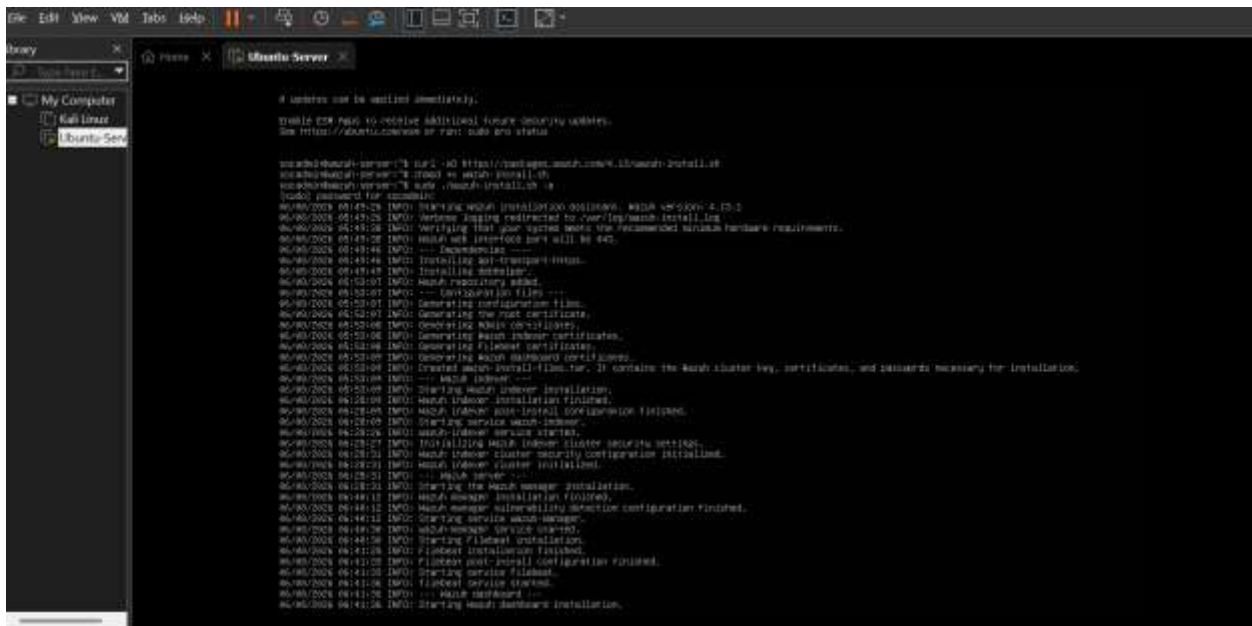


Figure 4.7 Dashboard Installation

4.4 Wazuh Installation Verification:

After the installation was completed, the Wazuh environment was verified to ensure that the required services were operational. Successful service operation confirmed that the Wazuh server environment was ready to receive information from monitored endpoints.

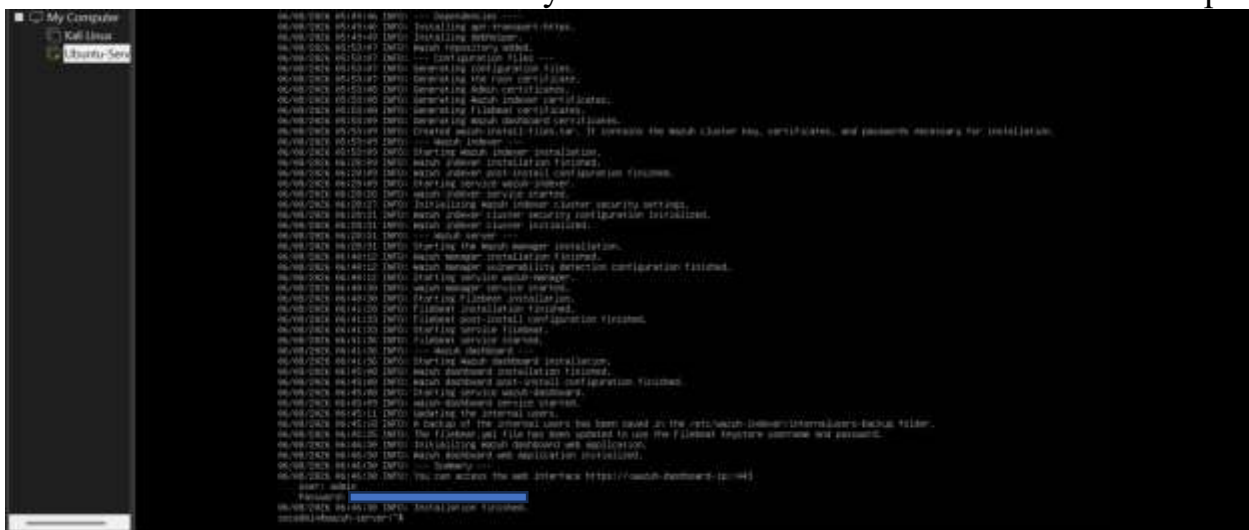


Figure 4.8 Wazuh Installation Completed

4.5 Wazuh Dashboard Access:

After completing the server deployment, the Wazuh Dashboard was accessed through a web browser using the configured Wazuh Dashboard address. The Dashboard provided access to the monitoring and analysis features used throughout the project.


```

06/06/2026 06:40:50 INFO: Starting Filebeat installation.
06/06/2026 06:41:28 INFO: Filebeat installation finished.
06/06/2026 06:41:33 INFO: Filebeat post-install configuration finished.
06/06/2026 06:41:33 INFO: Starting wazuh filebeat.
06/06/2026 06:41:36 INFO: Filebeat service started.
06/06/2026 06:41:36 INFO: --- Wazuh Dashboard ---
06/06/2026 06:41:36 INFO: Starting Wazuh dashboard installation.
06/06/2026 06:45:09 INFO: Wazuh dashboard installation finished.
06/06/2026 06:45:09 INFO: Wazuh dashboard post-install configuration finished.
06/06/2026 06:45:09 INFO: Starting wazuh dashboard.
06/06/2026 06:45:09 INFO: Wazuh dashboard service started.
06/06/2026 06:45:11 INFO: Updating the internal users.
06/06/2026 06:45:18 INFO: A backup of the internal users has been saved in the /etc/wazuh-indexer/internalusers-backup folder.
06/06/2026 06:45:35 INFO: The filebeat.yml file has been updated to use the Filebeat keystone username and password.
06/06/2026 06:46:30 INFO: Initializing Wazuh dashboard web application.
06/06/2026 06:46:30 INFO: Wazuh dashboard web application initialized.
06/06/2026 06:46:39 INFO: --- Summary ---
06/06/2026 06:46:39 INFO: You can access the web interface https://wazuh-dashboard-ip:443
Users: 4
Password:
06/06/2026 06:46:39 INFO: Installation finished.
wazuh@wazuh-server:~$

```

Figure 4.9 Dashboard Credentials



Figure 4.10 Login page

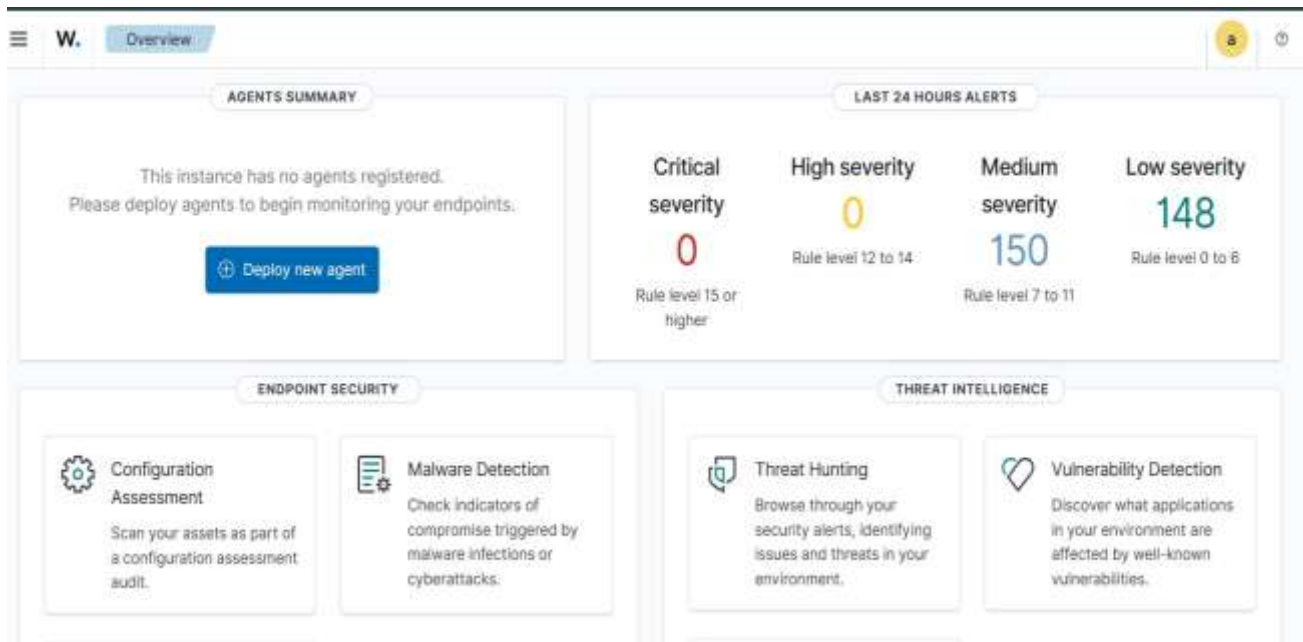


Figure 4.11 Dashboard Homepage

4.6 Windows Wazuh Agent Installation:

Wazuh agent(version 4.13.4) is downloaded from official source.

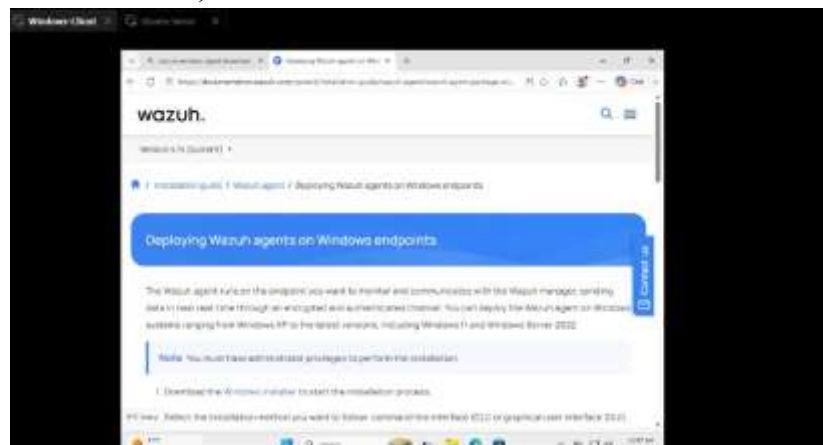


Figure 4.12 Agent downloaded

To monitor the Windows 11 Pro endpoint, the Wazuh Agent was installed on the Windows virtual machine. The agent acts as the endpoint-side component of the Wazuh

architecture and is responsible for collecting relevant Windows security information and communicating with the Wazuh Manager.

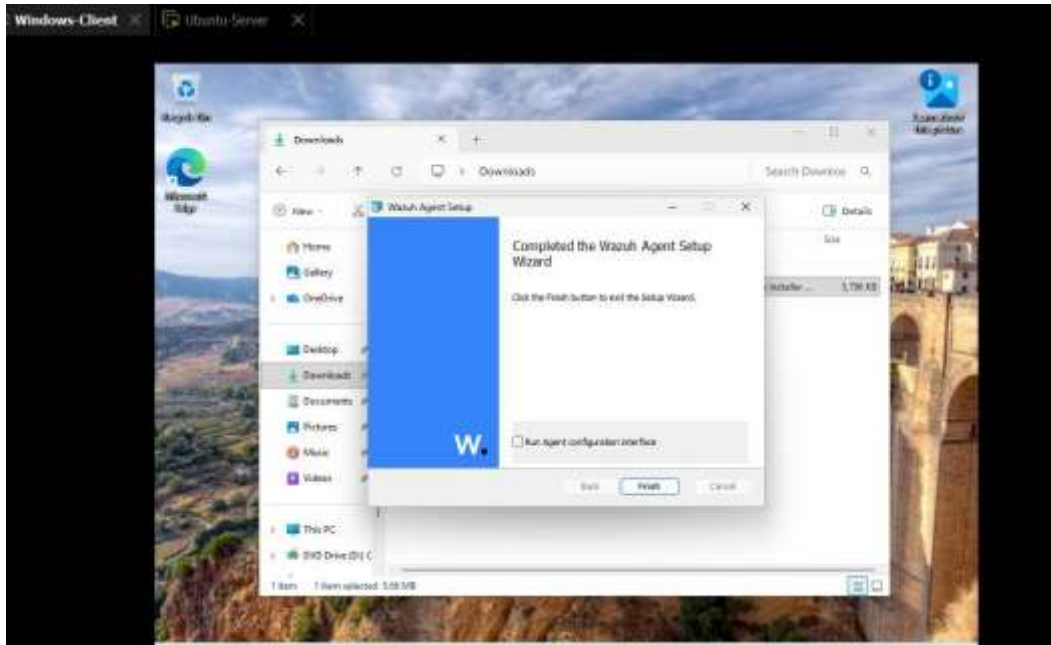


Figure 4.13 Agent Installation

4.7 Wazuh Agent Configuration:

During the agent configuration, the Windows endpoint was configured to communicate with the Wazuh server running on Ubuntu Server. The server's network information was specified so that the agent could establish communication with the central Wazuh Manager.

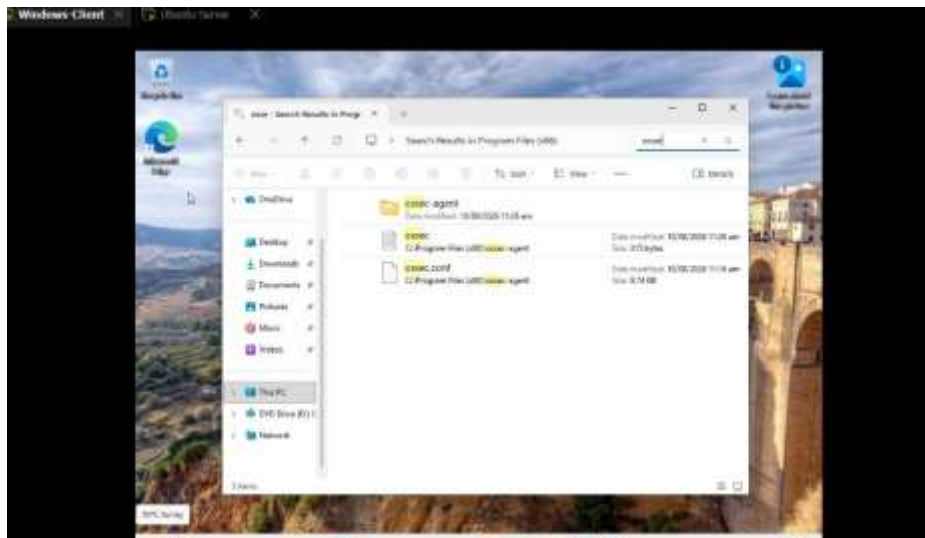


Figure 4.14 Agent Configuration file

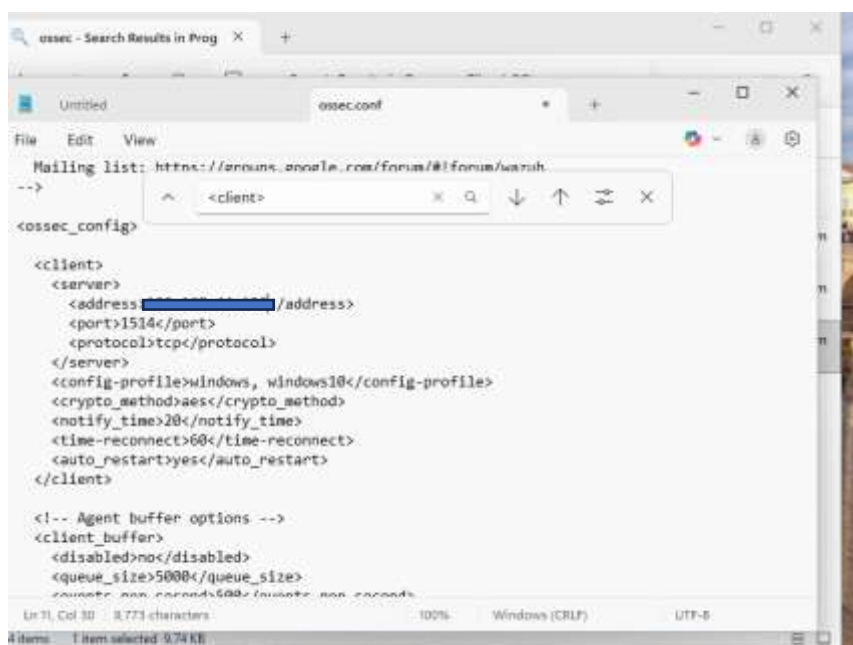


Figure 4.15 IP configured

4.8 Wazuh Agent Service:

After installation and configuration, the Wazuh Agent service was started on the Windows endpoint. The service was required to remain operational so that security information could be collected and transmitted to the Wazuh server.

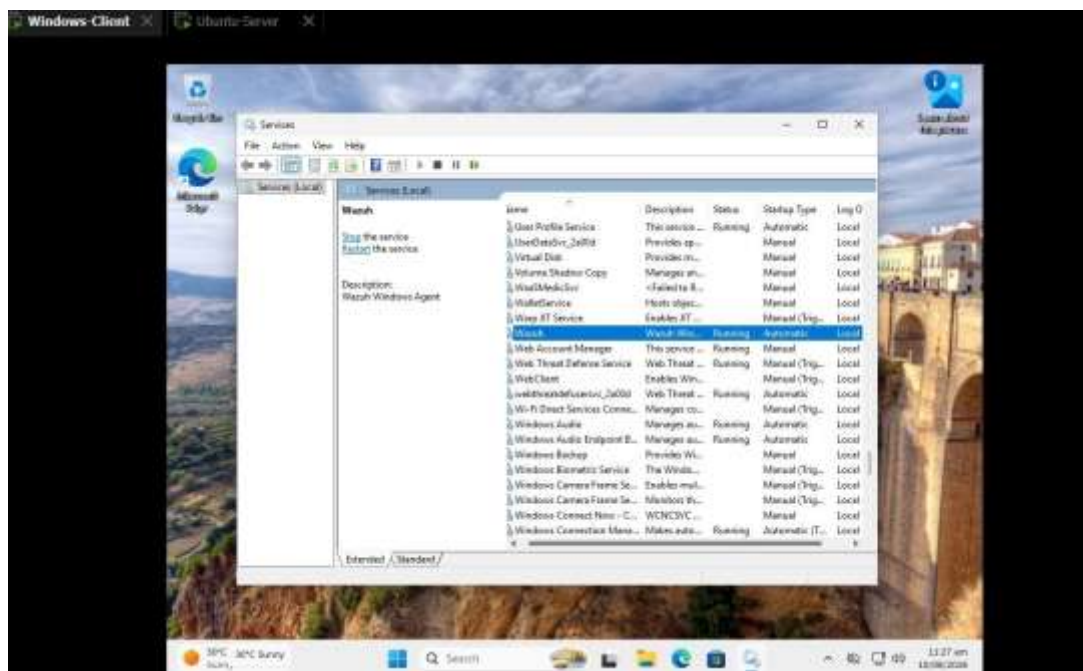


Figure 4.16 wazuh agent service running

4.9 Agent Registration and Activation:

After configuring the Windows Agent, the endpoint was registered with the Wazuh Manager. The Wazuh Dashboard was then checked to confirm that the Windows endpoint appeared as an active agent. The active status indicated that the endpoint had successfully established communication with the Wazuh monitoring infrastructure.

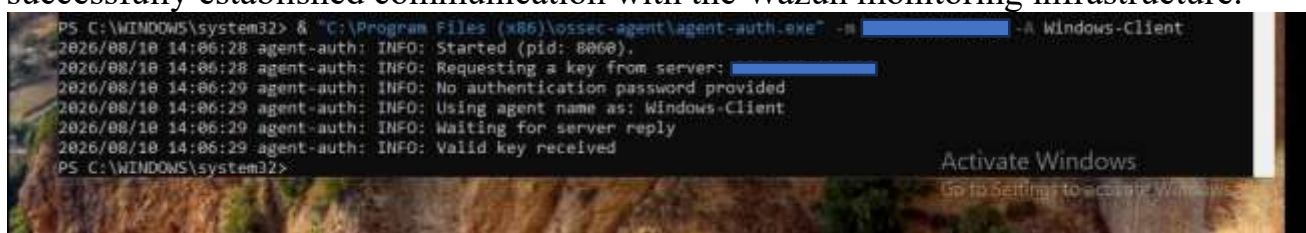


Figure 4.17 Agent Authentication

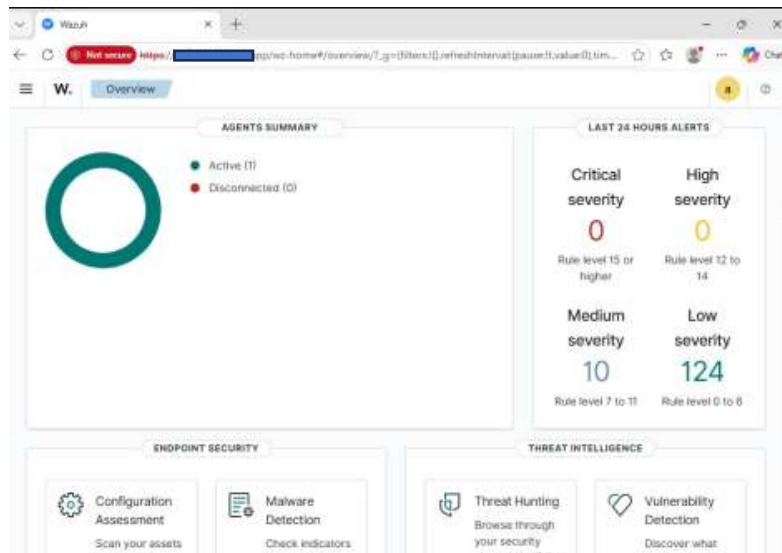


Figure 4.18 Agent Active

4.10 Agent Communication Verification:

Network connectivity between the Windows endpoint and the Wazuh Server was verified through a successful TCP connection. This confirmed that the endpoint was able to establish network-level communication with the Wazuh monitoring infrastructure.

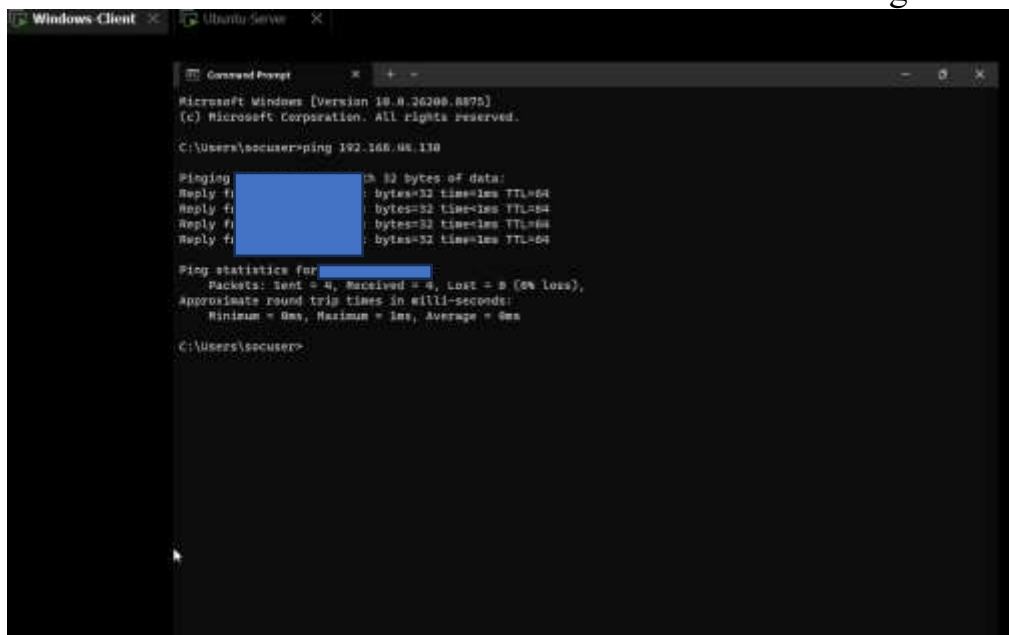


Figure 4.19 Successful connectivity

5 Windows Security Assessment

5.1 Windows Update Status:

Windows Update was checked to determine whether the endpoint had the latest available security and system updates. Keeping the operating system updated is important because security updates may address known vulnerabilities that could otherwise be exploited by attackers.



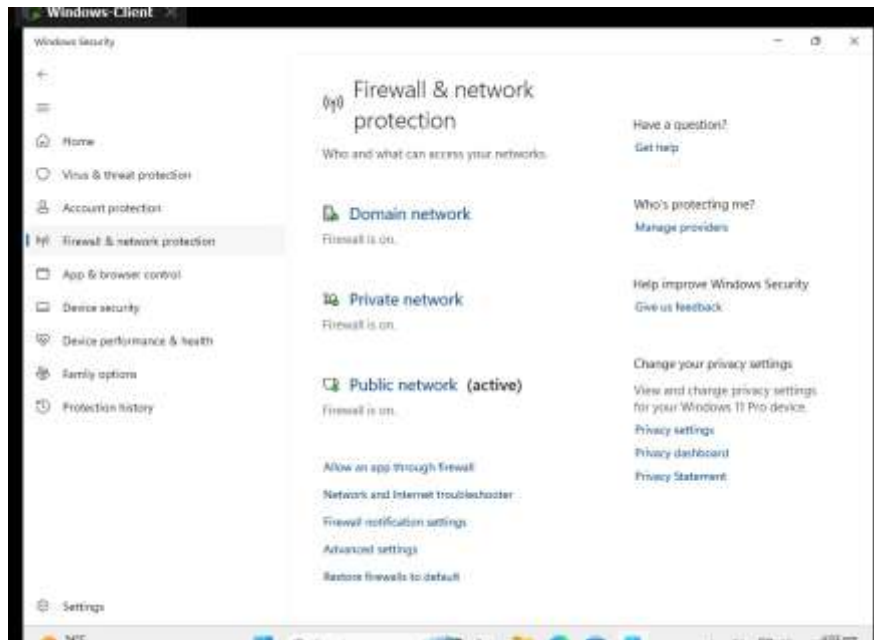
Figure 5.1 Windows Update Status

Finding:

The Windows endpoint was found to be up to date at the time of assessment.

5.2 Windows Firewall Assessment:

Windows Firewall was examined to determine the protection status of the network interfaces on the Windows endpoint. The firewall provides network-level protection by controlling inbound and outbound network traffic according to configured rules.



Figure

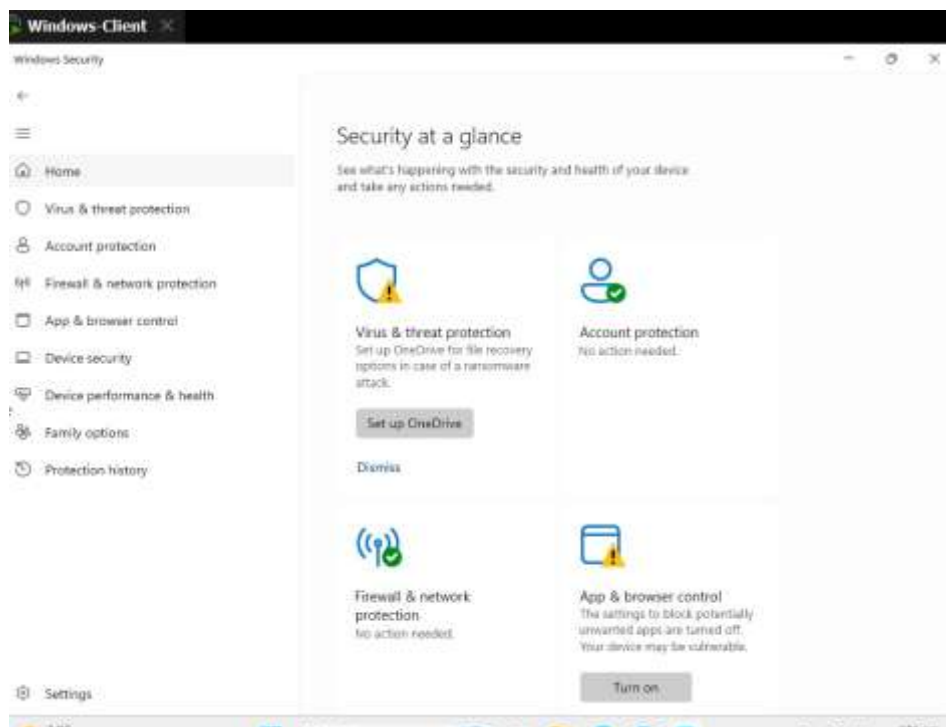
5.2 Firewall Status

Finding:

Windows Firewall was enabled during the initial security assessment.

5.3 Windows Defender:

Microsoft Defender Antivirus was checked to determine the current malware protection status of the Windows endpoint. Defender provides built-in protection against malware and other potentially unwanted threats.



Figure

5.3 Defender Security status

Findings:

Microsoft Defender/Windows Security was active on the Windows endpoint. The security dashboard showed that account protection and firewall/network protection required no immediate action. However, the Virus & threat protection section displayed a notification related to OneDrive file-recovery options for ransomware protection, while App & browser control also indicated that some protection settings were turned off. Therefore, the endpoint had active built-in security controls, but not all optional protection features were fully configured.

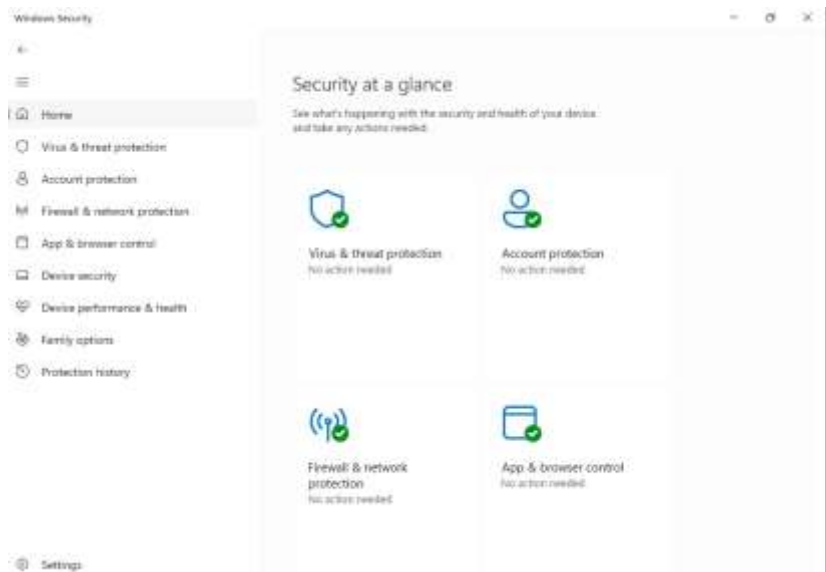
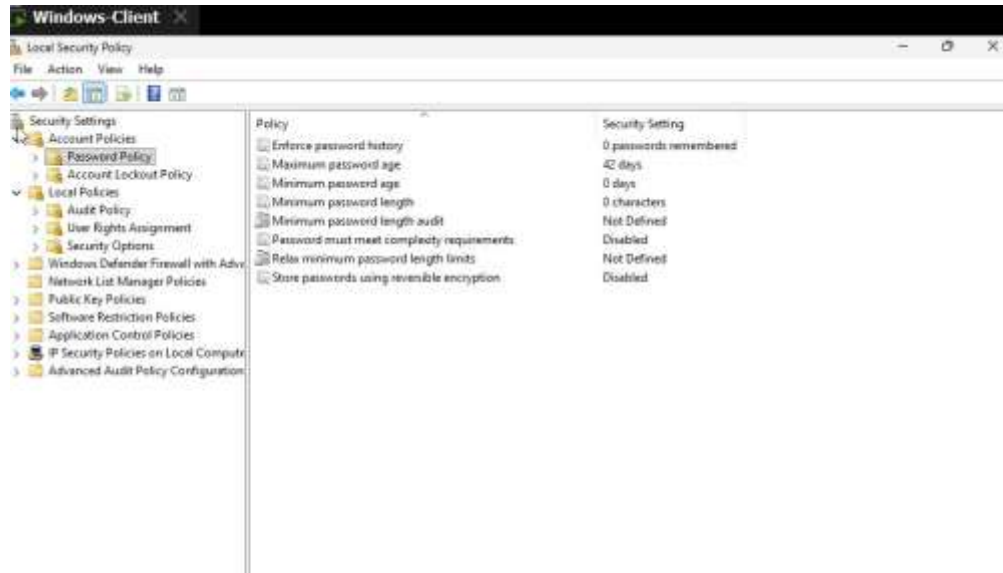


Figure 5.4 After assessment

5.4 Password Policy:

The Windows password policy was reviewed to determine the configured authentication requirements. Password policies are important because weak password requirements can make accounts more susceptible to password guessing and brute-force attacks.



Figure

5.5 Password Policy

Findings:

The initial password policy contained several weaknesses. The minimum password length was configured as 0 characters and password complexity requirements were

disabled. Password history was also configured to remember 0 previous passwords, while the minimum password age was set to 0 days. These settings provided limited protection against weak or repeatedly reused passwords. The maximum password age was configured as 42 days, and reversible password encryption was disabled.

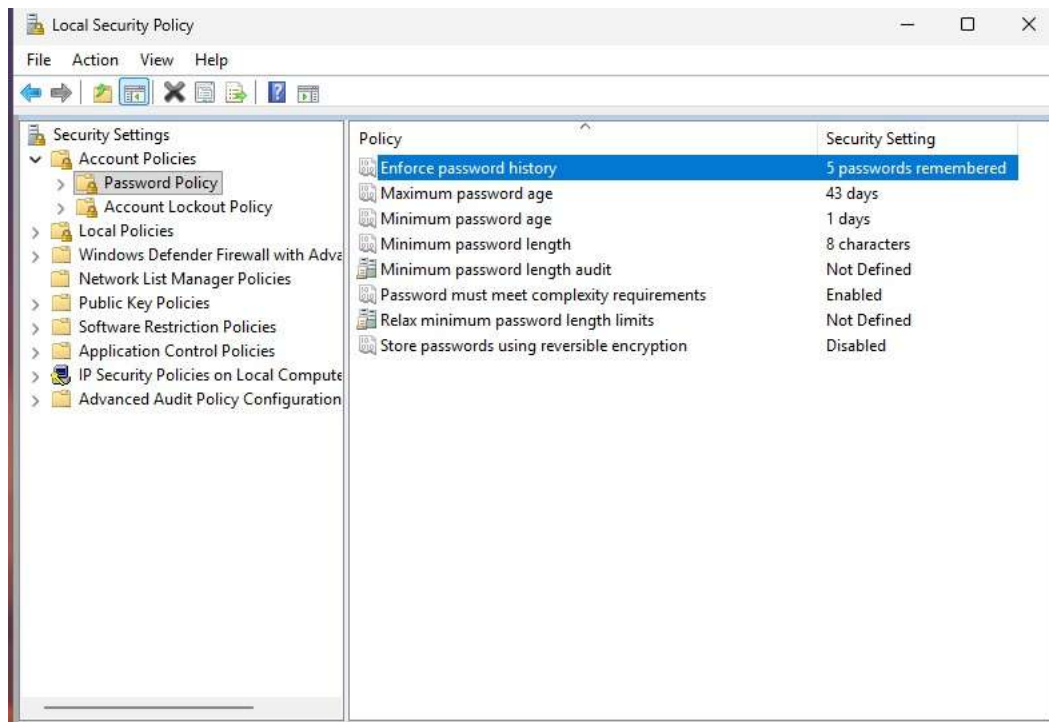


Figure 5.6 After Assessment

The identified password-policy weaknesses were addressed by enforcing a minimum password length of 8 characters, enabling password complexity requirements, retaining five previous passwords in password history, and setting a minimum password age of one day. The maximum password age remained at 42 days, while reversible password encryption remained disabled. The password policy was strengthened to provide better protection against weak and repeatedly reused passwords. The updated configuration improved the authentication security baseline of the Windows endpoint.

5.5 Running Services:

Running Windows services were reviewed to identify services currently operating on the endpoint. Unnecessary or insecure services can increase the attack surface because they may expose additional functionality or network interfaces.

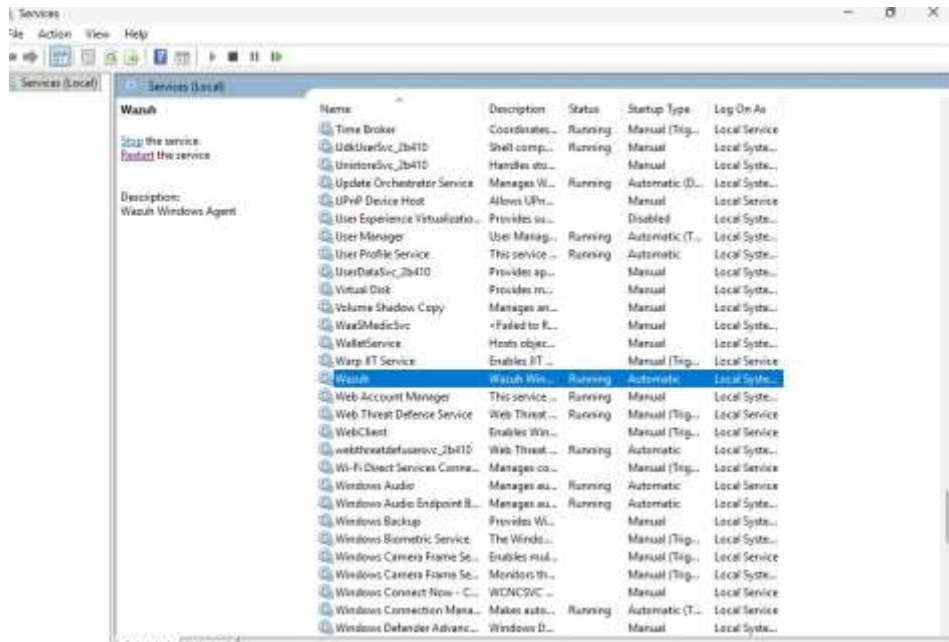


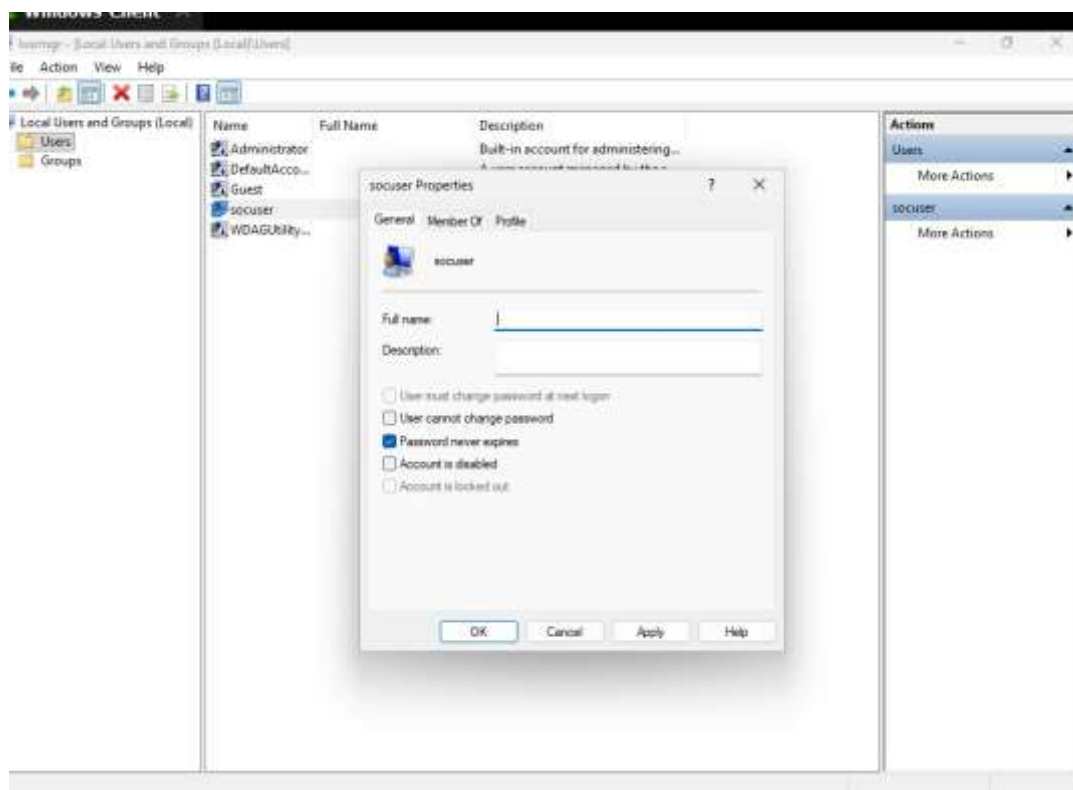
Figure 5.7 Wazuh Service Running

Findings:

Particular attention was given to the **Wazuh Agent** service because it is responsible for communication between the Windows Client and Wazuh Manager. The Wazuh Agent service was confirmed to be running.

5.6 User Account Assessment:

Local user accounts were examined to identify the accounts configured on the Windows endpoint and to understand their privilege levels. Reviewing user accounts is important because unnecessary or highly privileged accounts can increase the attack surface of a system.



Figure

5.8 User Account

Findings:

After assessment ,it is found that all other accounts are disabled except SOCUSER account and it has administrative privileges.No other suspicious account exist or enabled.

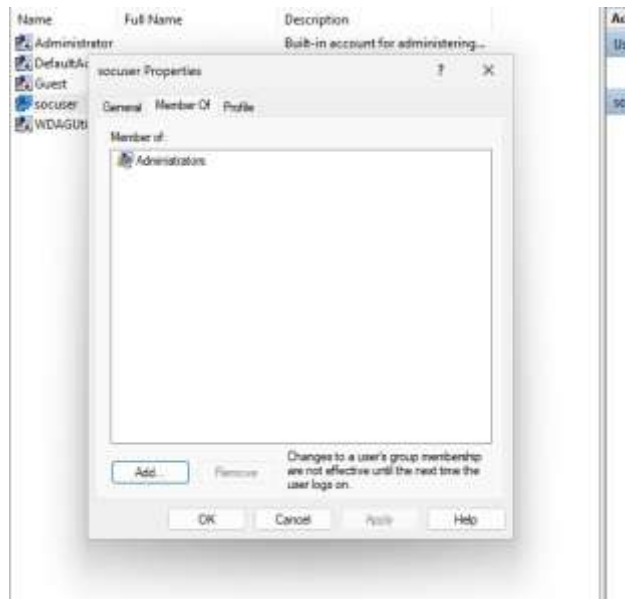
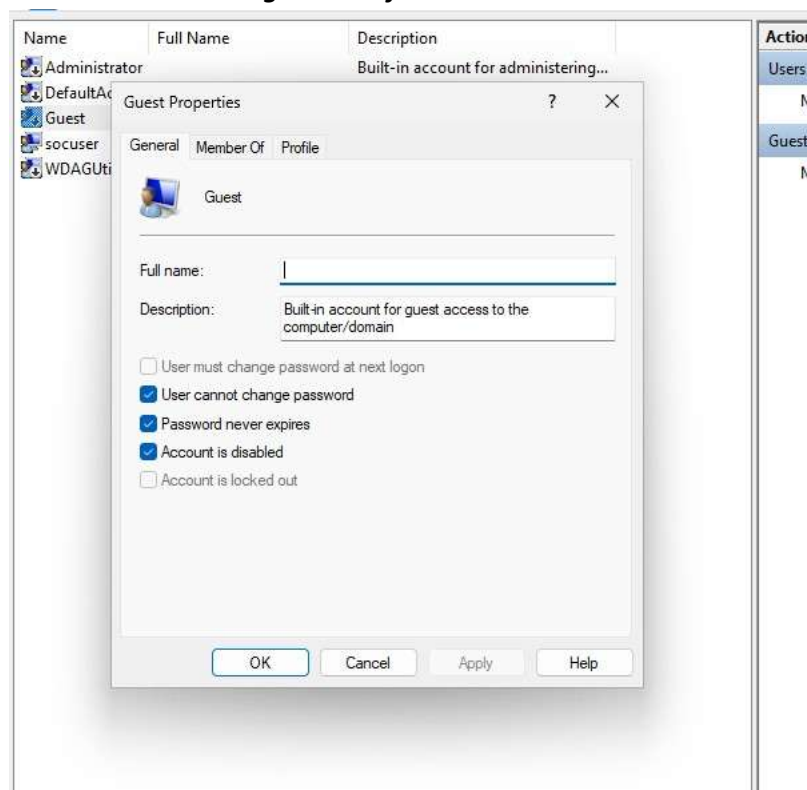


Figure 5.9 After assessment



Figure

5.10 Guest account disabled

5.7 Installed Software Assessment:

The installed applications list was reviewed to identify software present on the Windows endpoint. The visible applications are primarily standard Microsoft applications, including Sound Recorder, Sticky Notes, Terminal, Weather, Web Media Extensions,

and Xbox. The **Wazuh Agent 4.13.1** is also installed as the endpoint monitoring component used in this project. No obviously unauthorized or suspicious third-party application was identified in the displayed list. this Windows 11 VM was created specifically for my project, it is a relatively clean lab endpoint. I haven't installed a bunch of random third-party software on it.

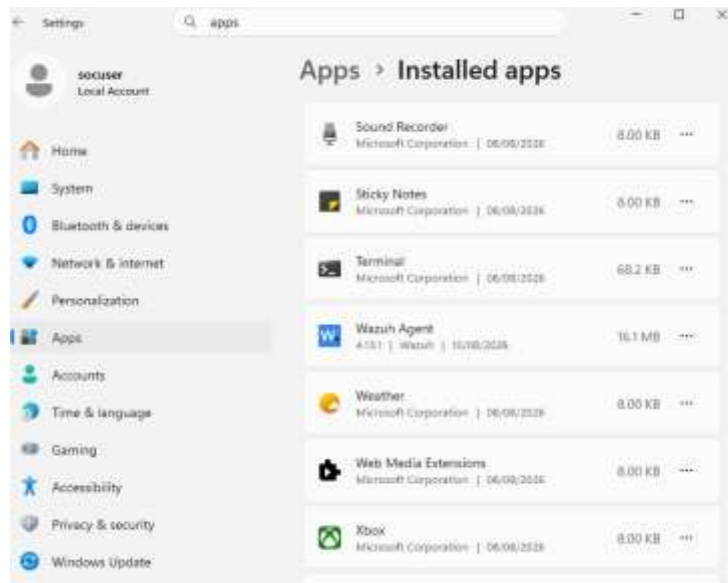


Figure 5.11 Wazuh Software installed

5.8 Open\Listening Ports Assessment:

Kali Linux was used as a security assessment machine within the isolated virtual laboratory. Nmap was used to identify open and listening network ports on the Windows endpoint. Port scanning helps identify network services that are exposed and therefore potentially increase the attack surface.

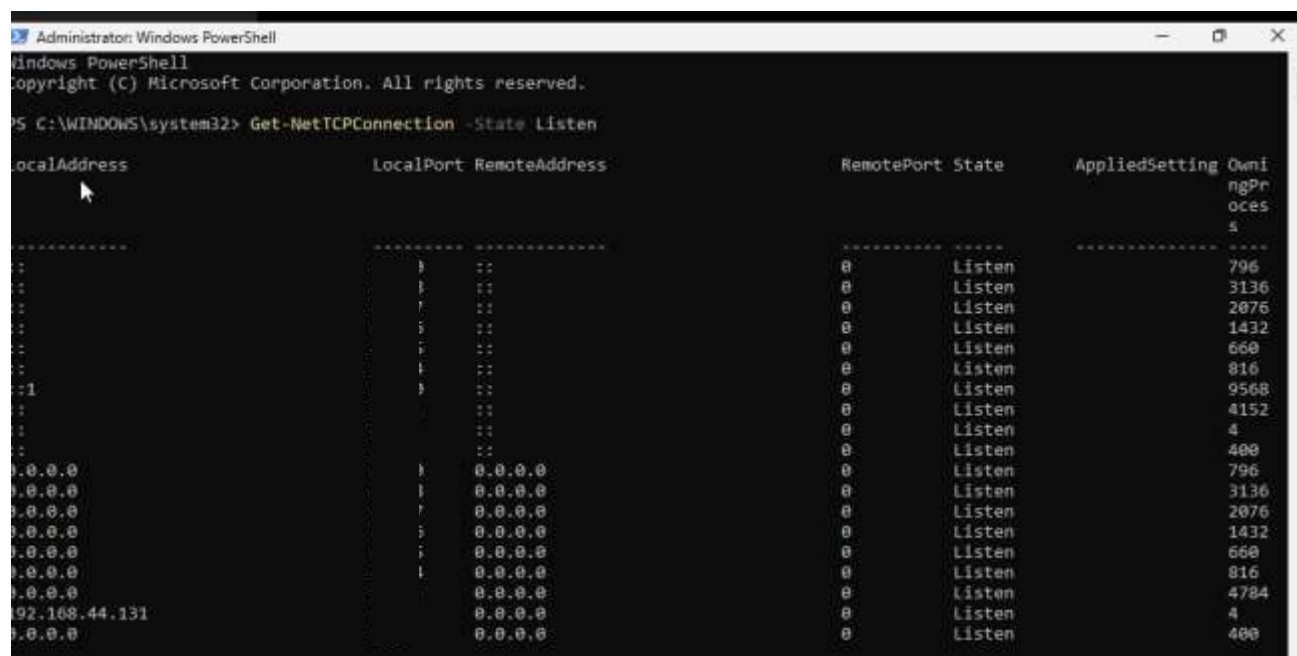
```
(kali@kali)-[~]
$ nmap [redacted]
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-13 08:51 +0500
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 72.00% done; ETC: 08:52 (0:00:06 remaining)
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 72.20% done; ETC: 08:52 (0:00:06 remaining)
Nmap scan report for [redacted]
Host is up (0.00039s latency).
All 1000 scanned ports on [redacted] are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: [redacted] (VMware)

Nmap done: 1 IP address (1 host up) scanned in 21.80 seconds
```

Figure 5.12 Nmap scan

The Nmap scan is conducted on target host **192.168.44.131** confirmed that host is active/online. However, all 1000 scanned TCP ports were identified as filtered (no response). This indicates that an active host firewall (Windows Defender Firewall) is blocking scan packets and preventing direct port discovery.

Another way, we used PowerShell also for checking the Listening Ports.



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-NetTCPConnection -State Listen

LocalAddress      LocalPort RemoteAddress      RemotePort State      AppliedSetting OwningProcess
-----
:::               :         :::               :         Listen      :         796
:::               :         :::               :         Listen      :         3136
:::               :         :::               :         Listen      :         2076
:::               :         :::               :         Listen      :         1432
:::               :         :::               :         Listen      :         660
:::               :         :::               :         Listen      :         816
:::               :         :::               :         Listen      :         9568
:::               :         :::               :         Listen      :         4152
:::               :         :::               :         Listen      :         4
:::               :         :::               :         Listen      :         400
0.0.0.0           :         0.0.0.0           :         Listen      :         796
0.0.0.0           :         0.0.0.0           :         Listen      :         3136
0.0.0.0           :         0.0.0.0           :         Listen      :         2076
0.0.0.0           :         0.0.0.0           :         Listen      :         1432
0.0.0.0           :         0.0.0.0           :         Listen      :         660
0.0.0.0           :         0.0.0.0           :         Listen      :         816
0.0.0.0           :         0.0.0.0           :         Listen      :         4784
192.168.44.131   :         0.0.0.0           :         Listen      :         4
0.0.0.0           :         0.0.0.0           :         Listen      :         400
```

It displayed all the active listening ports that are available.

5.9 Windows Event Log Assessment:

Windows Event Viewer was examined to review security and system events generated by the endpoint. Event logs provide information about authentication, account management, system activity, application activity, and other security-relevant operations.

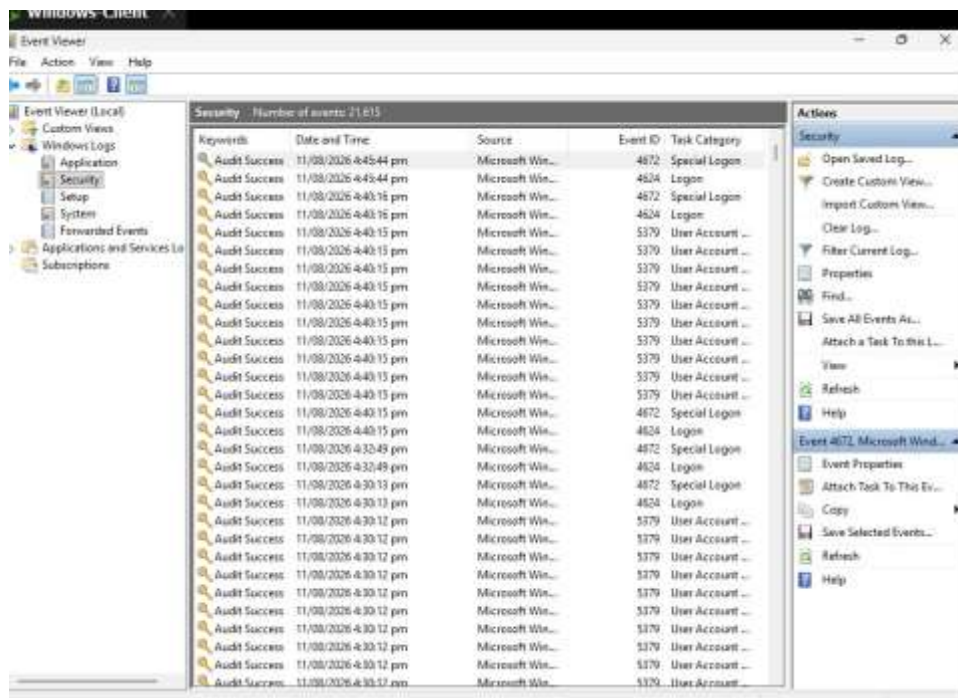


Figure 5.13 Security Logs

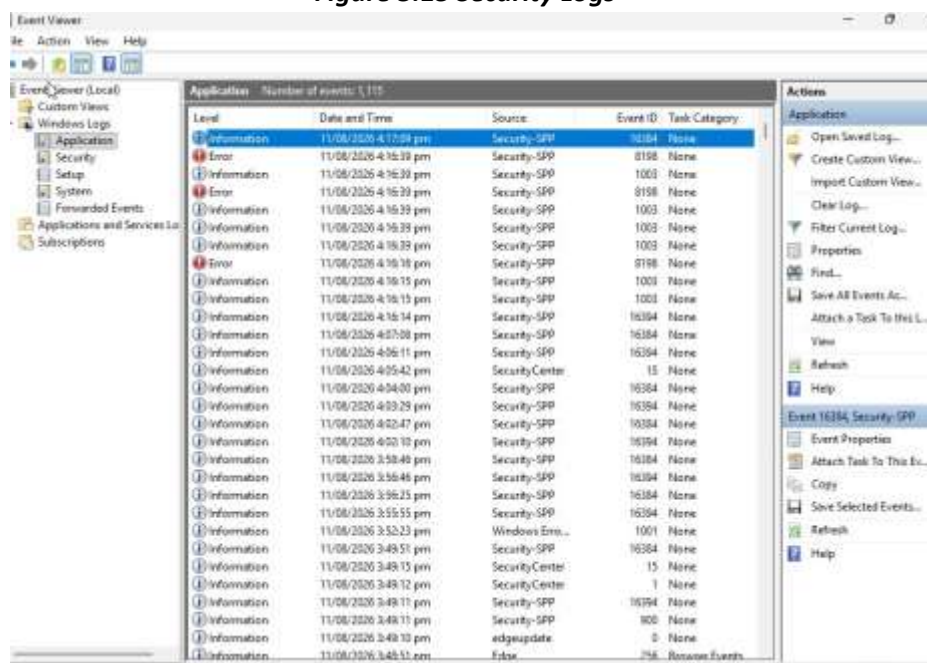


Figure 5.14 Application Logs

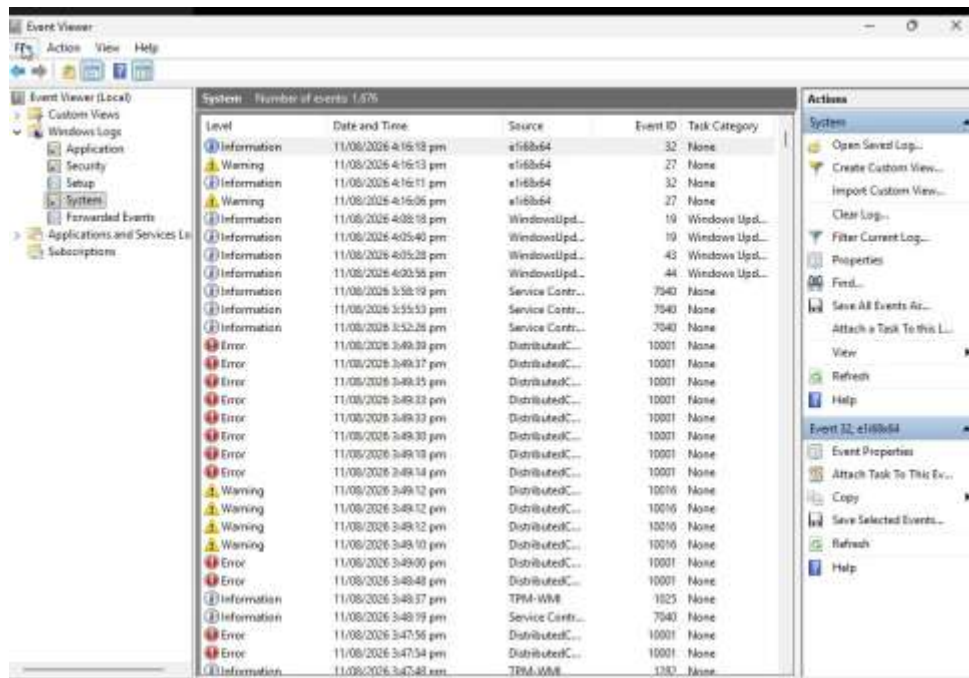


Figure 5.15 System Logs

Windows Event Viewer was reviewed to confirm that the Windows Client was generating system and security logs.

6 Wazuh Dashboard Monitoring

6.1 Dashboard Overview:

Dashboard provided a centralized view of the monitored Windows endpoint and allowed security events to be investigated without manually reviewing every event directly on the endpoint. It shows Wazuh Rules with severity Levels defined in ossec.conf file.

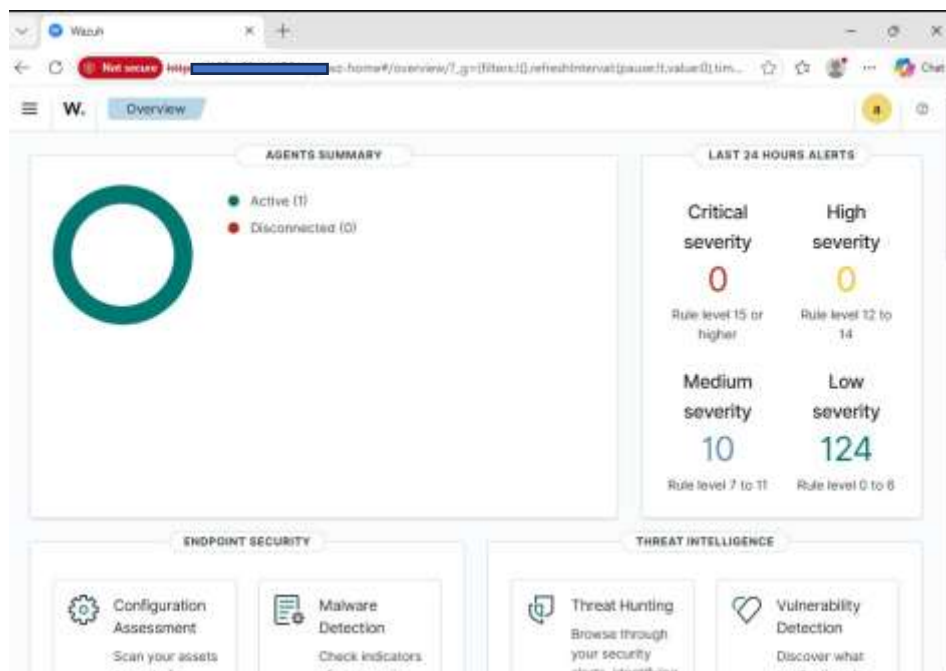


Figure 6.1 Dashboard Overview

Dashboard provides access to several sections:

- Agents Management
- Security Events
- FIM
- Alerts

6.2 Agents:

It provides the agent summary ,we can check active and disconnected agents.

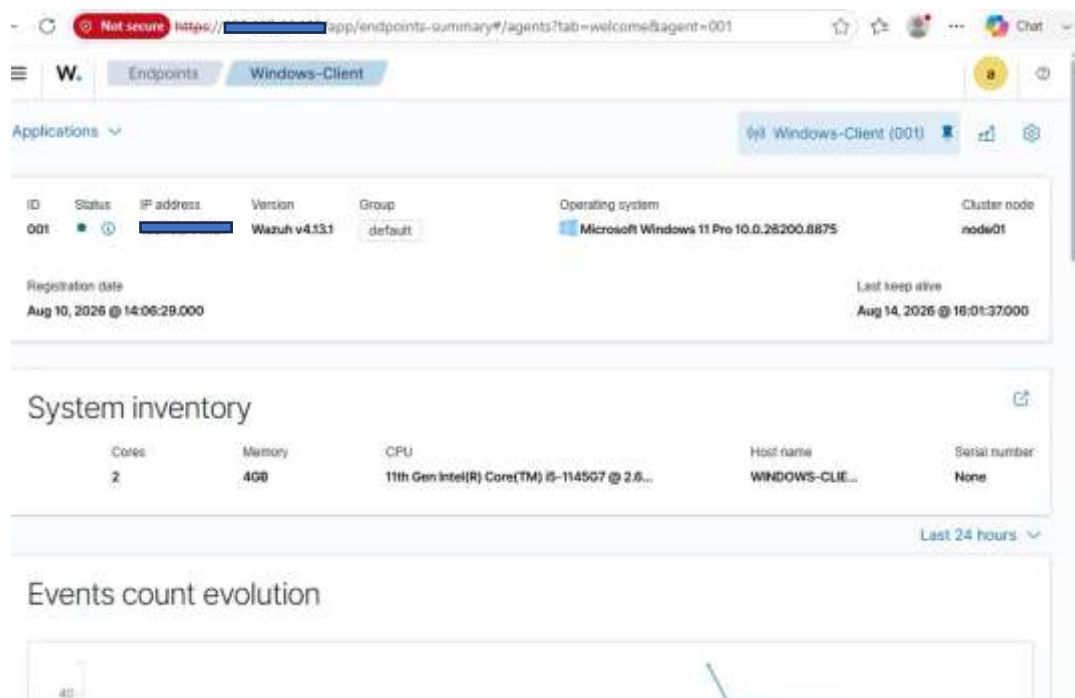


Figure 6.2 Agent Summary

6.3 Events:

Here we can check the events and we can filter these by eventIDs.

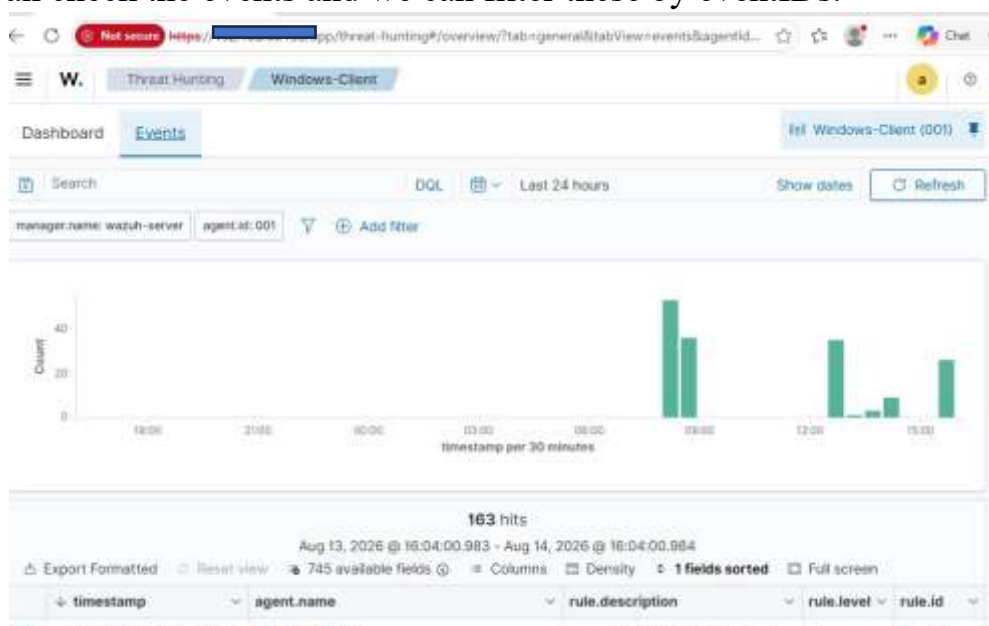


Figure 6.3 Event Section

6.4 FIM(File Integrity Monitoring):

FIM is used to monitor file changes to selected files/directories.

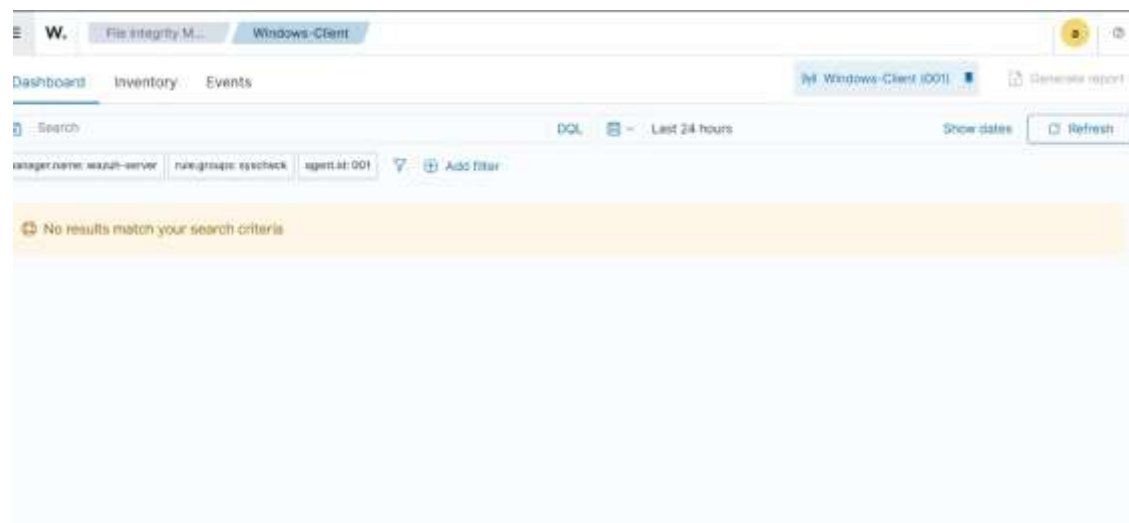


Figure 6.4 FIM dashboard

This is Empty because FIM is not Configured yet. This is in the next phase.

6.5 Alerts:

In this section, alerts are generated by monitoring systems. Each alert contains important information like agent name, agent id, the rule that triggered it, and severity level.

Level 0-3 represents informational events that indicate normal system activity.

Level 4-7 represents low severity events that require attention.

Level 8-11 represent medium-severity alerts that indicate suspicious behaviour.

Level 12-15 high severity alerts that signal potential security incidents.

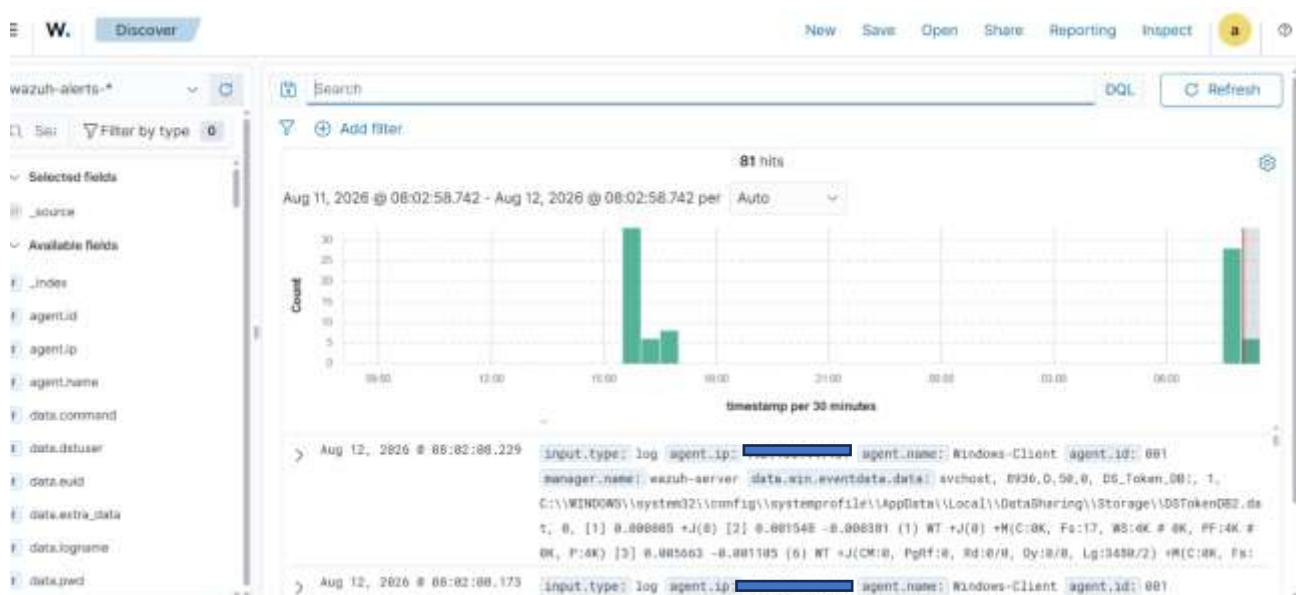


Figure 6.5 Alert Event Analysis

7 Practical Security Testing:

7.1 New User Account Creation:

Intentionaly , a user account is created to check Wazuh detectes it or not.

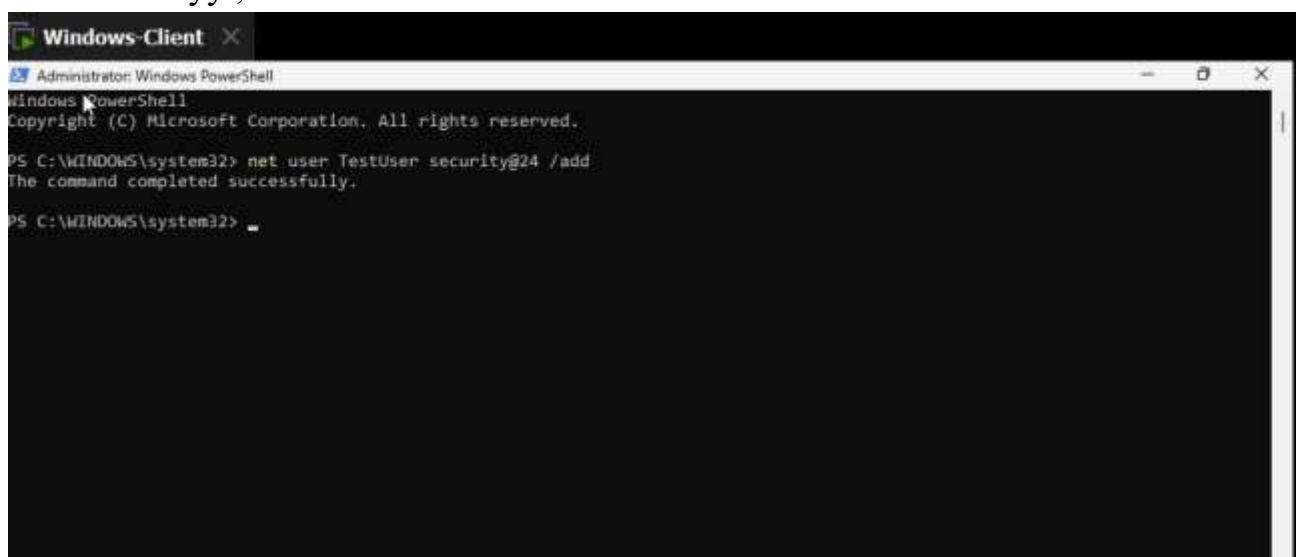


Figure 7.1 User created

A user account is created successfully using powershell.Event ID: 4720

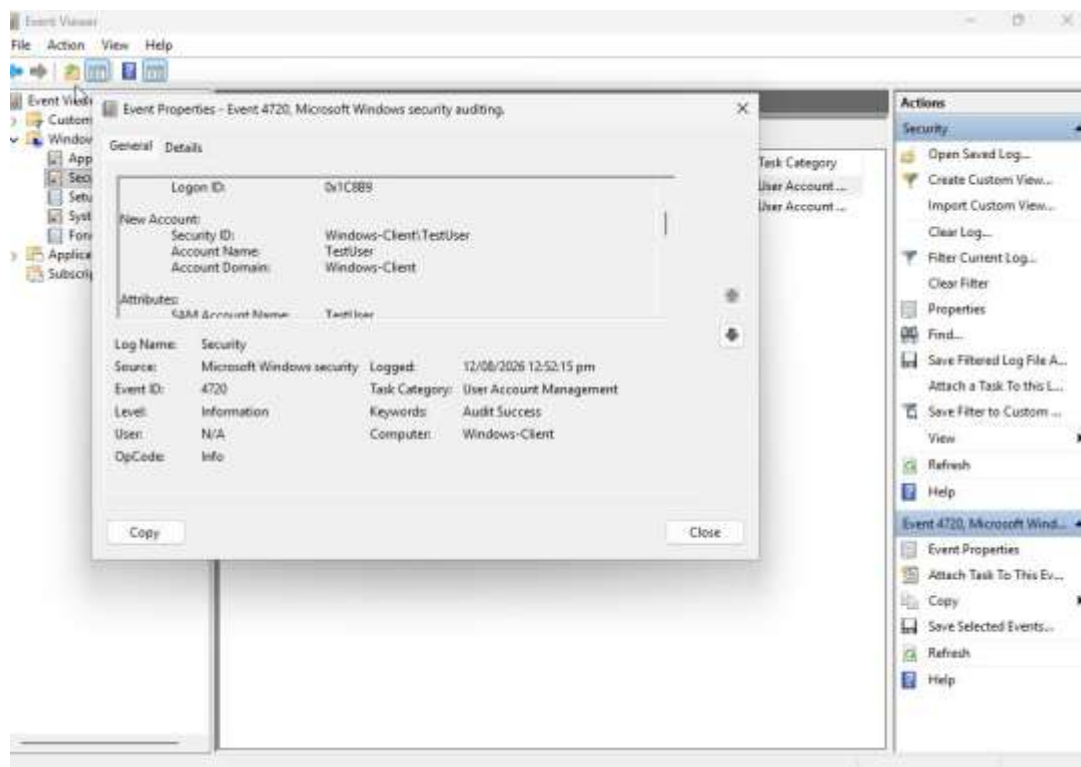


Figure 7.2 Window event

Event is successfully located Windows Event Viwer.

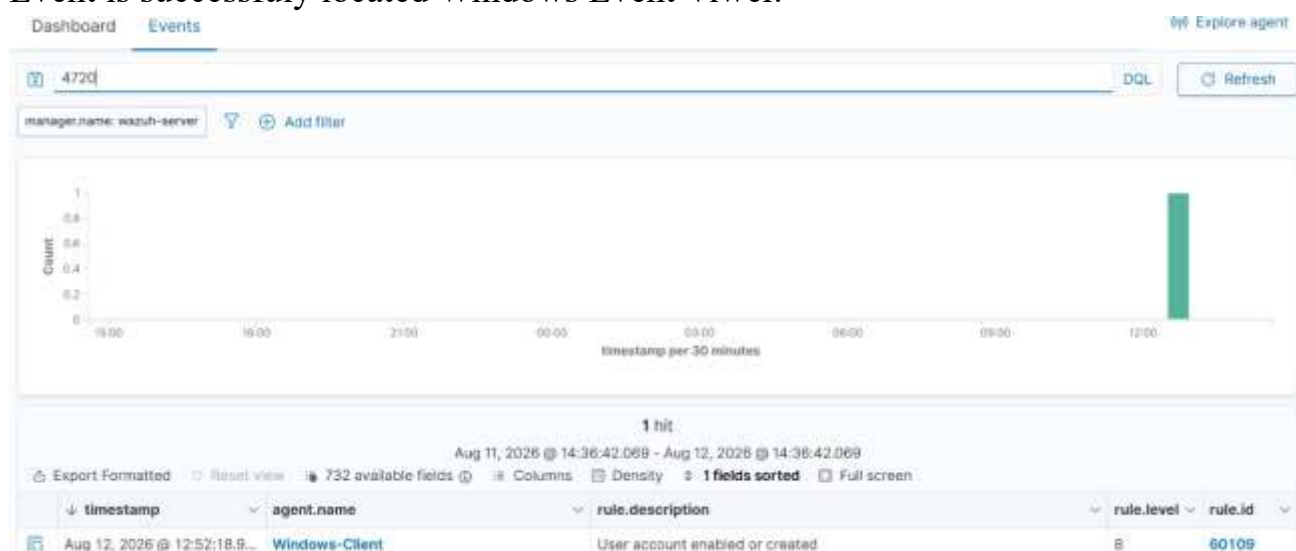


Figure 7.3 wazuh alert

Also Wazuh detects the event.

7.2 User Deletion:

Now that user is deleted. Event ID: 4726

```
last logon: never
Logon hours allowed: All
Local Group Memberships: *Users
Global Group memberships: *None
The command completed successfully.

PS C:\WINDOWS\system32> net user TestUser /delete
The command completed successfully.

PS C:\WINDOWS\system32>
```

Figure 7.4 user deleted

User account is deleted.

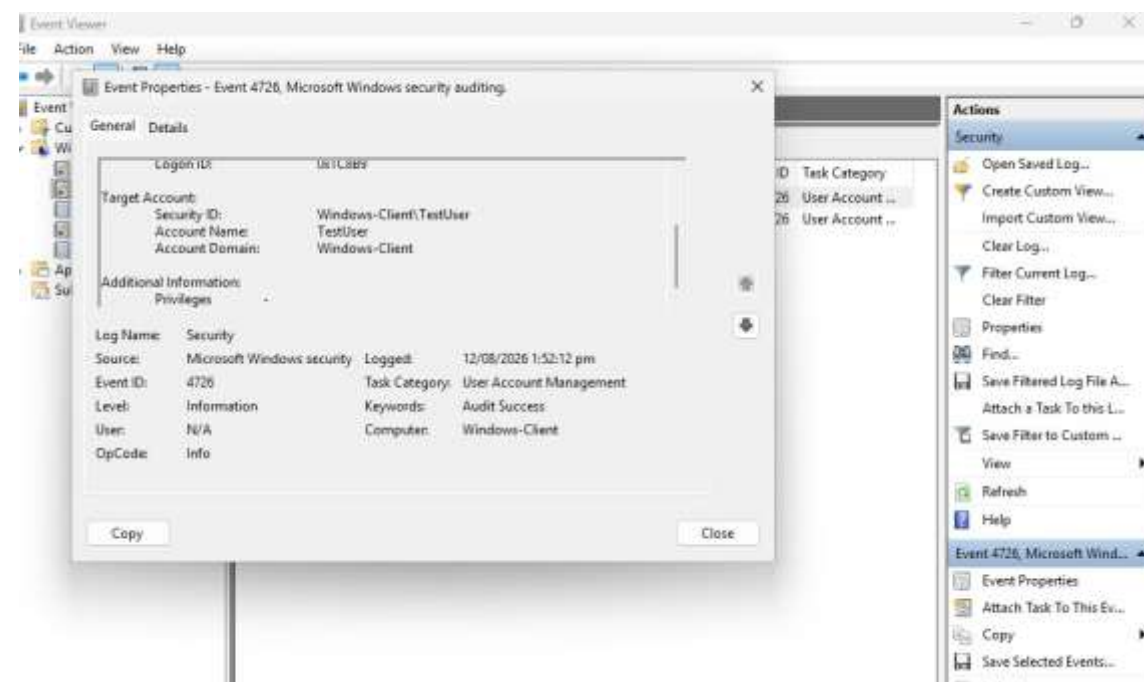


Figure 7.5 Windows Event viewer

In windows event viewr, event is detected.



Figure 7.6 Wazuh Deletion Detection

7.3 FIM(File Integrity Monitoring):

FIM is used to monitor file changes to selected files/directories.

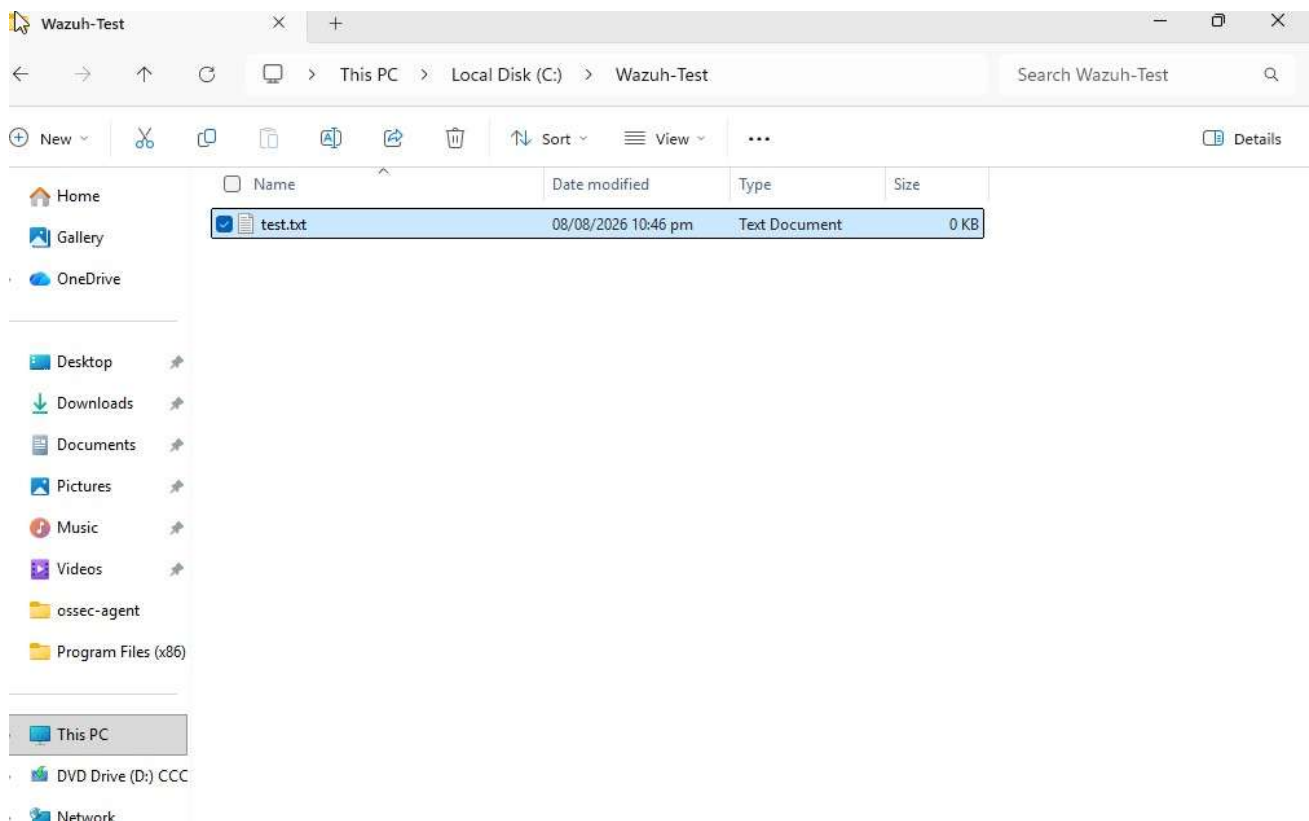


Figure 7.7 FIM test folder

```
<interval>5m</interval>
<max_eps>10</max_eps>
</synchronization>
<directories realtime="yes" report_changes="yes">C:\Wazuh-Test</directories>
</syscheck>
```

Figure 7.8 FIM configuration

The test folder is added in the SYSCHECK block as it controls File Integrity Monitoring in the ossec.conf file.

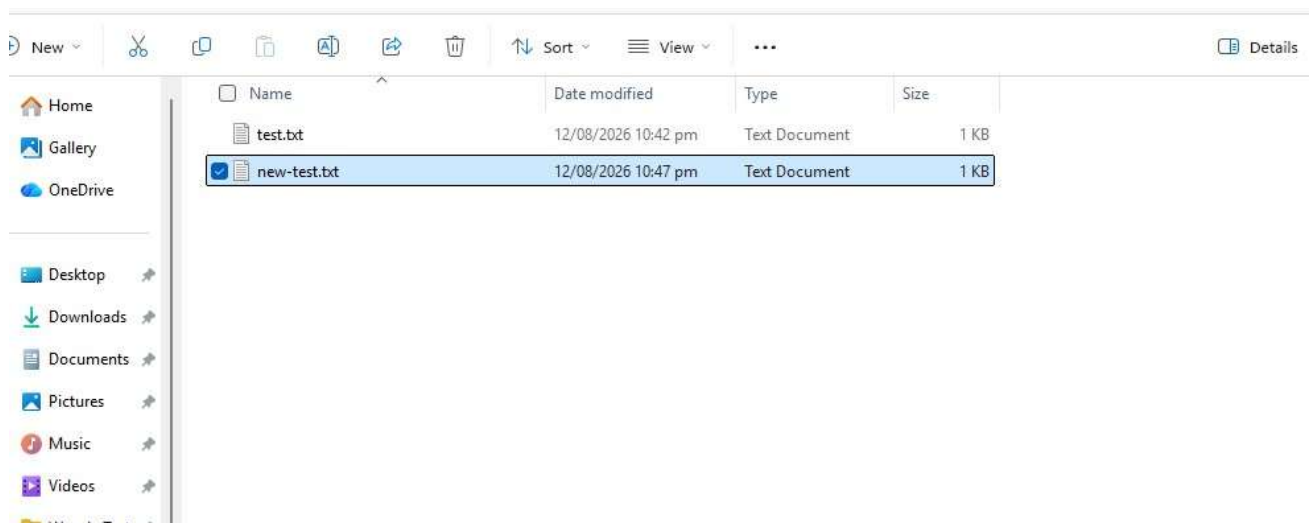


Figure 7.9 File created

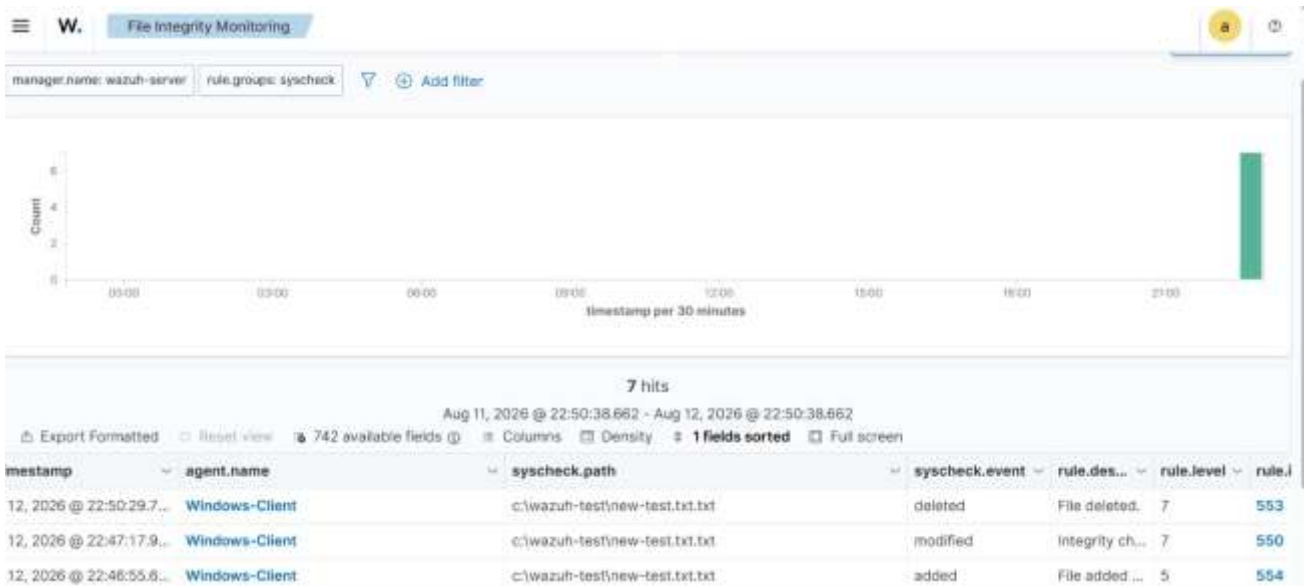


Figure 7.10 Wazuh Alert

7.4 Firewall Enable\Disable:

Firewall is intentionally modified for testing.

```
Select Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-NetFirewallProfile | Select Name,Enabled

Name      Enabled
-----
Domain    True
Private   True
Public     True

PS C:\WINDOWS\system32>
```

Figure 7.11 Initial status: enabled

```

PS C:\WINDOWS\system32> Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False
PS C:\WINDOWS\system32> Get-NetFirewallProfile | Select Name,Enabled

Name      Enabled
-----
Domain    False
Private   False
Public     False

PS C:\WINDOWS\system32>

```

Figure

7.12 Disabled

Firstly, the firewall is initially enabled then it was made disabled.



Figure 7.13 windows detected

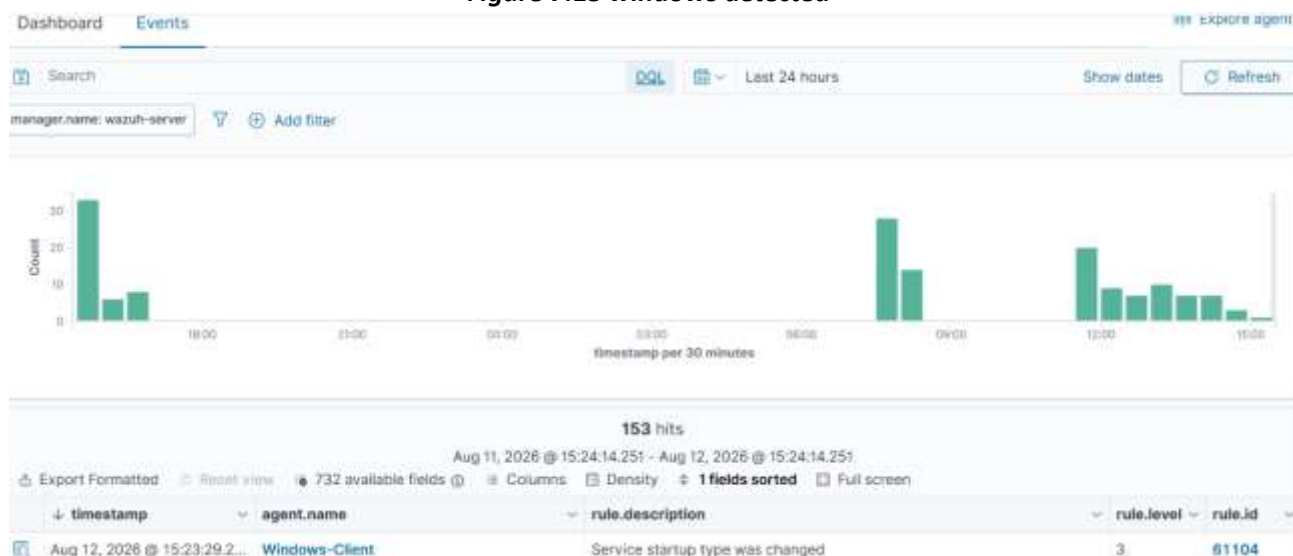


Figure 7.14 Wazuh Detection

Windows Firewall status change was successfully recorded in Windows Event Viewer, but the corresponding event is not generated by Wazuh security alert under the current default monitoring /rule configuration.

7.5 Successful login:

Controlled successful login attempt is performed on windows endpoint to generate authentication related security events.

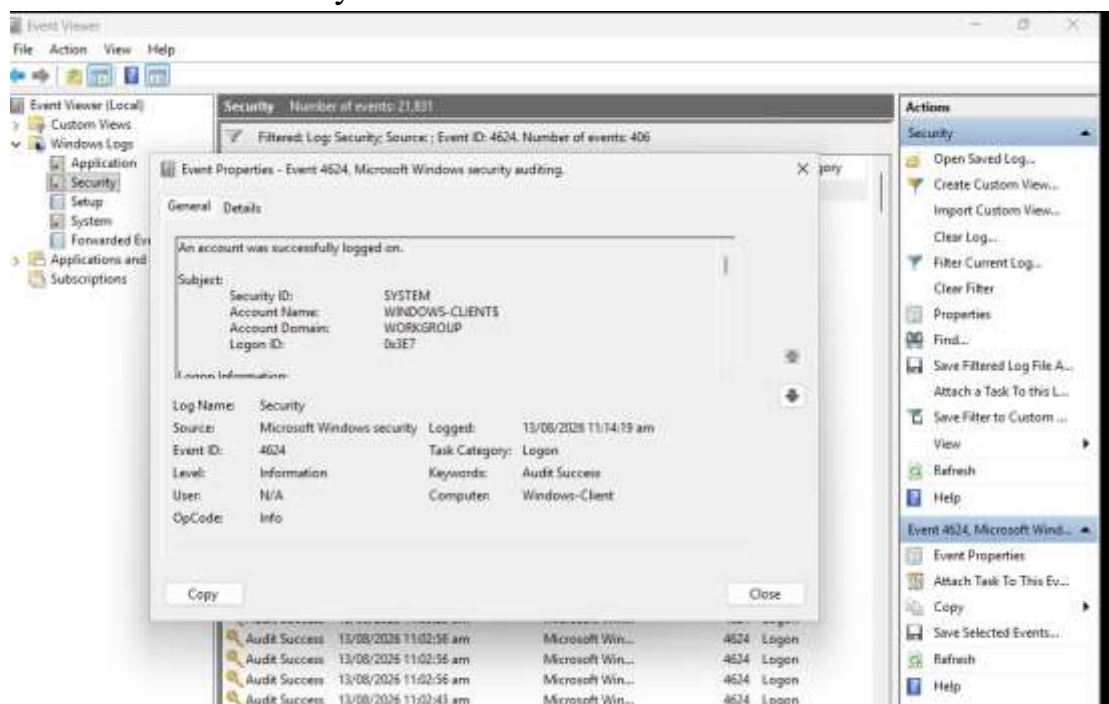


Figure 7.15 Window login detect

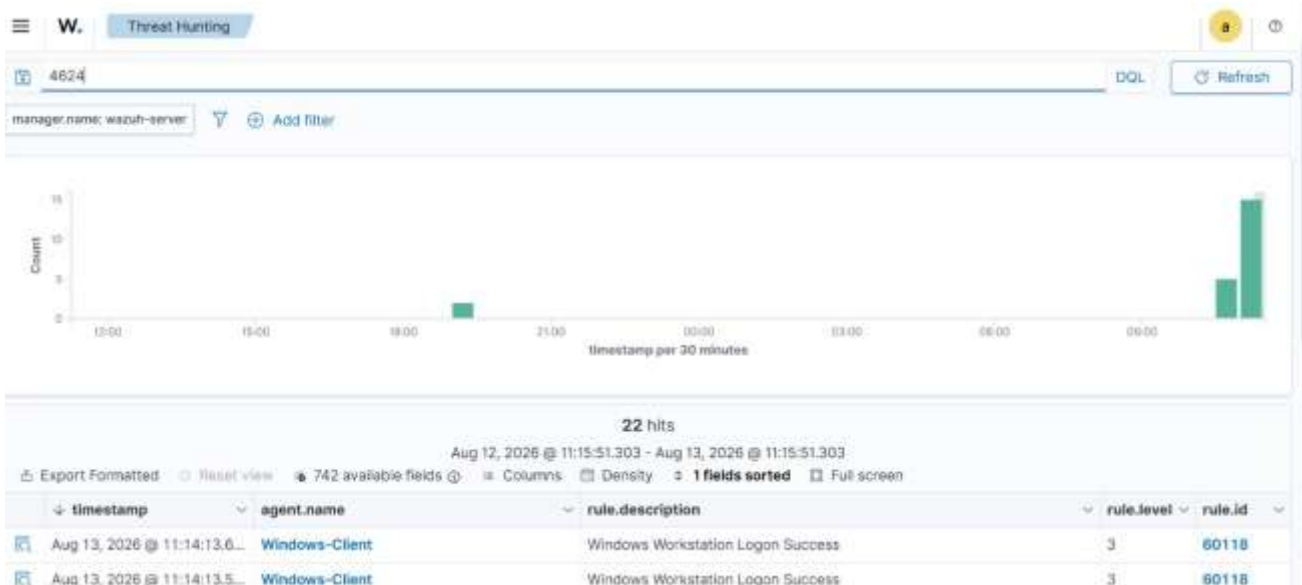


Figure 7.16 Wazuh Detection

Successful login event is found in both Windows Event Viewer and by Wazuh through Filtering.

7.6 Failed Login:

2-3 wrong passwords attempts are made intentionally to monitor activity.

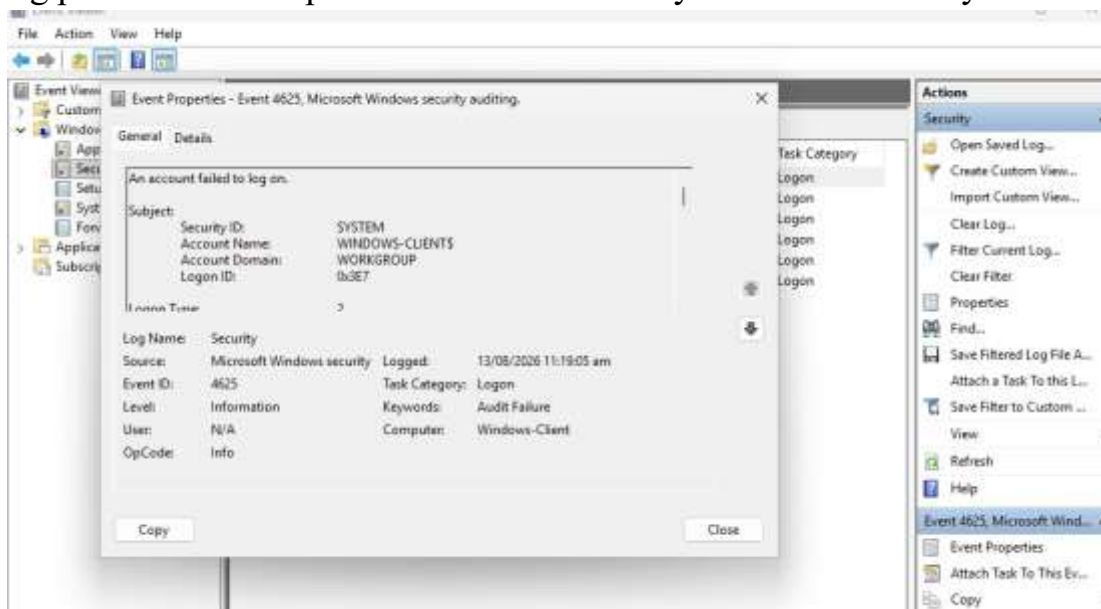


Figure 7.17 Windows Failed login

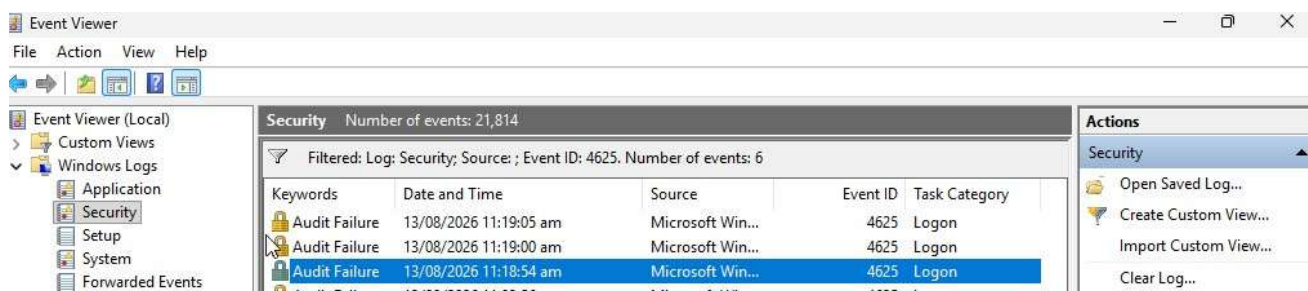


Figure 7.18 login attempts



Figure 7.19 Wazuh alert

Failed login attempts were successfully detected by windows and Wazuh which shows that alerts and events are monitored by the Wazuh.

8 Security Event Analysis:

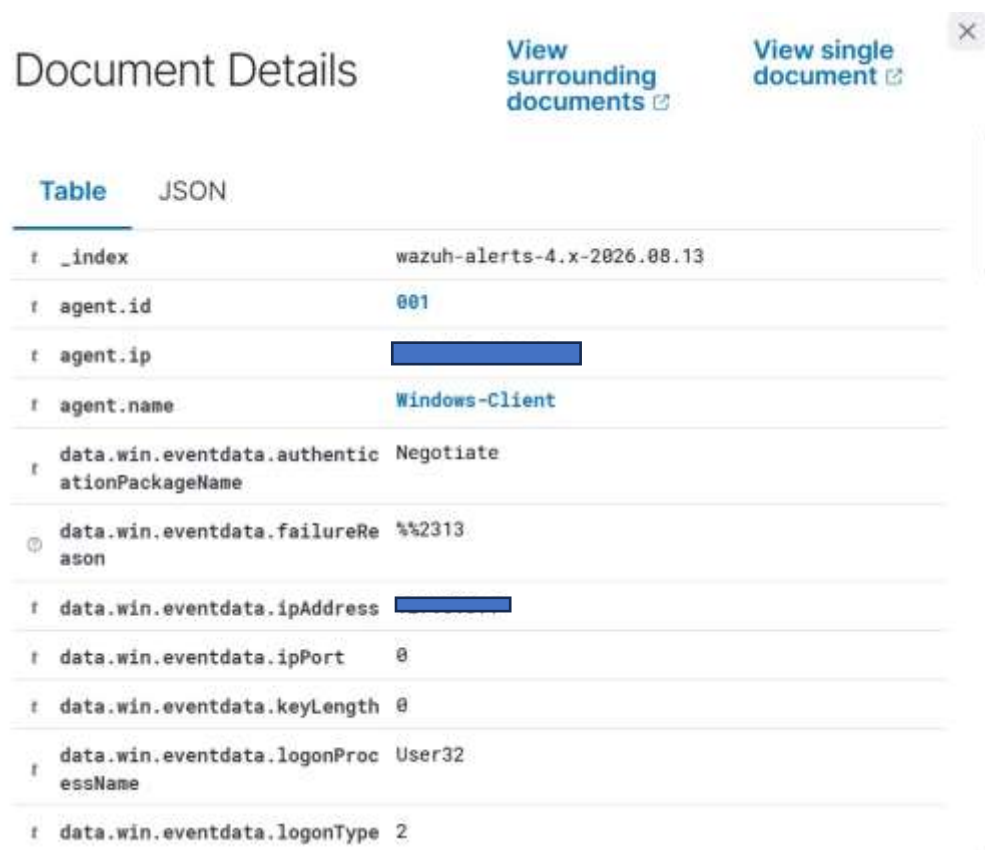
From all above observations, it is cleared that Wazuh agent is collecting event logs from the Windows VM and that events are successfully managed and analyzed by the Wazuh Server. Wazuh dashboard is showing all the monitored events and successfully generating the alerts.

Event: Any activity recorded by endpoints

Alert: when Wazuh rules are matched with an event and identify it as security relevant.

8.1 Failed Login Event Analysis:

A Controlled failed login attempt was already made for checking the Wazuh activity .so that failed login event is taken and analyzed and checked wether Wazuh generated an alert or not.



The screenshot displays the 'Document Details' interface for a Wazuh alert. At the top, there are two buttons: 'View surrounding documents' and 'View single document'. Below these, there are two tabs: 'Table' (selected) and 'JSON'. The table lists various fields related to the alert, including the index name, agent ID, agent IP, agent name, authentication package name, failure reason, IP address, IP port, key length, logon process name, and logon type. Some fields like the agent IP and IP address are redacted with blue boxes.

_index	wazuh-alerts-4.x-2026.08.13
agent.id	001
agent.ip	
agent.name	Windows-Client
data.win.eventdata.authenticationPackageName	Negotiate
data.win.eventdata.failureReason	%%2313
data.win.eventdata.ipAddress	
data.win.eventdata.ipPort	0
data.win.eventdata.keyLength	0
data.win.eventdata.logonProcessName	User32
data.win.eventdata.logonType	2

Figure 8.1 Alert Details 1

Document Details

[View surrounding documents](#)

[View single document](#)



<code>data.win.eventdata.processId</code>	<code>0x65c</code>	  
<code>data.win.eventdata.processName</code>	<code>C:\\Windows\\System32\\svchost.exe</code>	
<code>data.win.eventdata.status</code>	<code>0xc000006d</code>	
<code>data.win.eventdata.subStatus</code>	<code>0xc000006a</code>	
<code>data.win.eventdata.subjectDomainName</code>	<code>WORKGROUP</code>	
<code>data.win.eventdata.subjectLogonId</code>	<code>0x3e7</code>	
<code>data.win.eventdata.subjectUserName</code>	<code>WINDOWS-CLIENTS</code>	
<code>data.win.eventdata.subjectUserSid</code>	<code>S-1-5-18</code>	
<code>data.win.eventdata.targetDomainName</code>	<code>WINDOWS-CLIENT</code>	
<code>data.win.eventdata.targetUserName</code>	<code>socuser</code>	
<code>data.win.eventdata.targetUserSid</code>	<code>S-1-0-0</code>	

Figure 8.2 alert details 2

Document Details		View surrounding documents	View single document	×
data.win.system.channel	Security			
data.win.system.computer	Windows-Client			
data.win.system.eventID	4625			
data.win.system.eventRecordID	55052			
data.win.system.keywords	0x8010000000000000			
data.win.system.level	0			
data.win.system.message	"An account failed to log on." Subject: Security ID: S-1-5-18 Account Name: WINDOWS-CLIENT T\$ Account Domain: windowsrvip			
data.win.system.opcode	0			
data.win.system.processID	816			
data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}			
data.win.system.providerName	Microsoft-Windows-Security-Auditing			

Figure 8.3 alert details

Document Details		View surrounding documents	View single document	×
rule.gdpr	IV_35.7.d, IV_32.2			
rule.gpg13	7.1			
rule.groups	windows, windows_security, authentication_failured			
rule.hipaa	164.312.b			
rule.id	60122			
rule.level	5			
rule.mail	false			
rule.mitre.id	T1531			
rule.mitre.tactic	Impact			
rule.mitre.technique	Account Access Removal			
rule.nist_800_53	AU.14, AC.7			
rule.pci_dss	10.2.4, 10.2.5			
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3			
timestamp	Aug 13, 2026 @ 11:19:09.594			

Figure 8.4 important details

A complete and detailed analysis is made by the security analyst through investigating the event important information.

Important Details:

- Agent name
- Event ID
- Rule ID
- Rule Level
- Description
- Username
- Agent IP
- Failure Information
- Time stamp

All above details can be collected and analyzed through Wazuh dashboard,so it provides centralized security Event Monitoring of any endpoint.

8.1.1 Alert Generated:

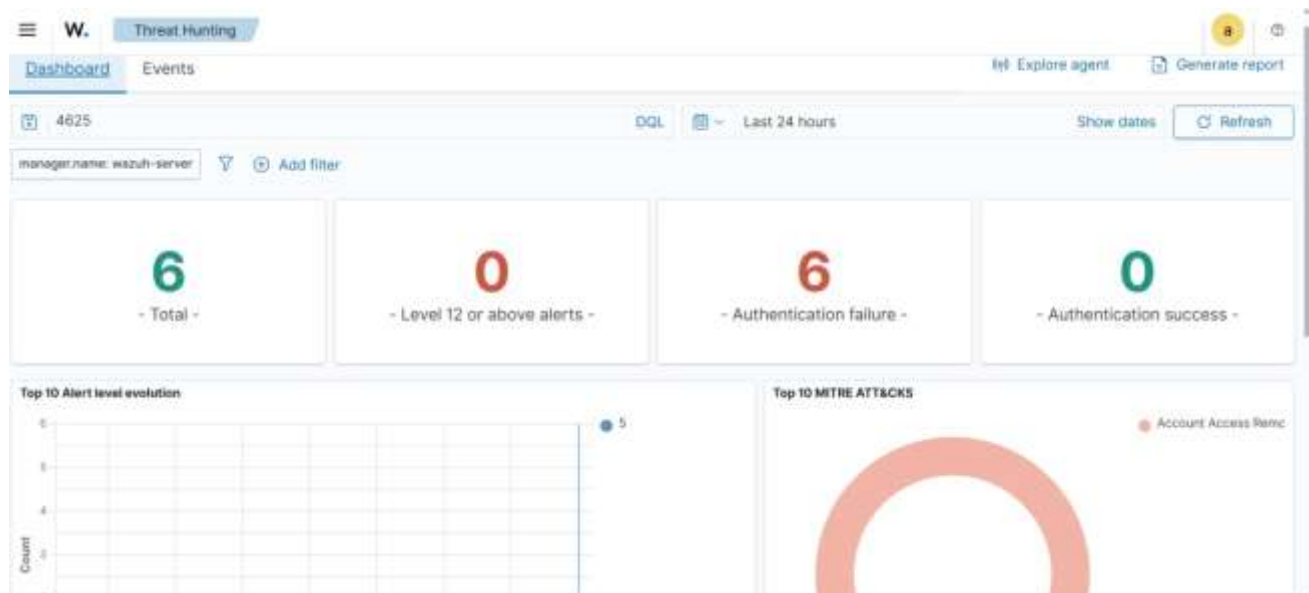


Figure 8.5 Alert generated

9 :Results and Findings:

Objective	Achievement
Deploy Wazuh SIE	Successfully implemented
Configure Wazuh Ma	Completed
Connect Windows end	Successfully connected
Install Wazuh Age	Successfully installed and configured
Verify agent communi	Successfully verified
Collect Windows securit	Successfully demonstrated
Monitor failed login a	Successfully demonstrated
Perform File Integrity M	Successfully tested
Assess Windows security	Completed
Investigate events through	Successfully demonstrated

10 Recommendations

Monitor More Endpoints:

In a production environment, additional Windows and Linux endpoints should be connected to Wazuh to provide organization-wide visibility.

Implement Role-Based Access:

Access to the Wazuh Dashboard should be controlled using appropriate user roles and permissions to prevent unauthorized access to security information.

Configure Appropriate Alert Rules:

Wazuh detection rules should be reviewed and customized according to the organization's security requirements to improve the identification of relevant security events and reduce unnecessary alerts.

Regular Security Policy Review:

Windows password policies, account lockout policies, firewall settings, and other security controls should be reviewed periodically.

Maintain Regular Updates:

The Wazuh components, Windows endpoints, and other security software should be kept updated to reduce exposure to known vulnerabilities.

Establish Alert Investigation Procedures:

A production SOC should establish procedures for triaging, investigating, documenting, and responding to security alerts generated by the SIEM.

11 Limitations

The implemented environment was developed as a controlled virtual laboratory rather than a production enterprise environment. Therefore, the project had several limitations.

1. Limited number of endpoints

The environment contained a single Windows client endpoint, whereas a production SOC would normally monitor many endpoints and servers.

2. Controlled testing

Security events were generated intentionally for testing purposes. Real-world environments contain a much larger variety of normal and malicious activities.

3. Limited network architecture

The project was implemented using virtual machines and a laboratory network rather than a large enterprise network containing multiple network segments and security devices.

4. Limited threat scenarios

The project focused on selected security-monitoring scenarios and did not attempt to simulate every possible attack technique.

5. Resource limitations

The virtual environment was constrained by the available system resources, which limited the scale of the deployment.

12 Conclusion:

The Windows Security Assessment and Monitoring using Wazuh SIEM project successfully demonstrated how a centralized SIEM solution can be used to monitor and analyze security-related activity on a Windows endpoint. The Wazuh Agent provided communication between the Windows client and the Wazuh Manager, allowing relevant endpoint information and security events to be centrally monitored through the Wazuh Dashboard.

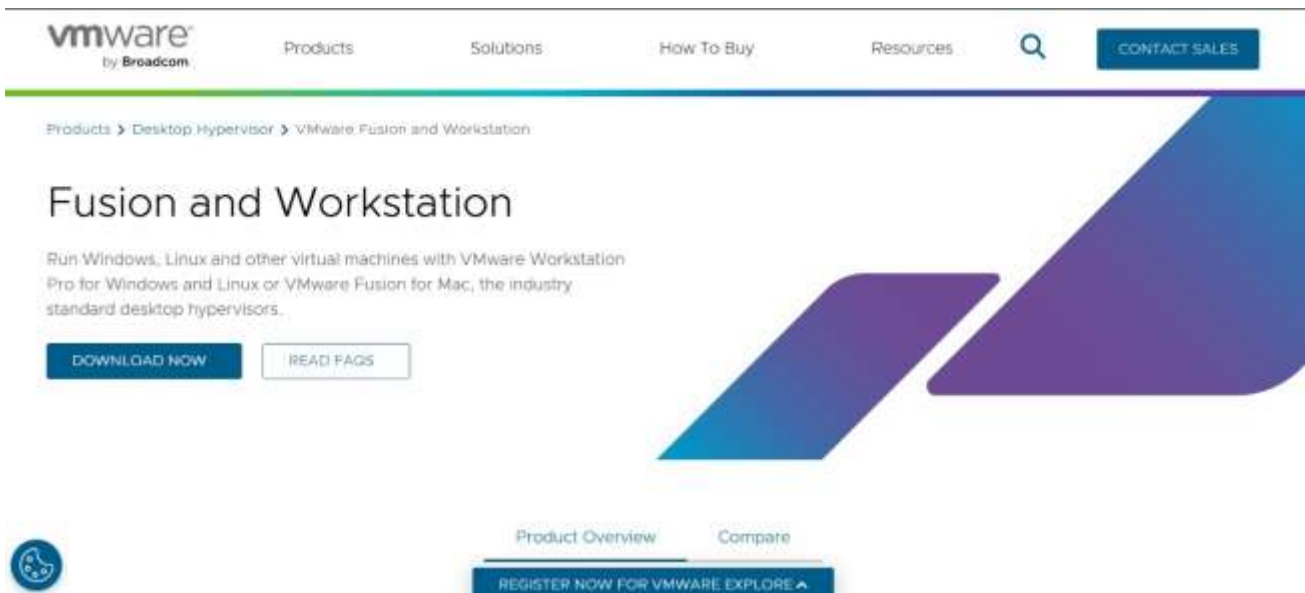
The project combined endpoint security assessment with centralized monitoring and demonstrated practical scenarios including Windows security-event monitoring, failed authentication detection, file-integrity monitoring, and security-event investigation. The results confirmed that Wazuh can provide useful centralized visibility into endpoint activity within a controlled laboratory environment.

The implemented solution can serve as a foundation for a larger security-monitoring environment containing multiple endpoints, additional log sources, customized detection rules, and automated incident-response capabilities.

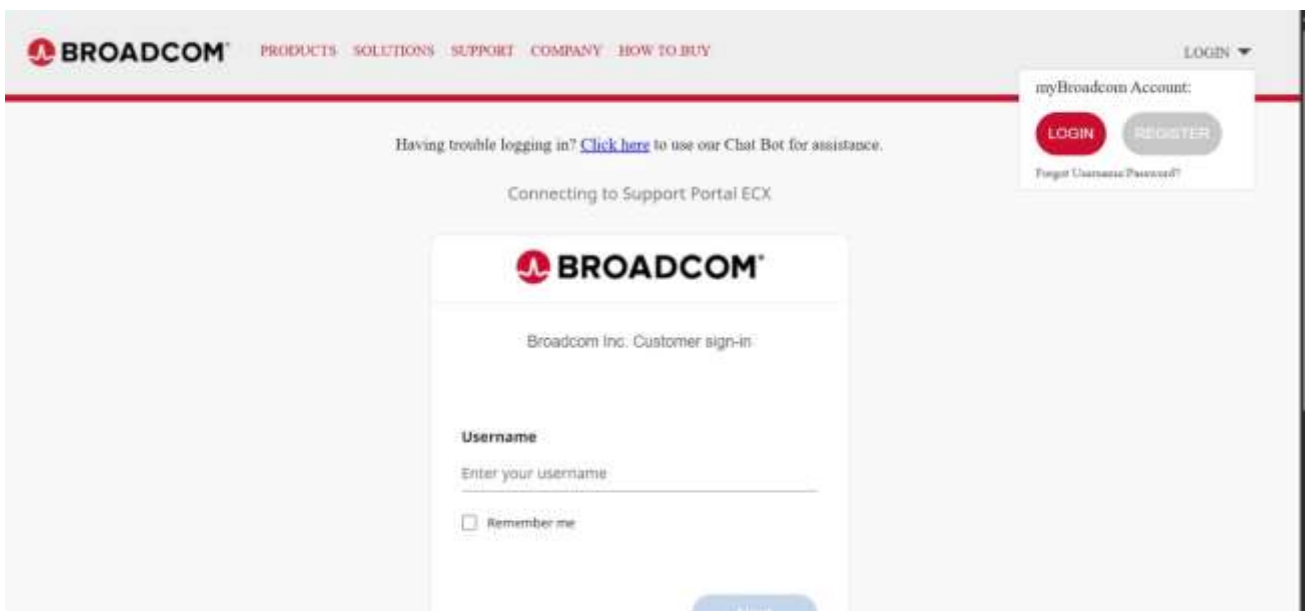
Appendices

13 Appendix A: VMware Setup

Firstly, go to the main official website for downloading the VMware Workstation.



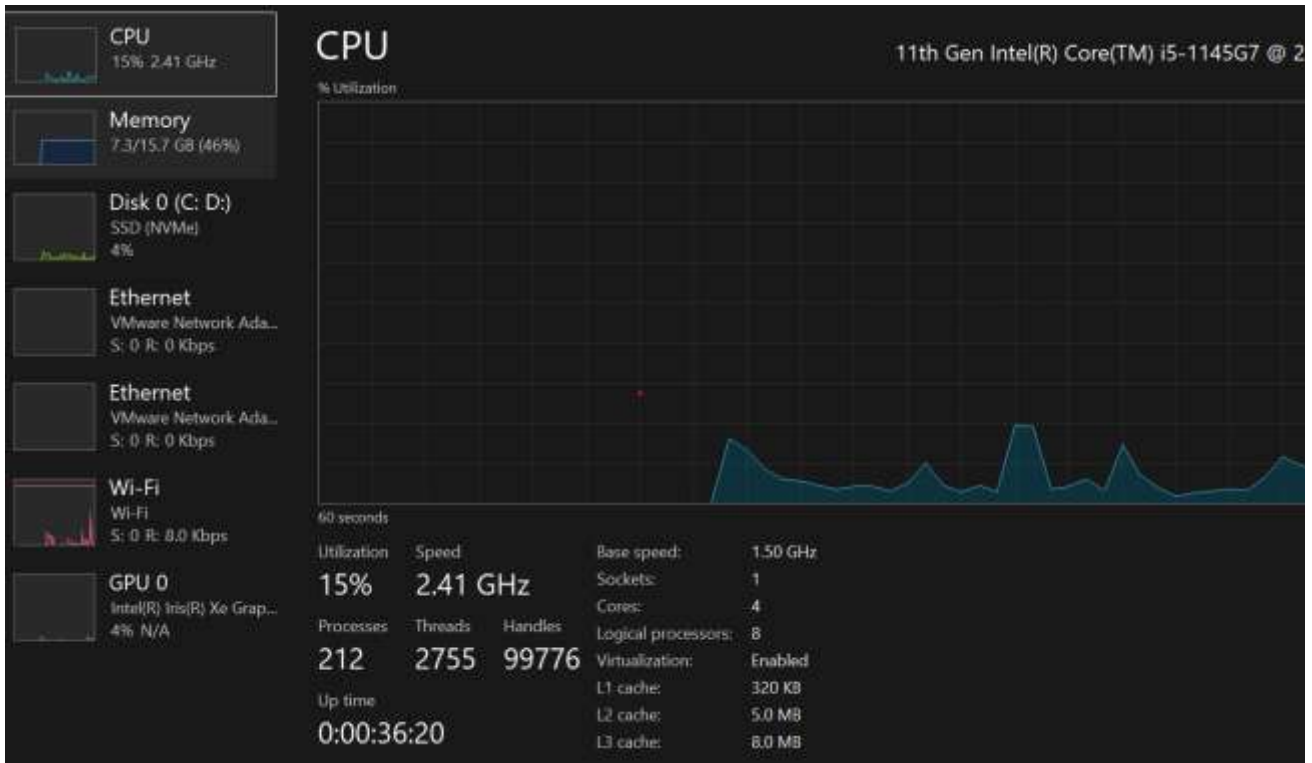
Then when you click Download now button, a website appears with name of Broad.com Go register on it and Download the latest version,i installed VMware Workstation 17 Pro.



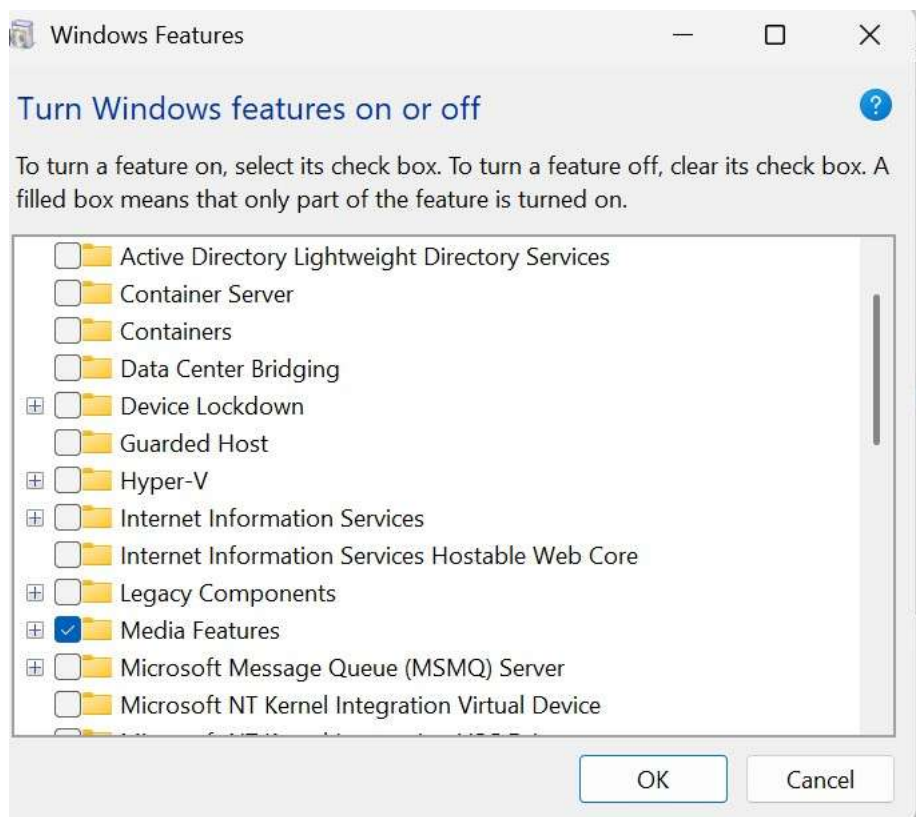
Now click on the downloaded file and simply install it without changing any settings and it will be installed successfully.I already installed VMware workstation.

14 Appendix B: Kali Linux Setup

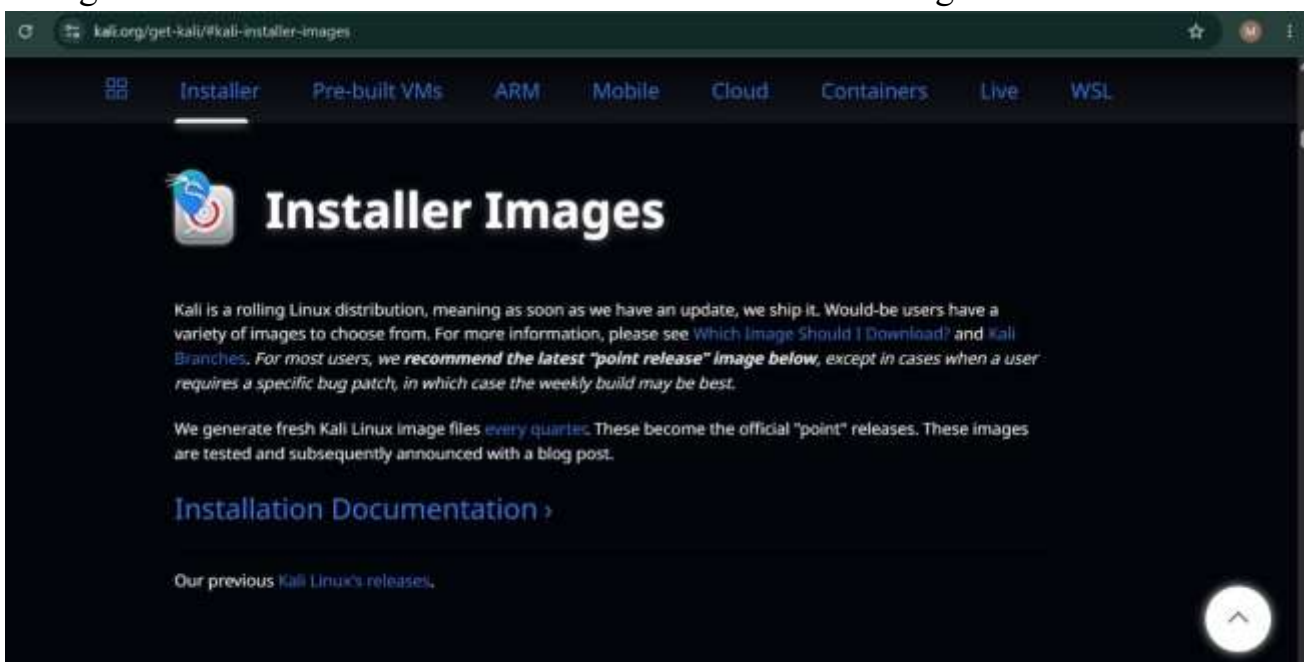
Firstly, make sure your CPU has Virtualization feature Enabled. Without it we cannot install VM.



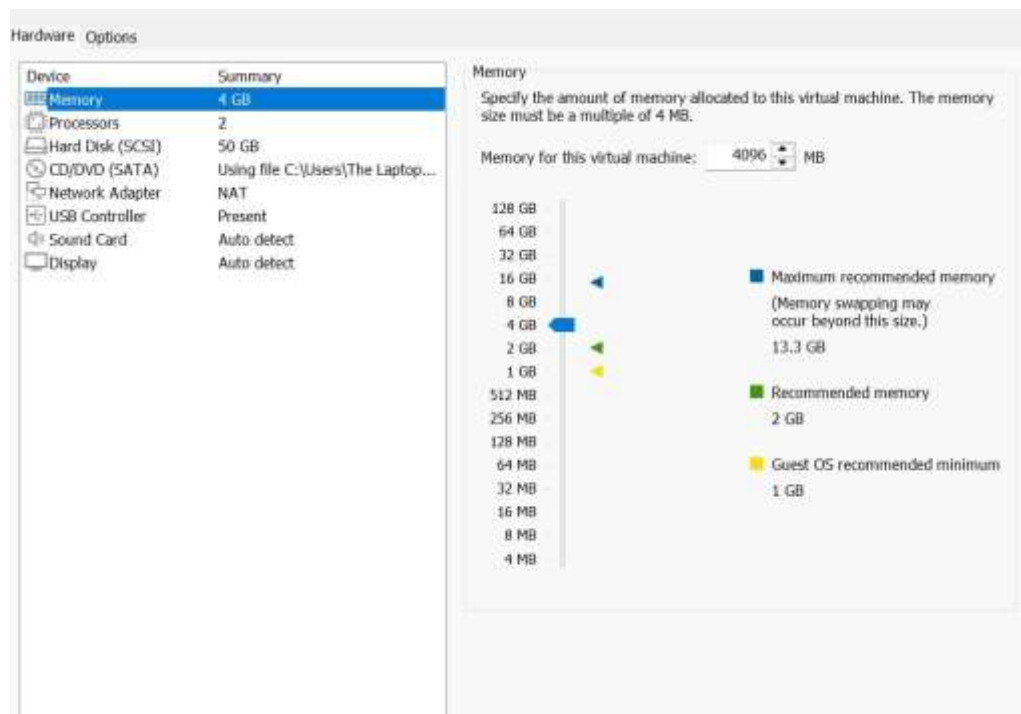
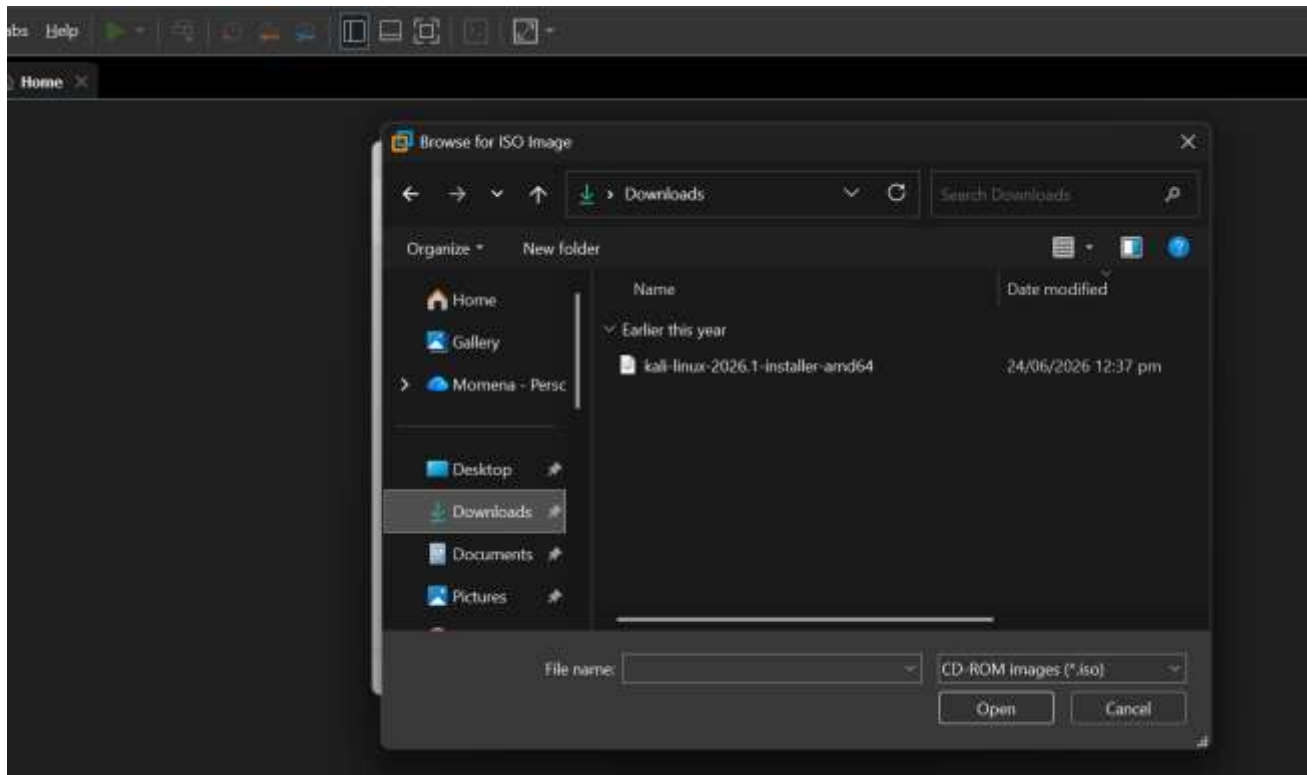
Then next open Windows features settings and unchecked the Hyper-v box, and other VM related features i.e. Virtual machine platform and others.

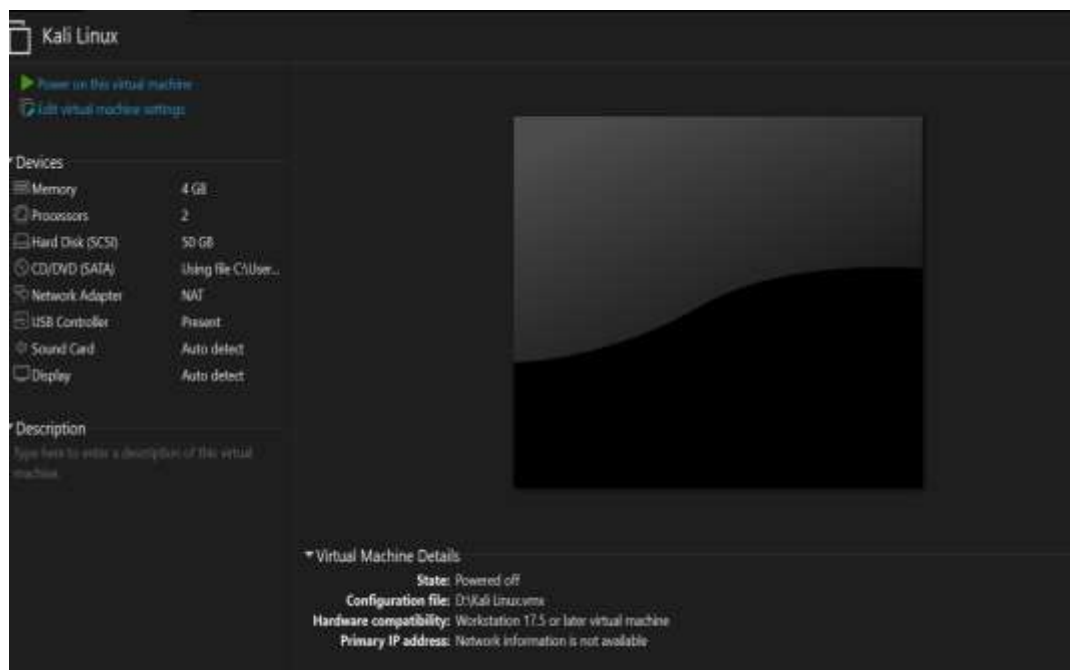


Then go to official website and download the Installer ISO image offline download.



After downloading it, open vmware workstation and click create a virtual machine then browse the downloaded ISO file and choose latest Debian version and configure all its configuration settings and click finish.

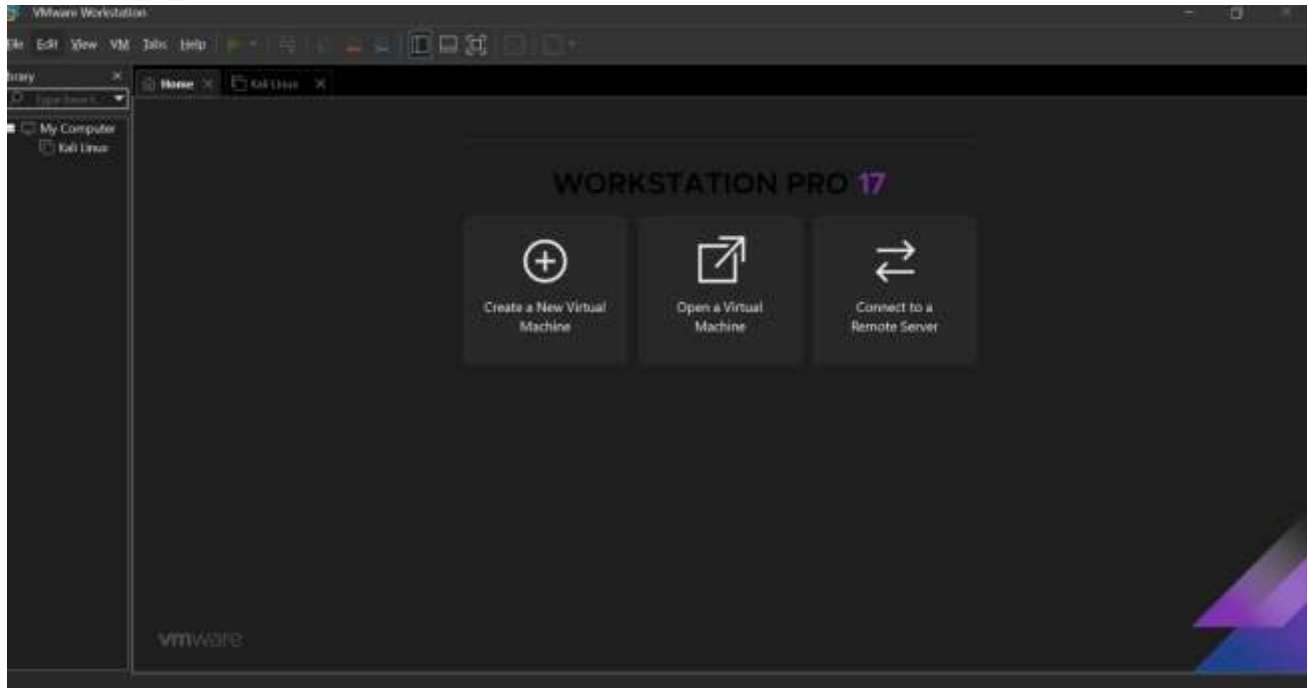


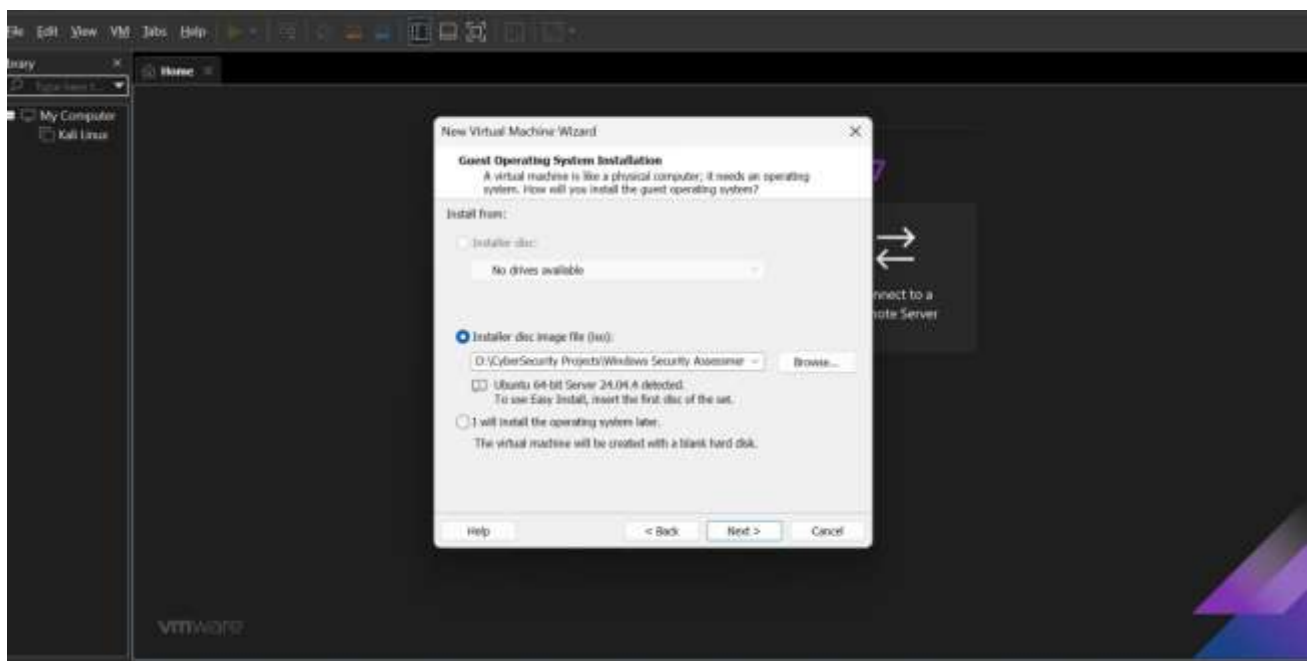
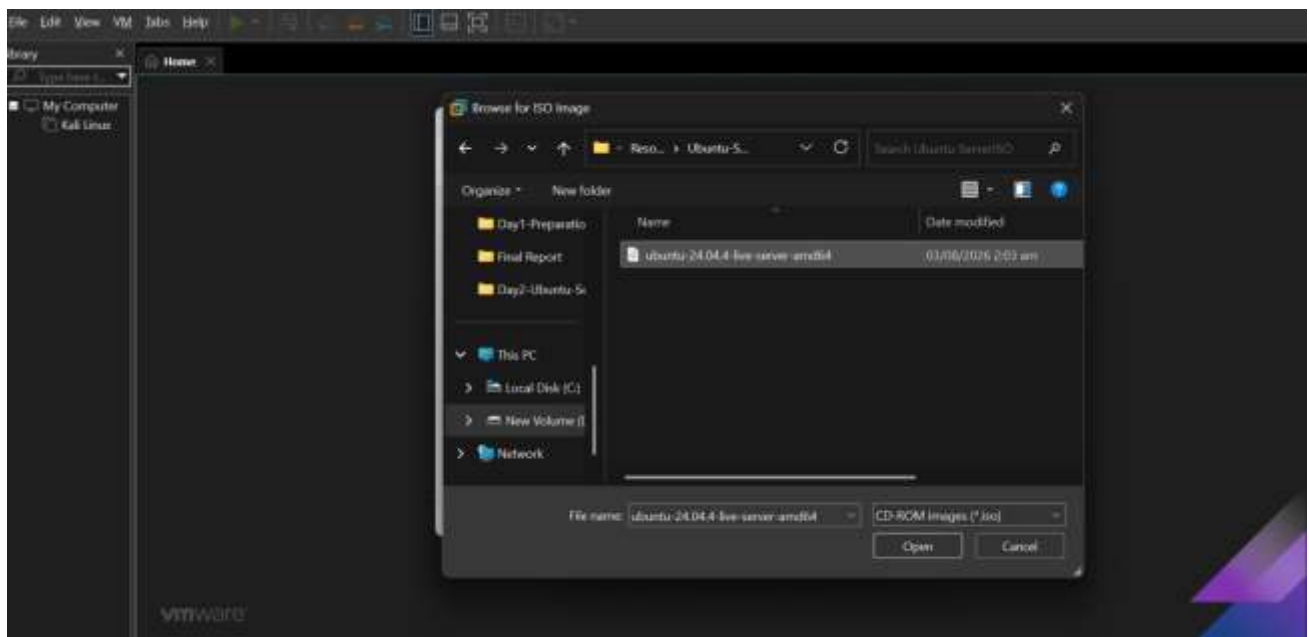


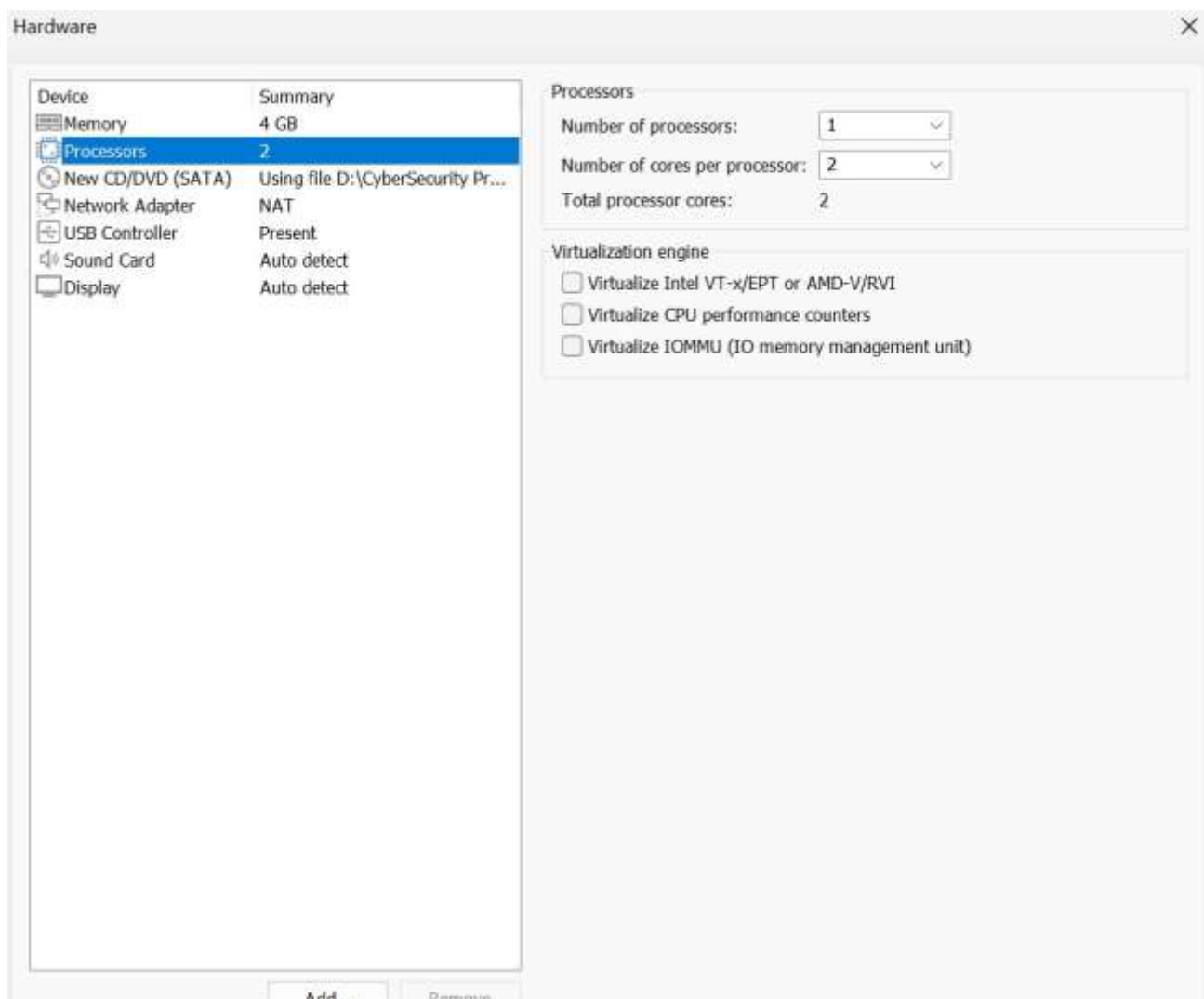
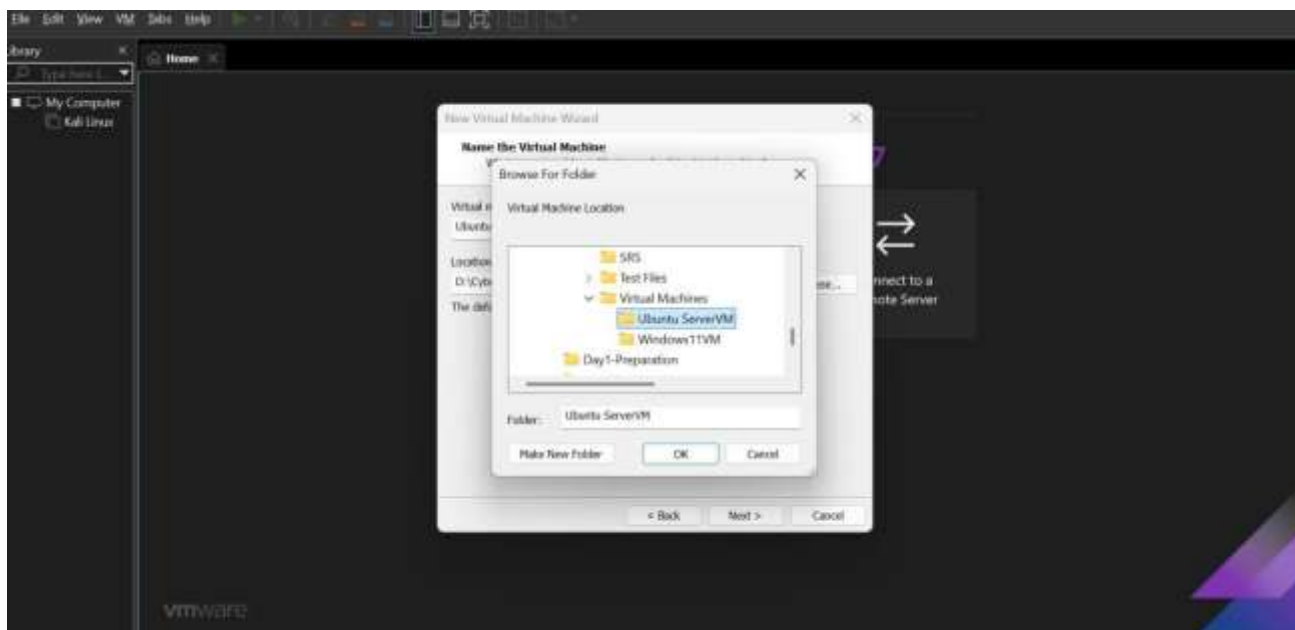
When you power on this vm,choose GUI then select language ,country and keyboardthen installation will started and ask for hostname and a password for kali linux.then partition disk screen will appear choose Guided use entire disk then installation will started. Once everything is done it will ask for Reboot and once its done kali will open and ask for credential, enter them and start.

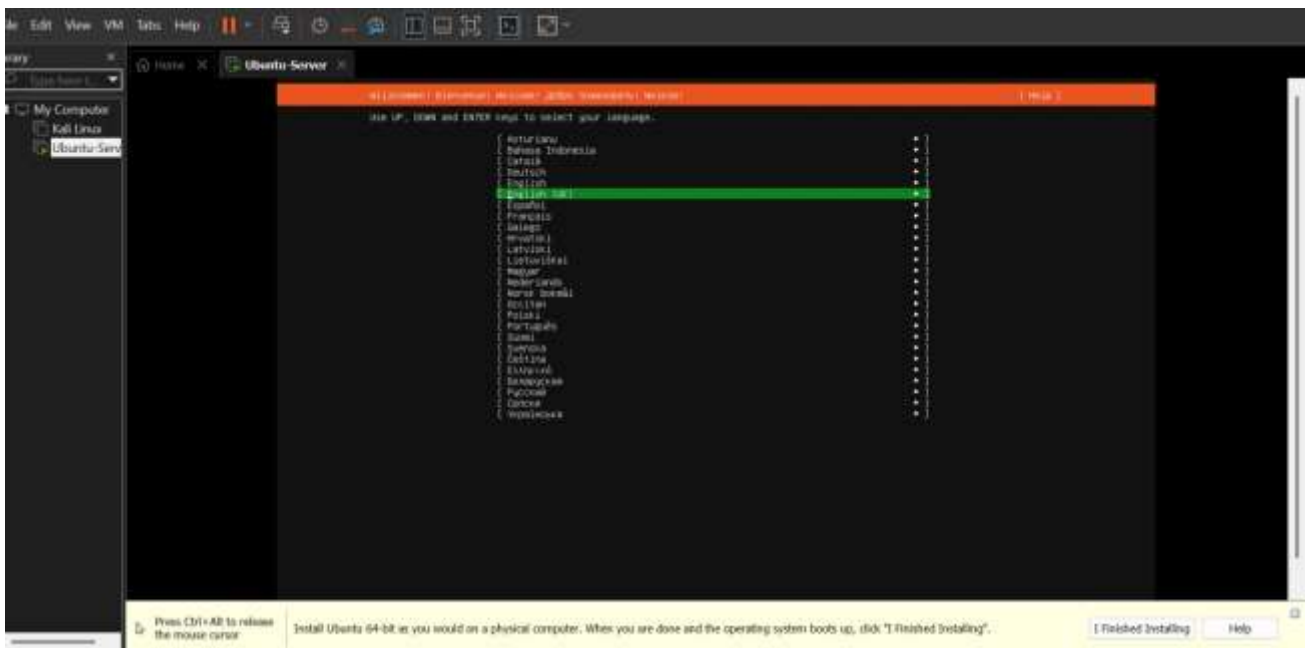
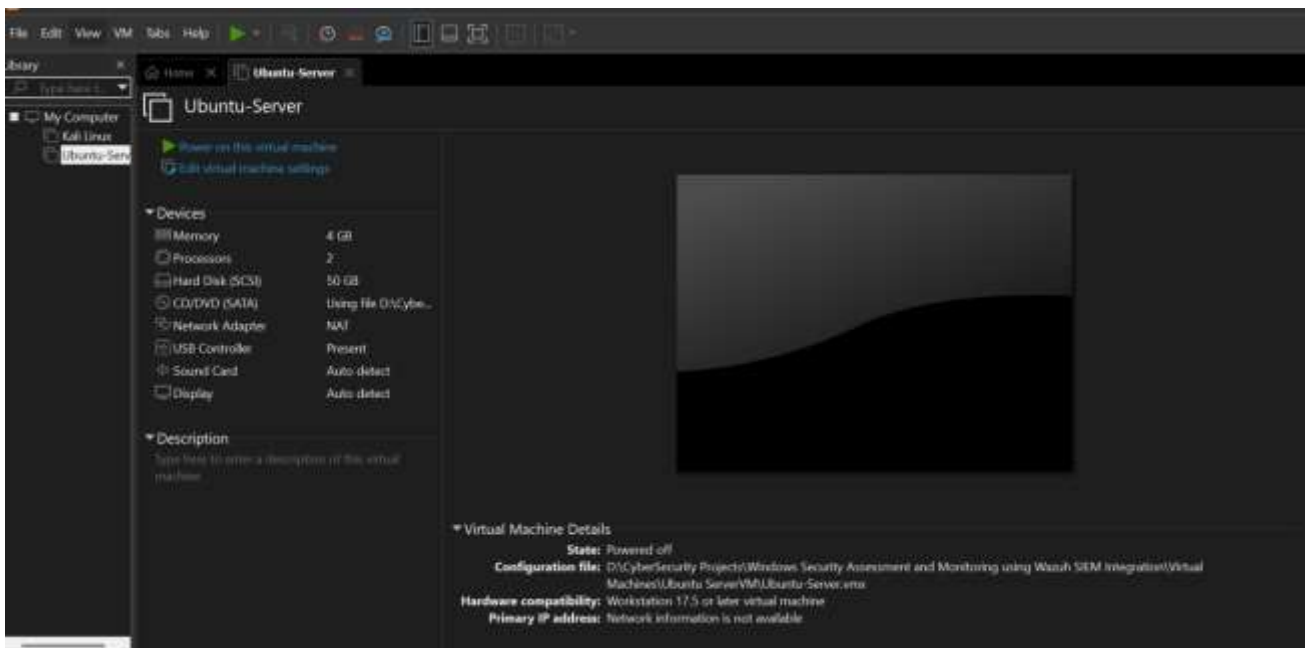


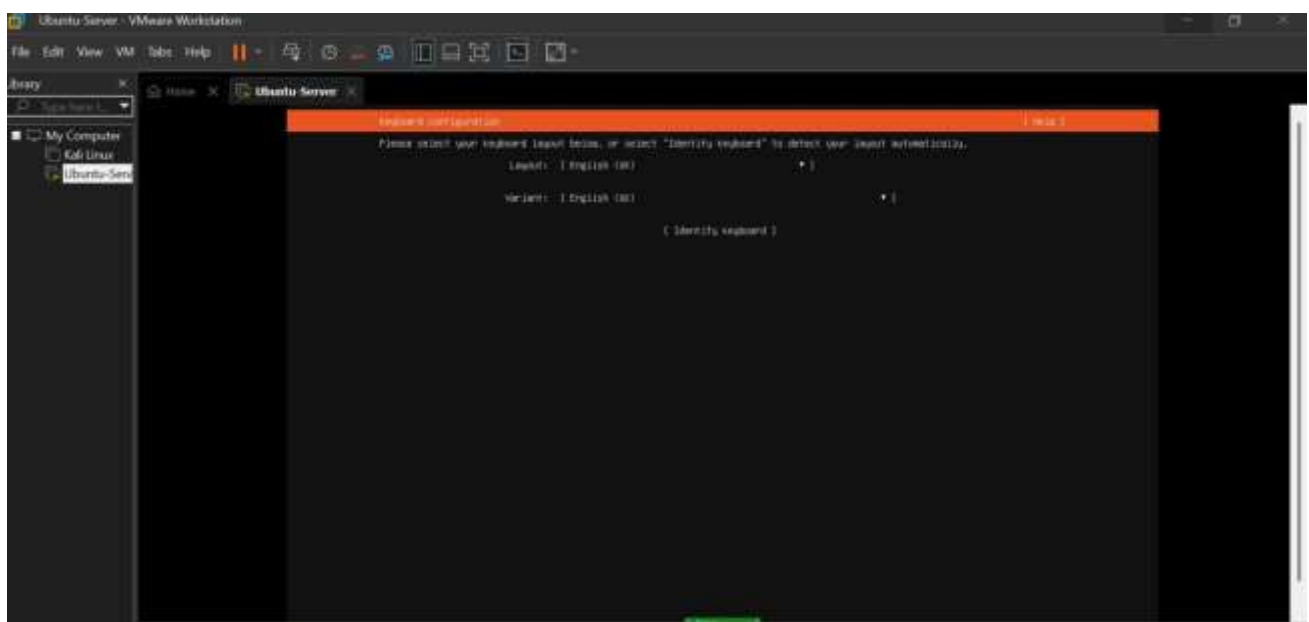
15 Appendix C: Ubuntu Server

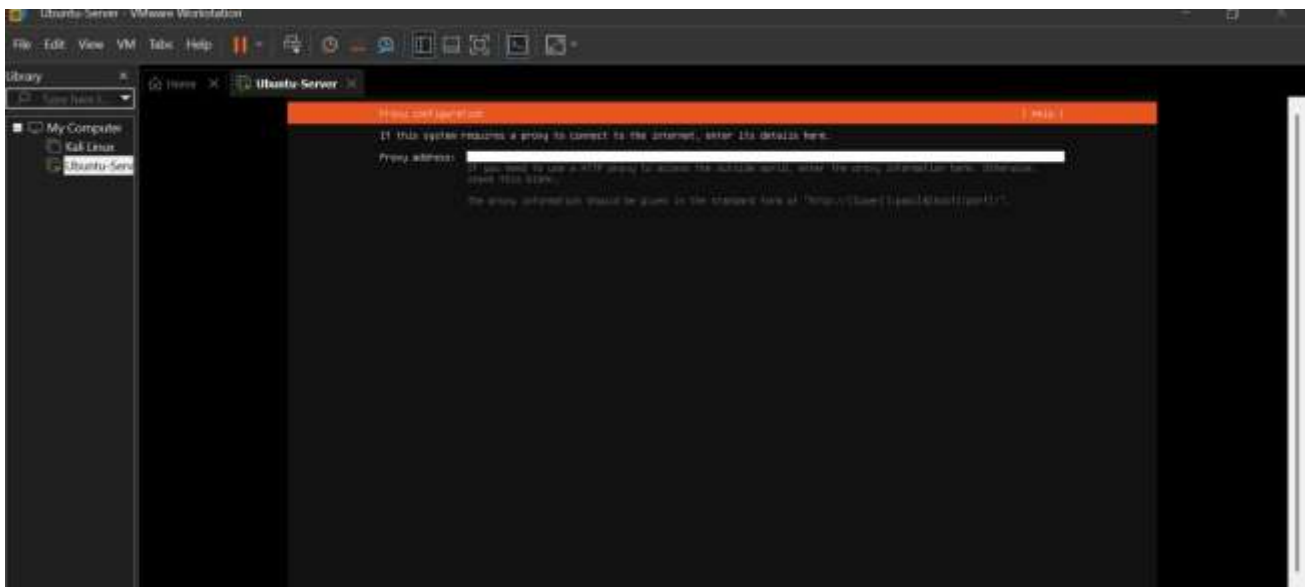
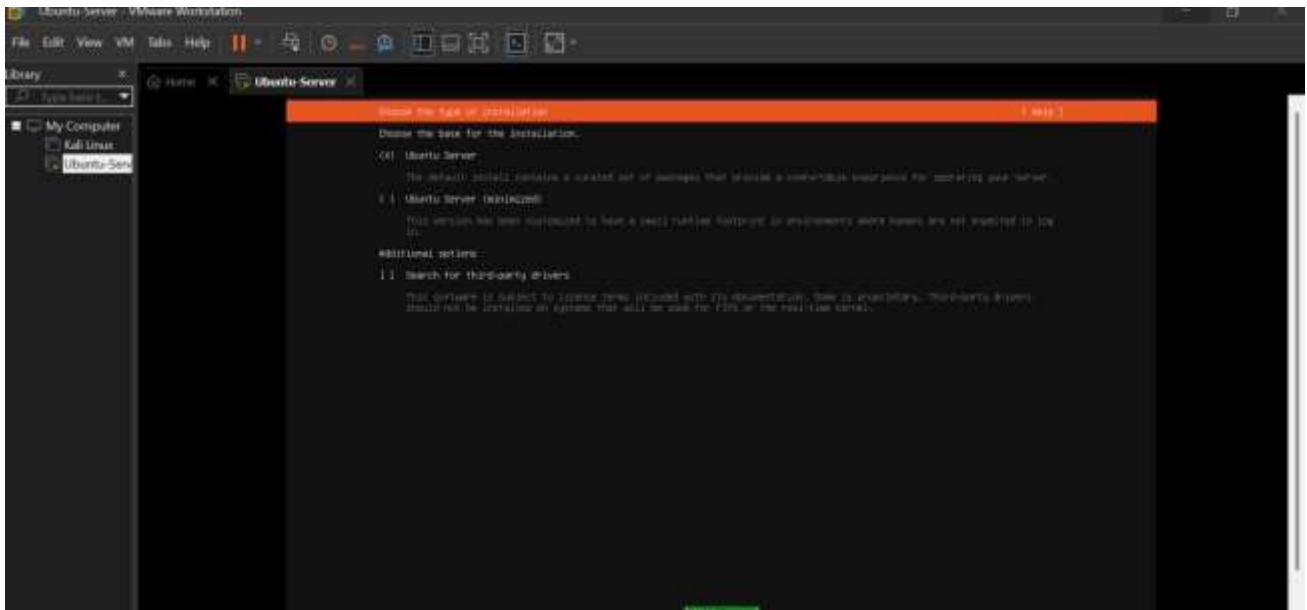


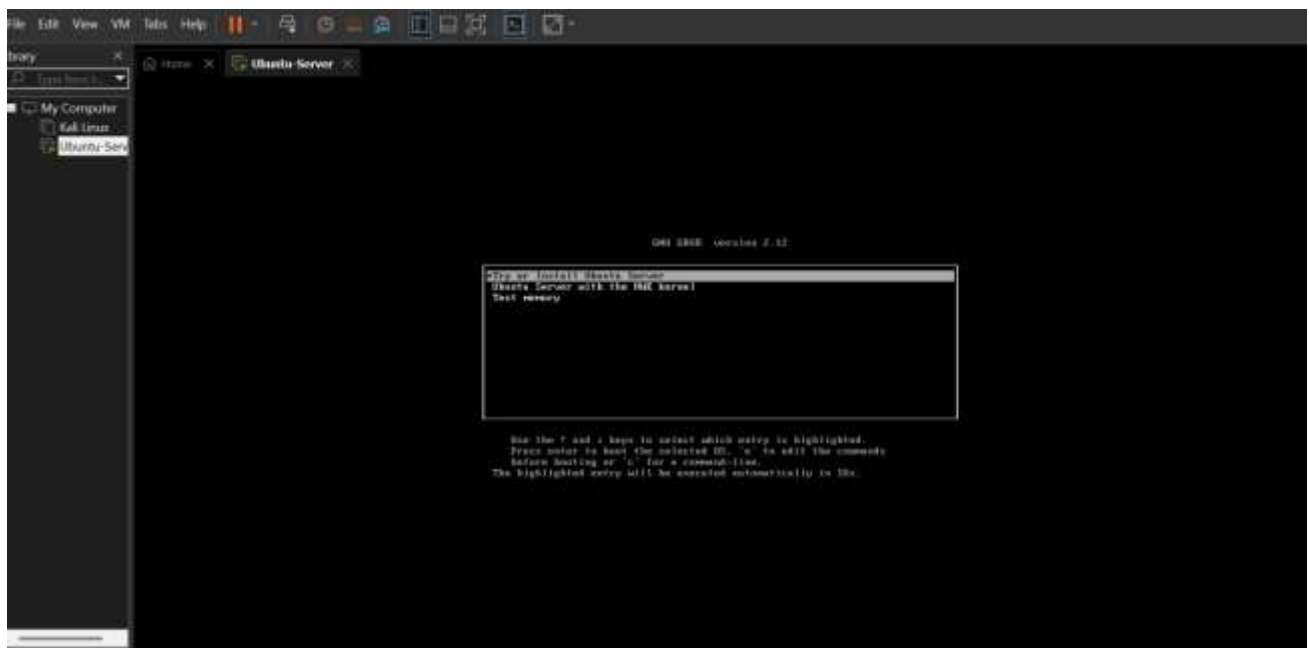


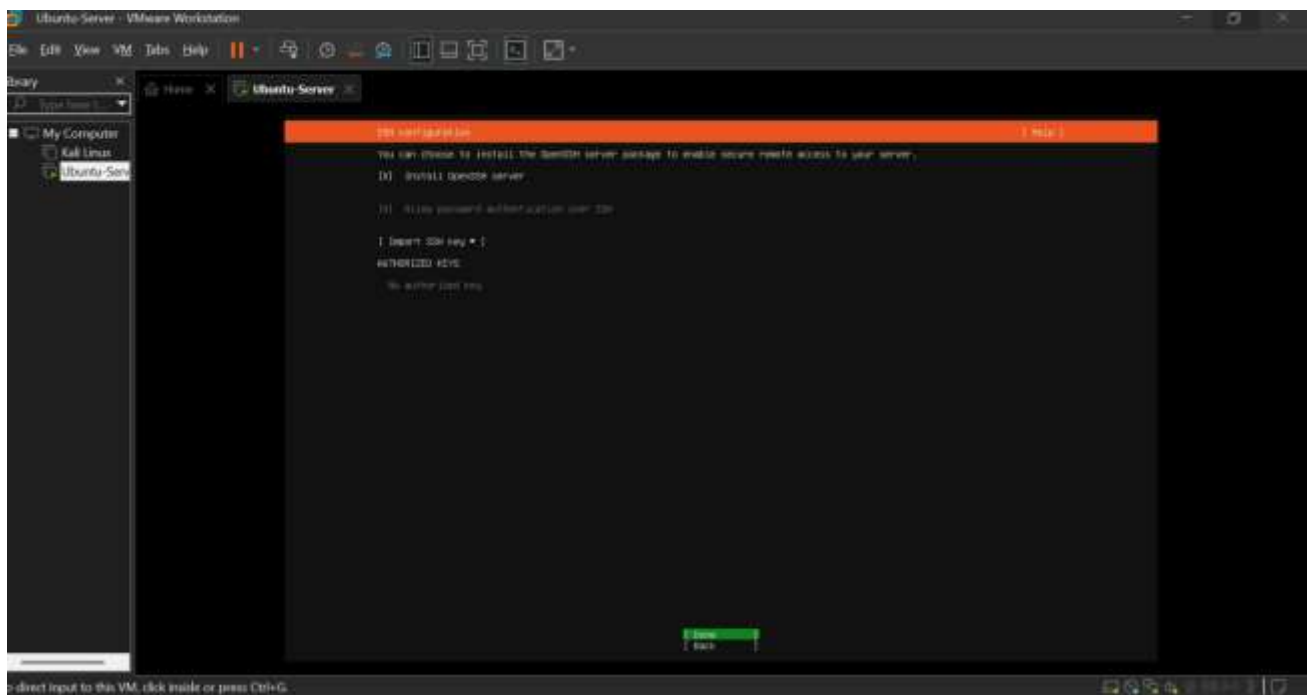
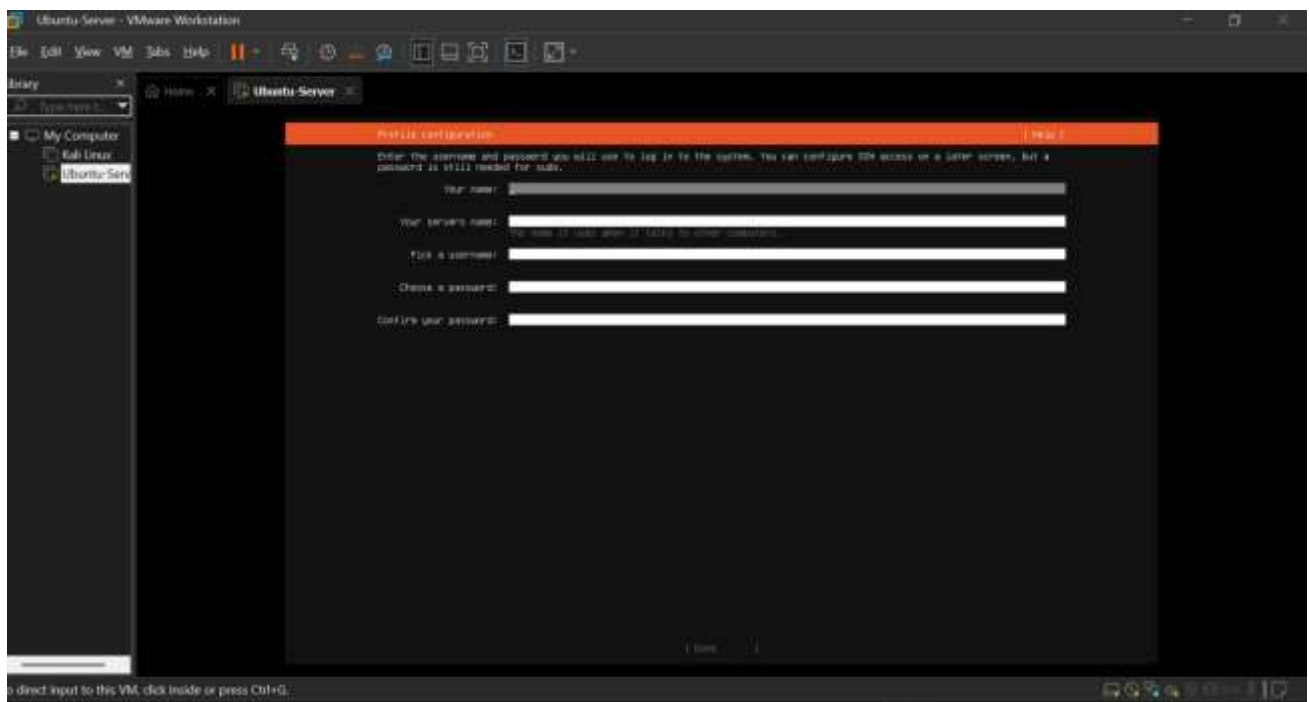


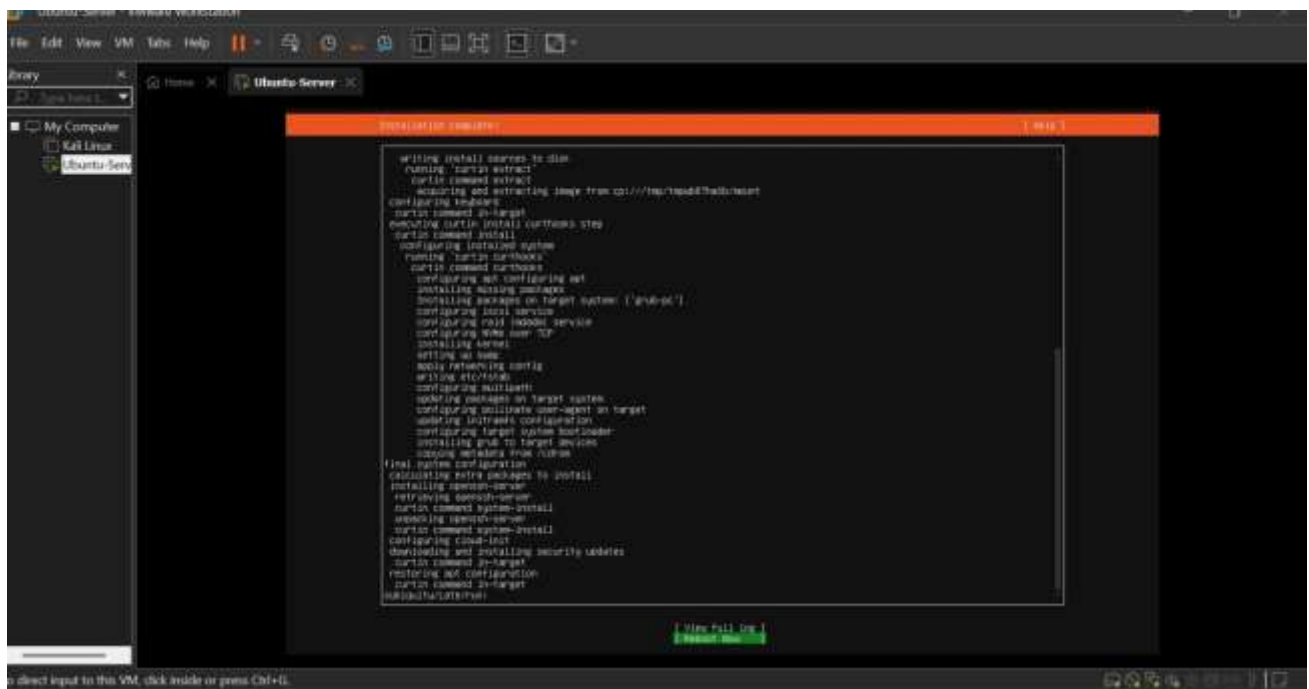
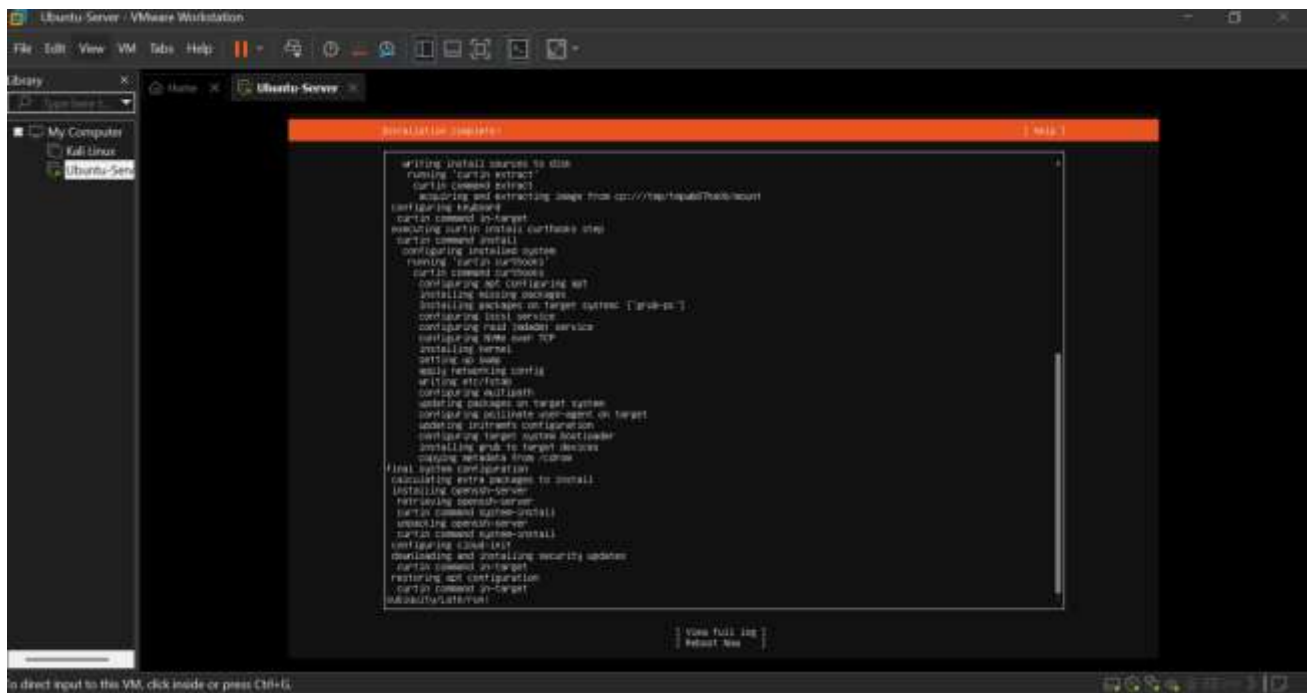


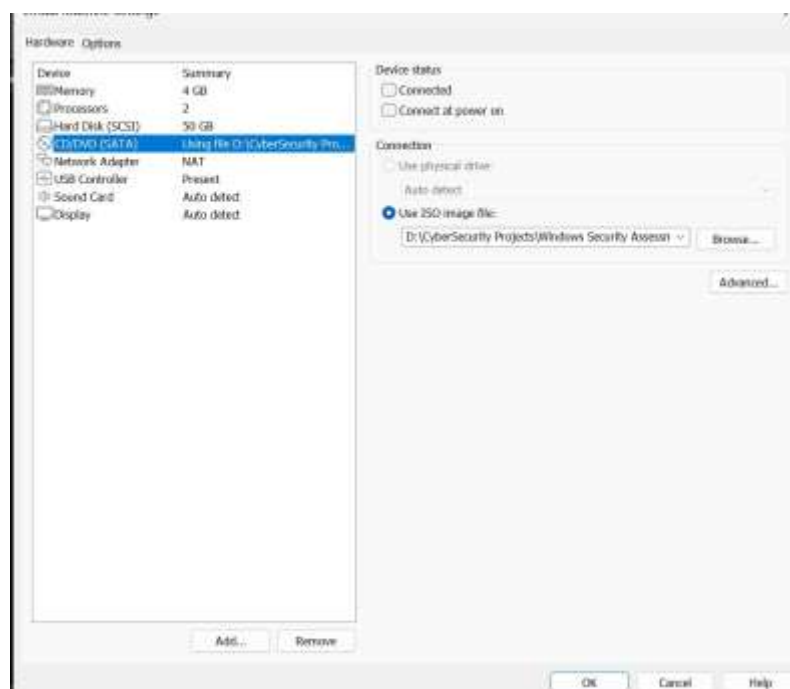
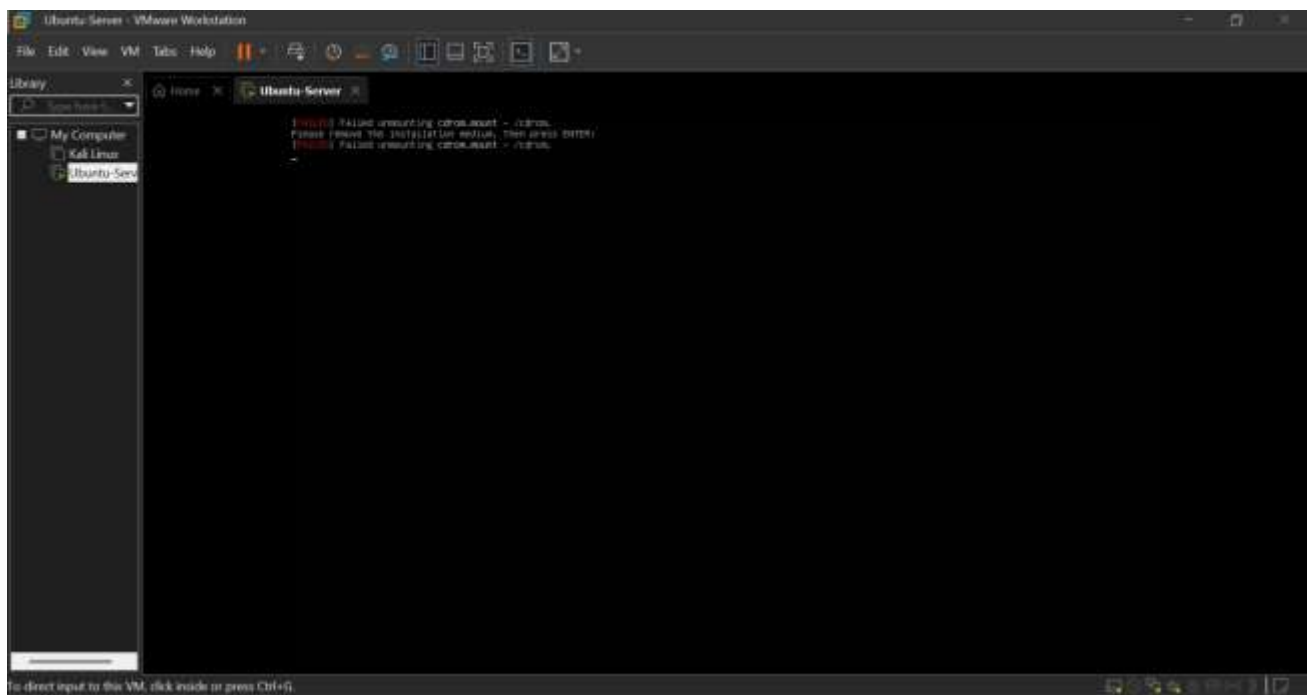


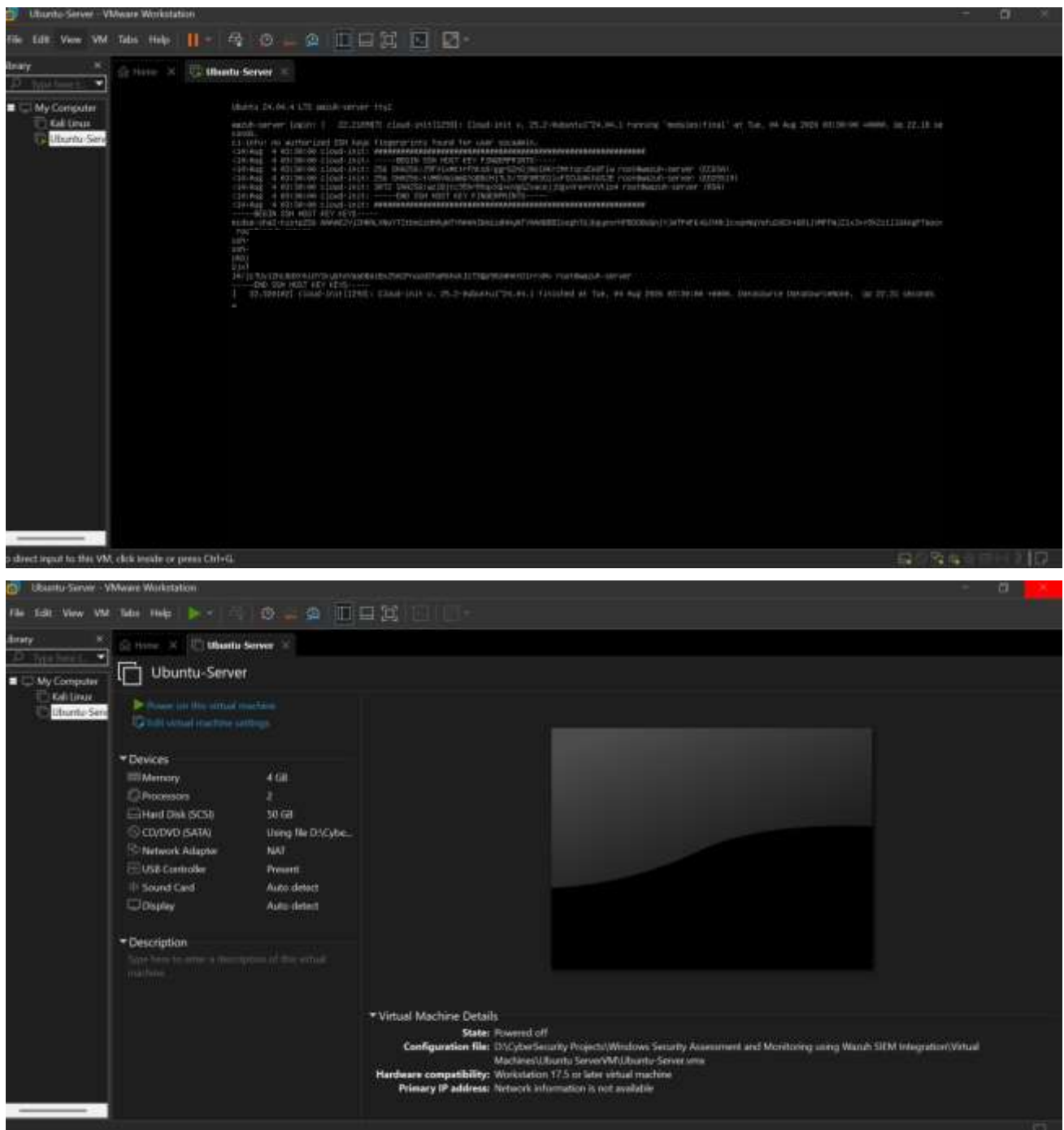




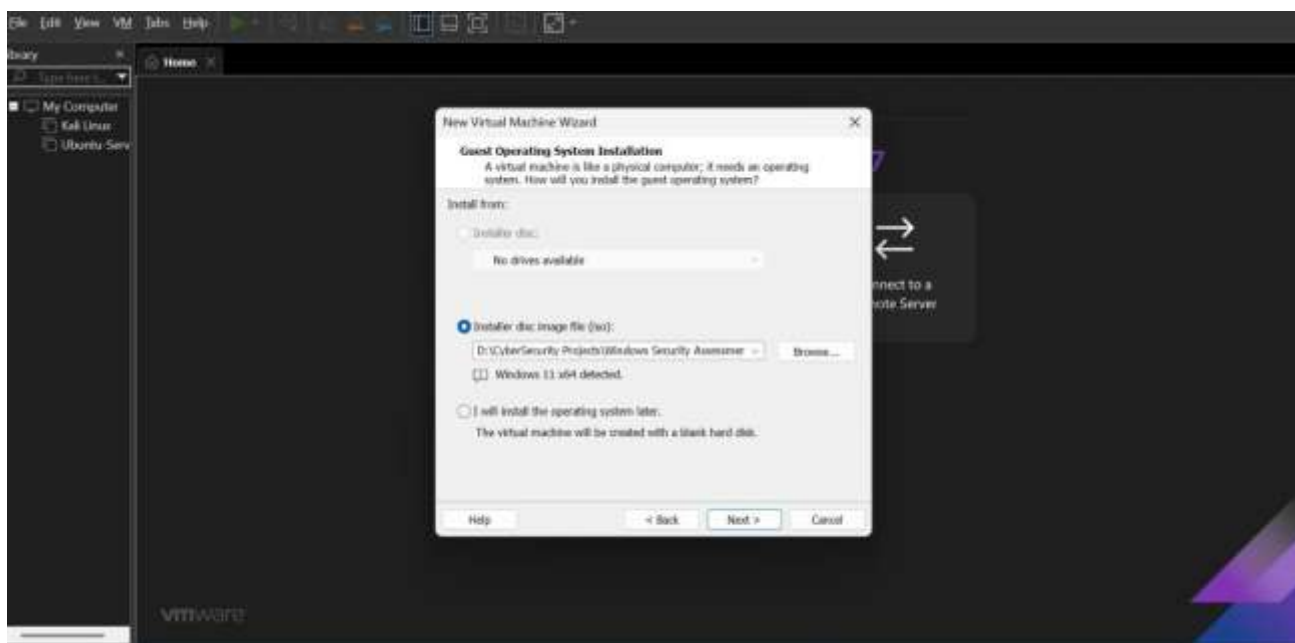
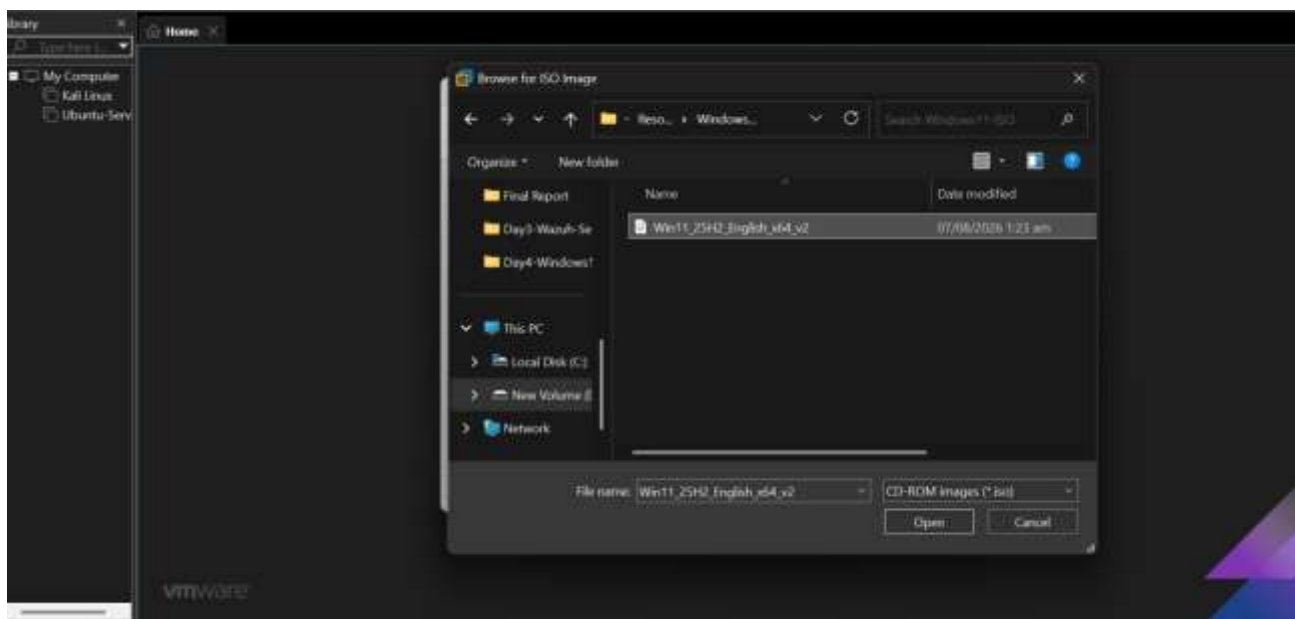
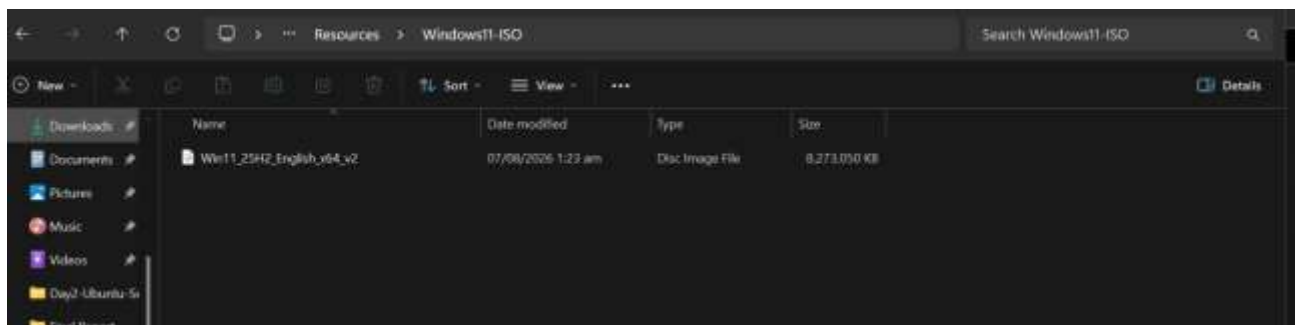


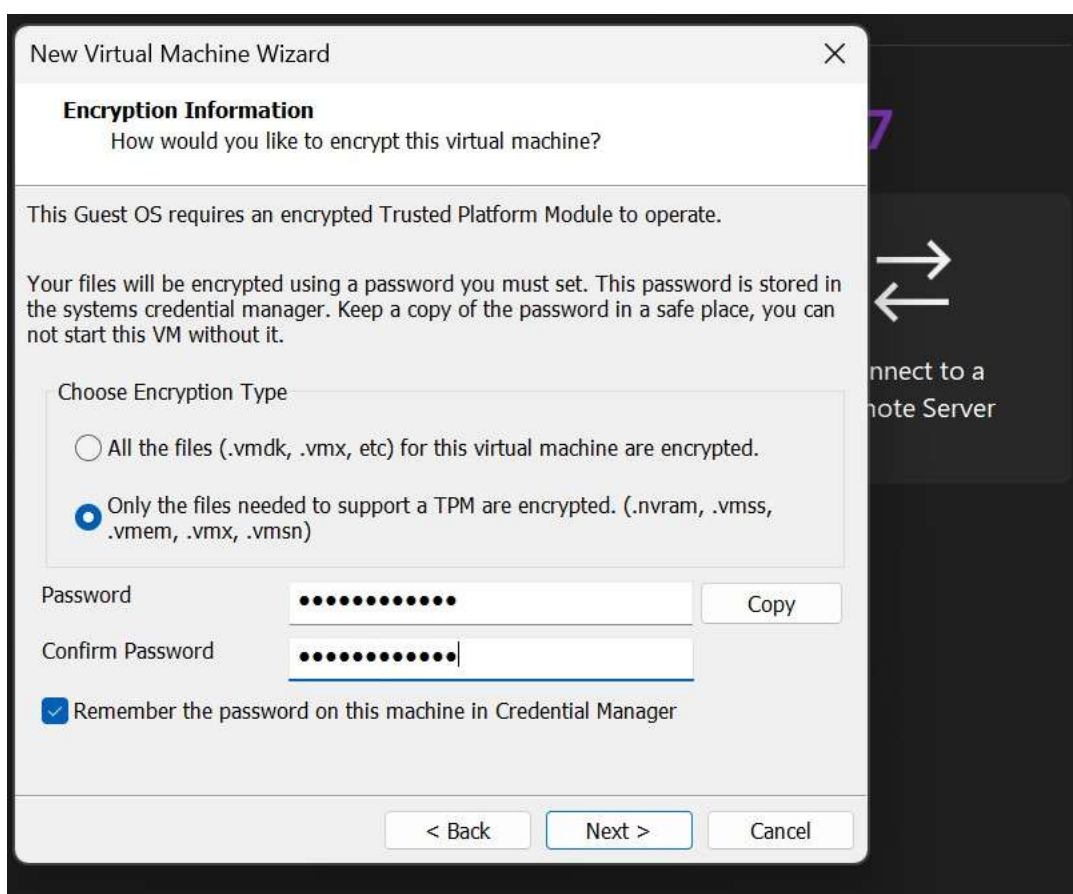
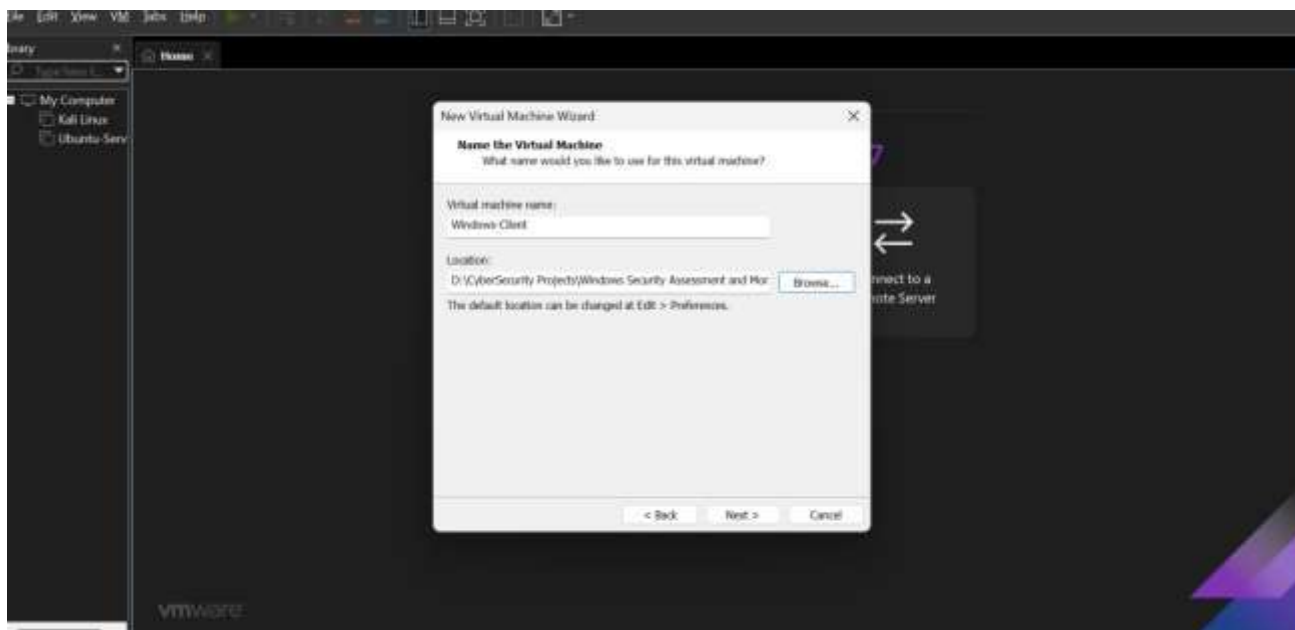


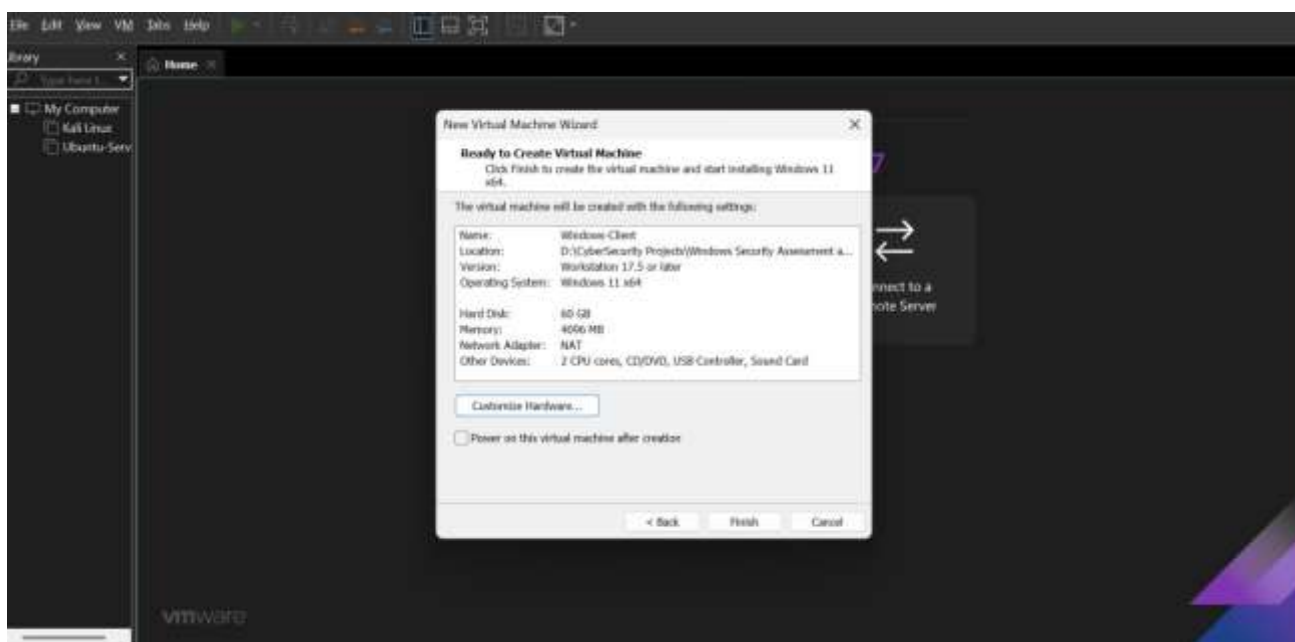
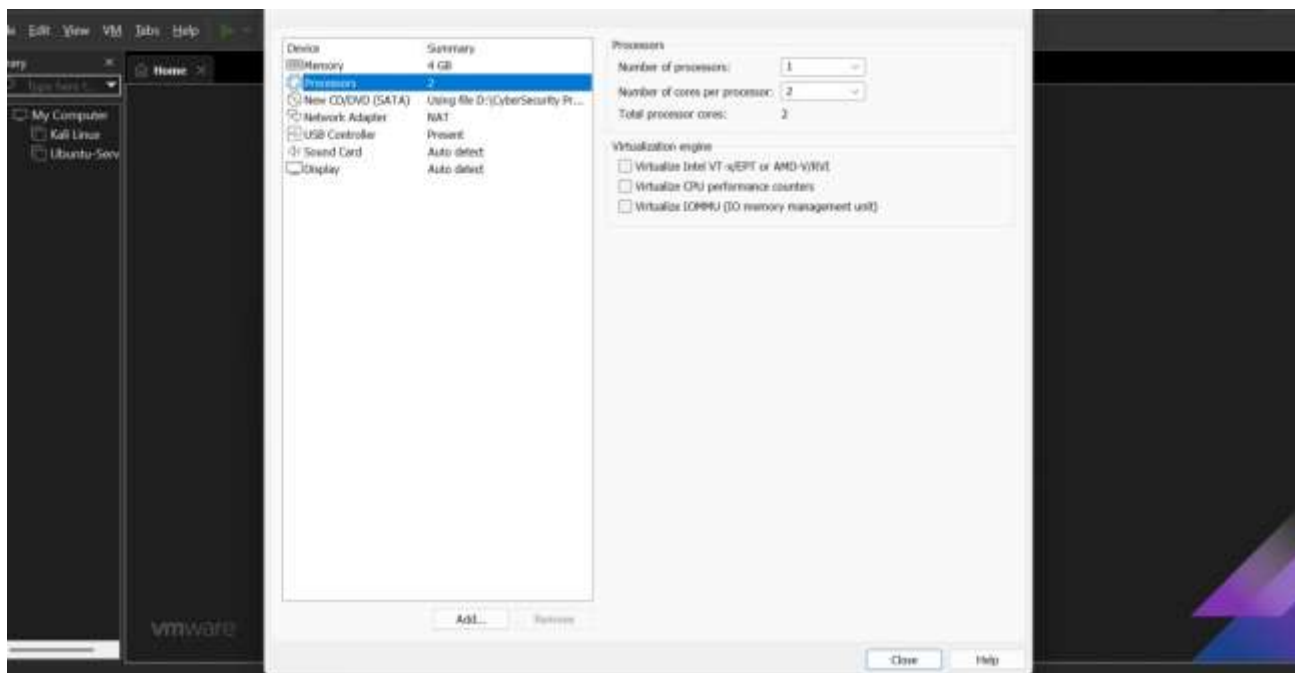


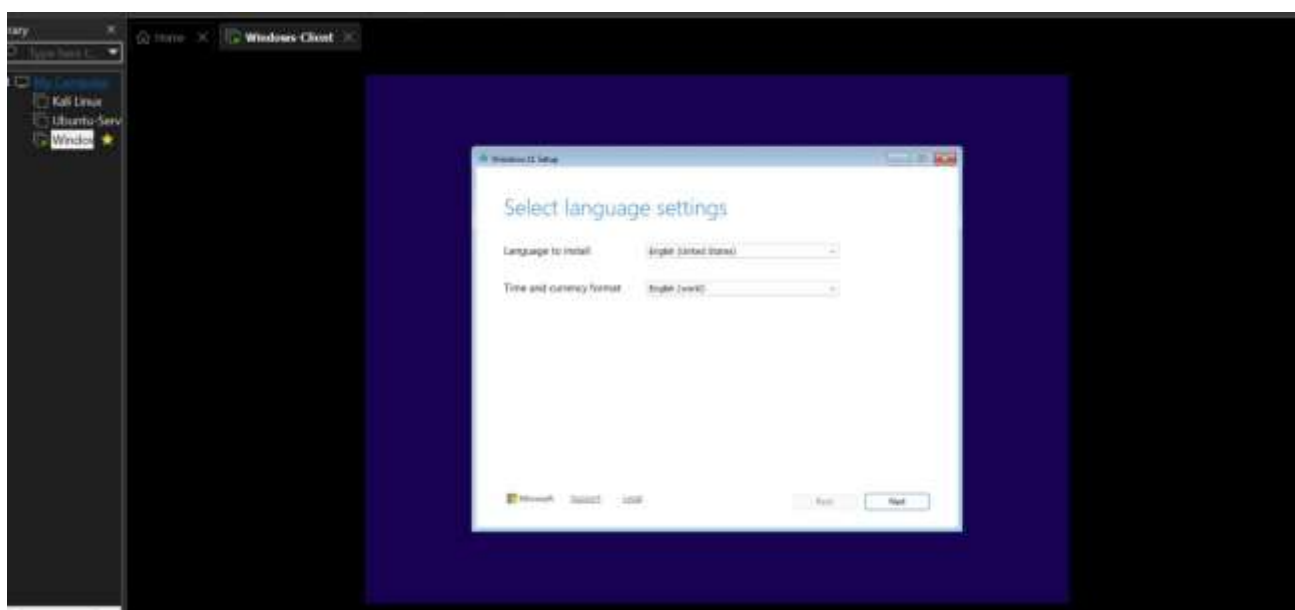
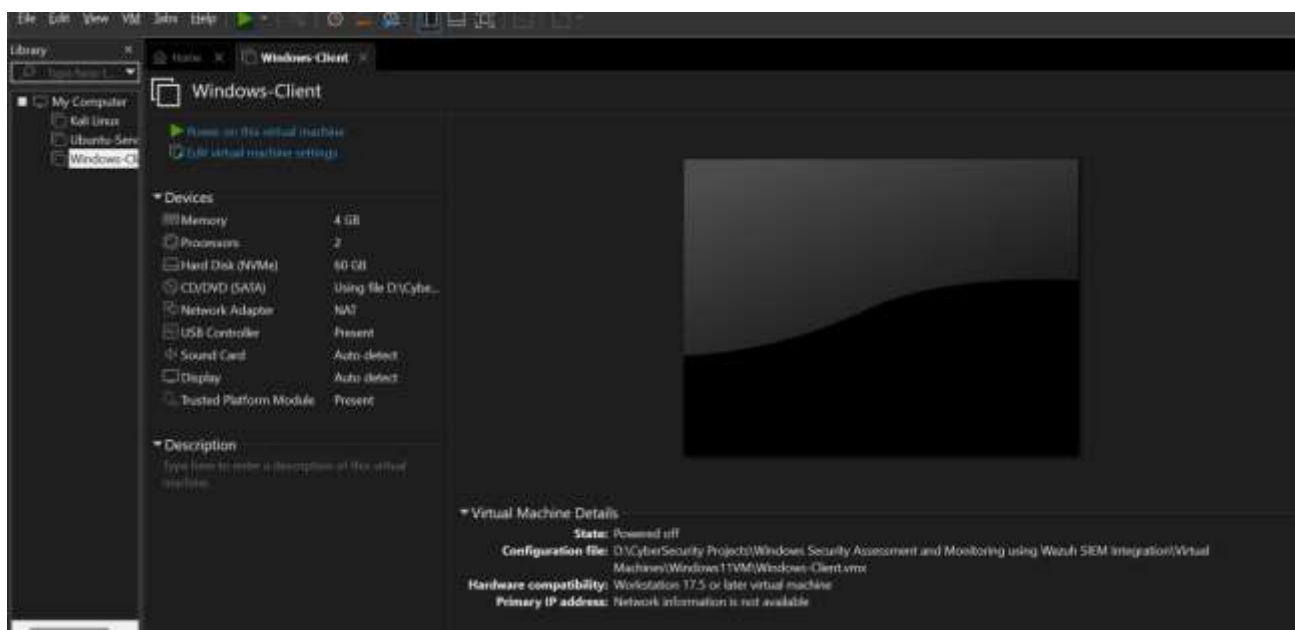


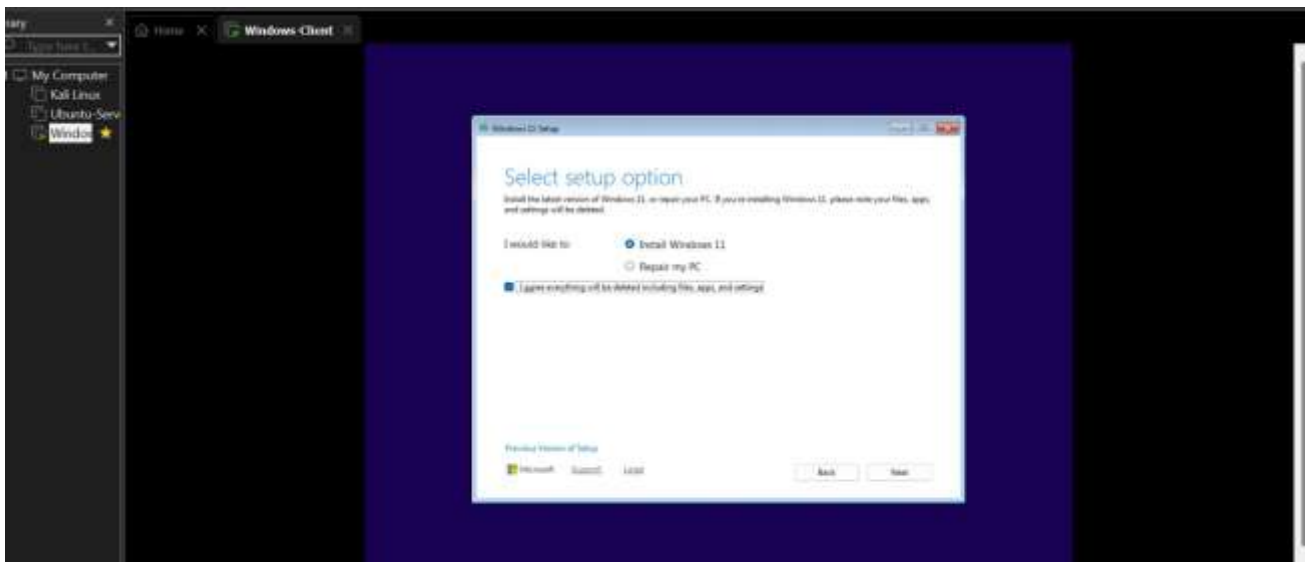
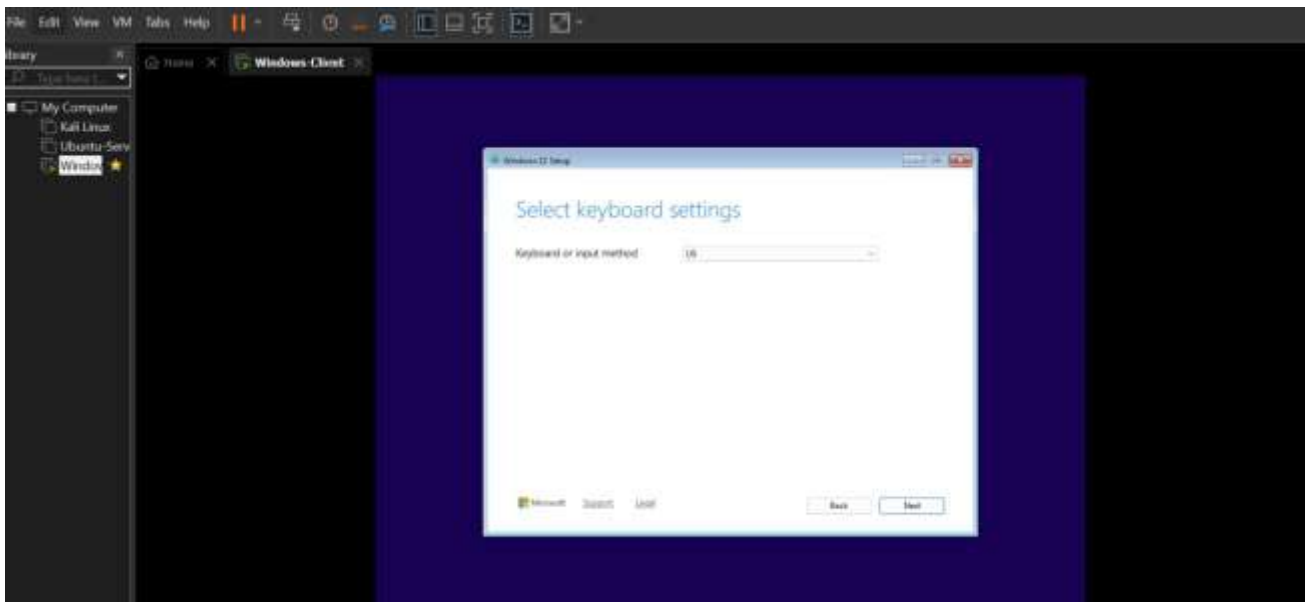
16 Appendix D: Windows VM

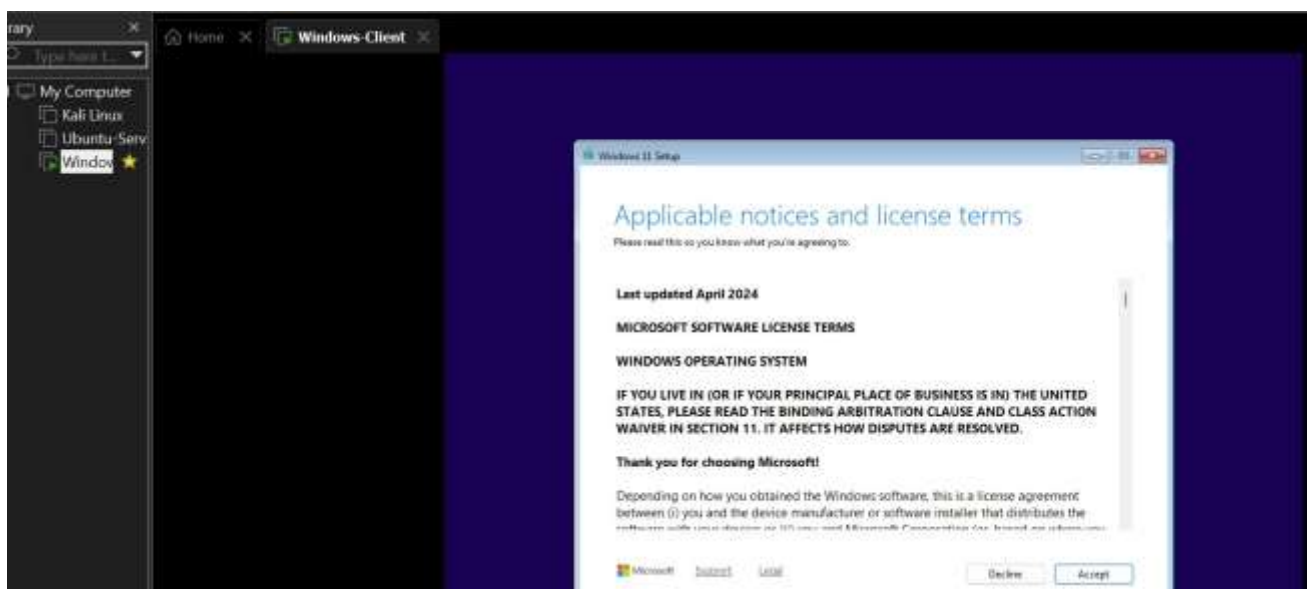
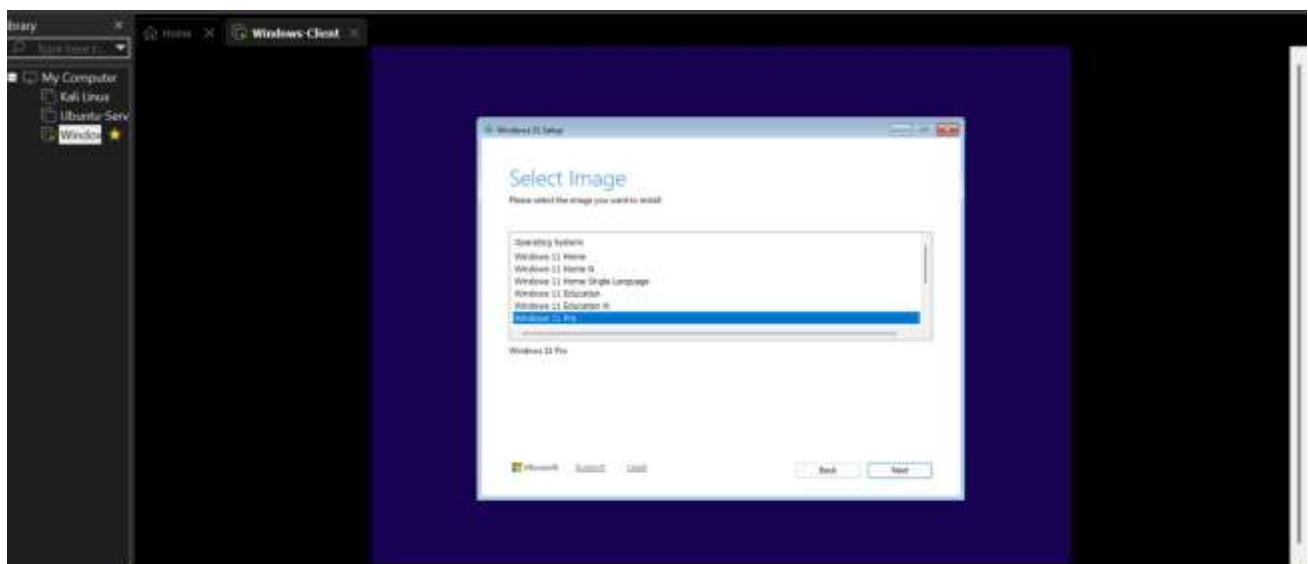
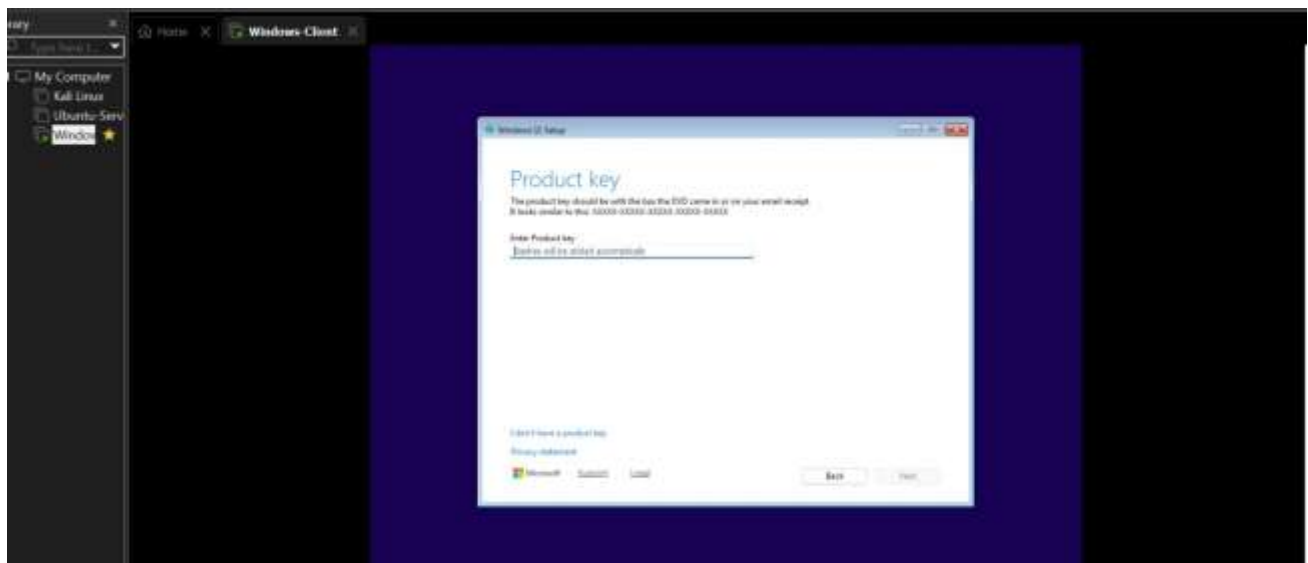


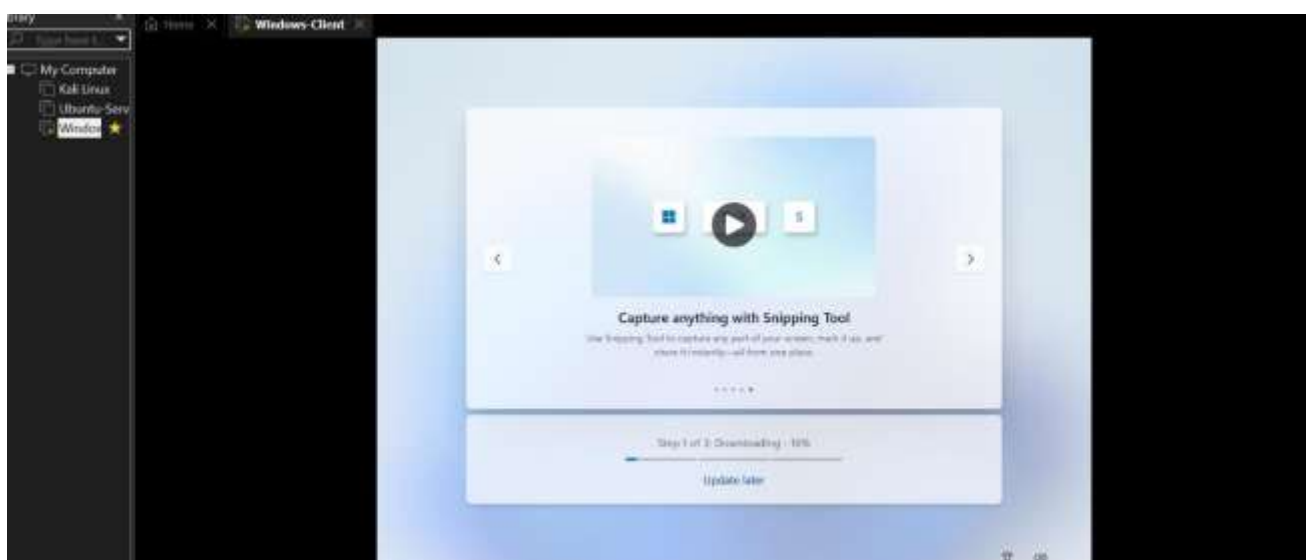
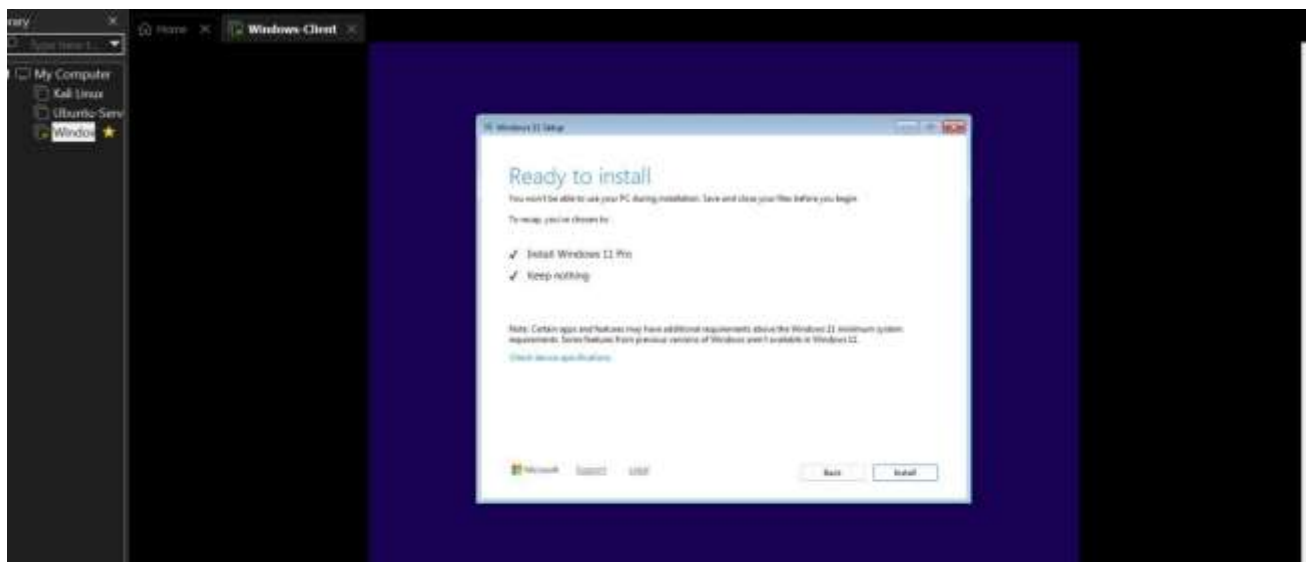


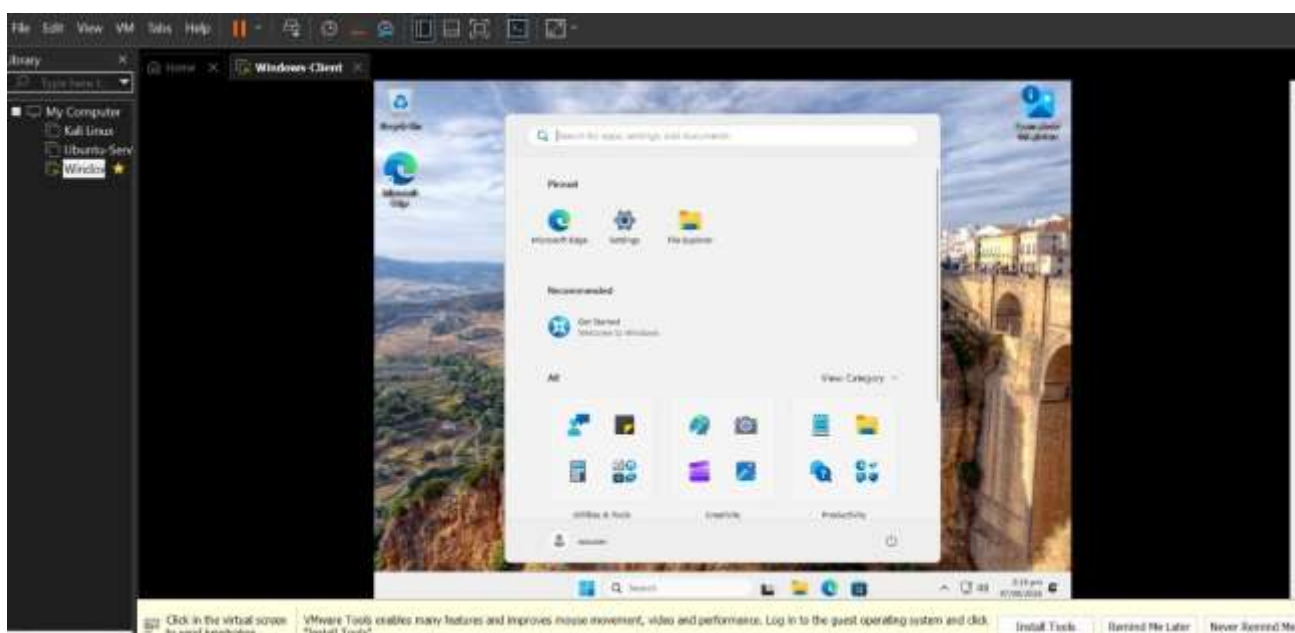
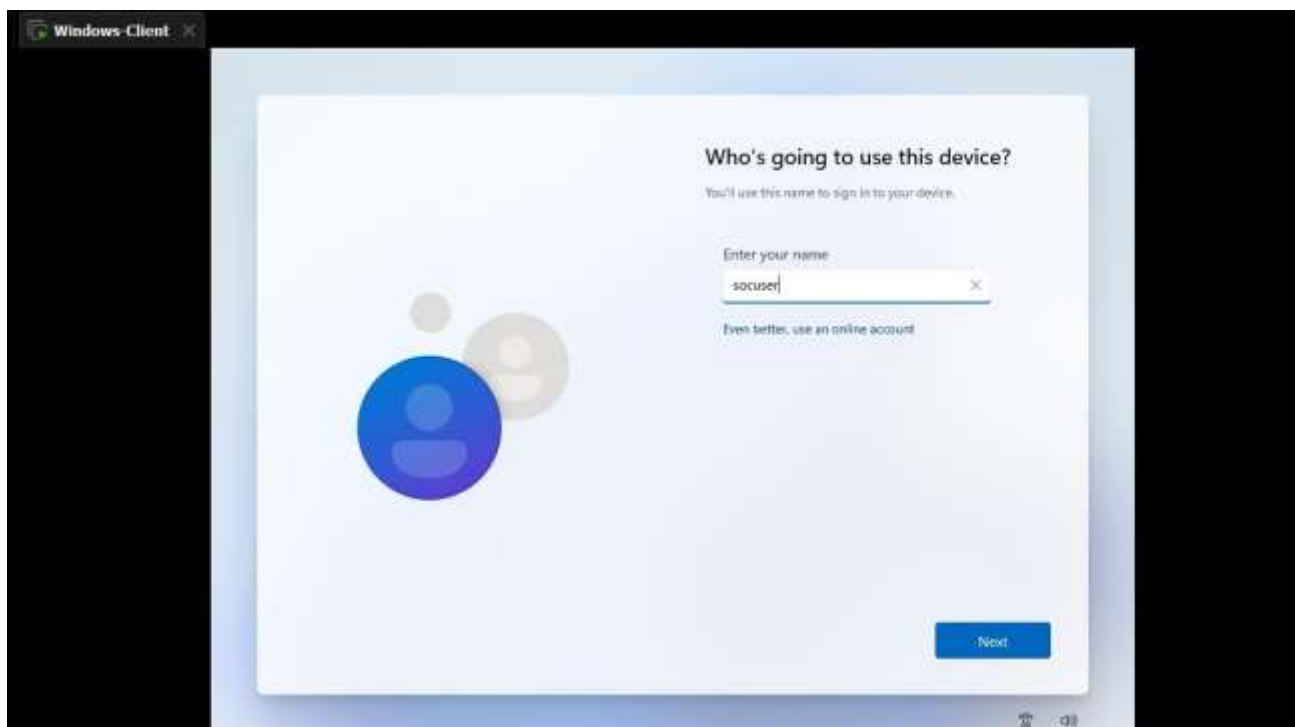


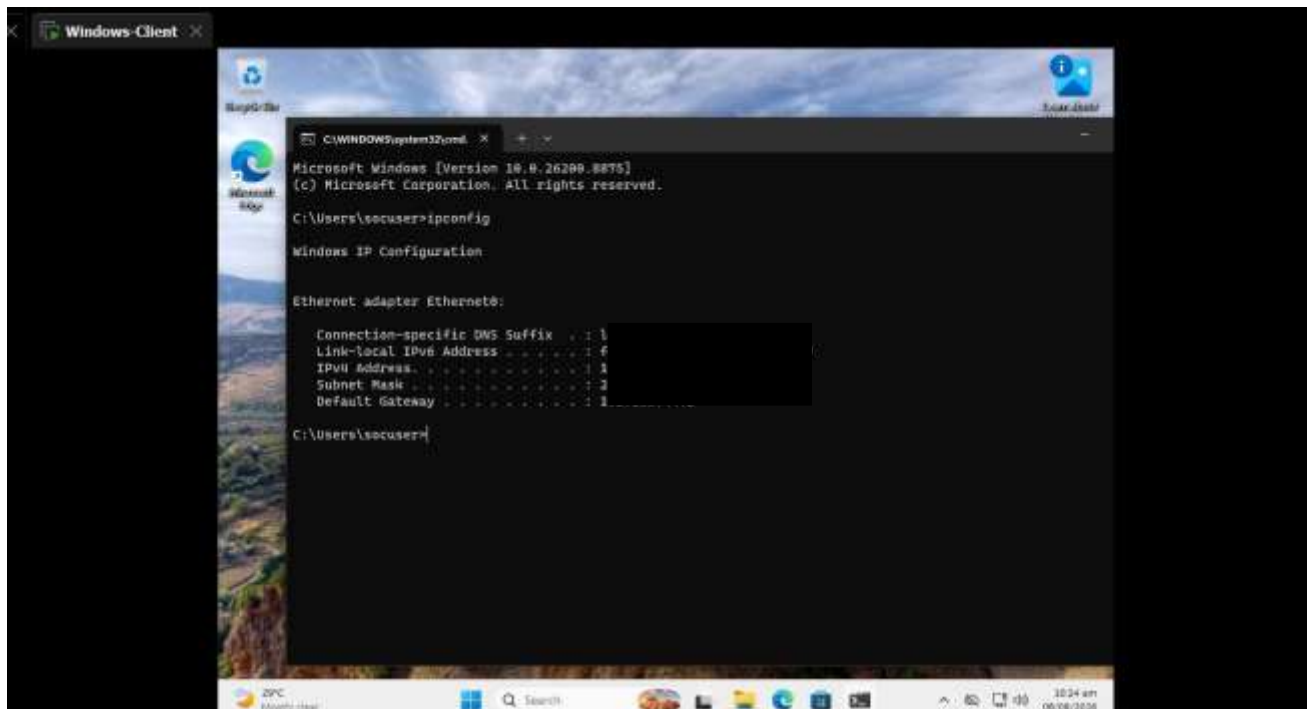
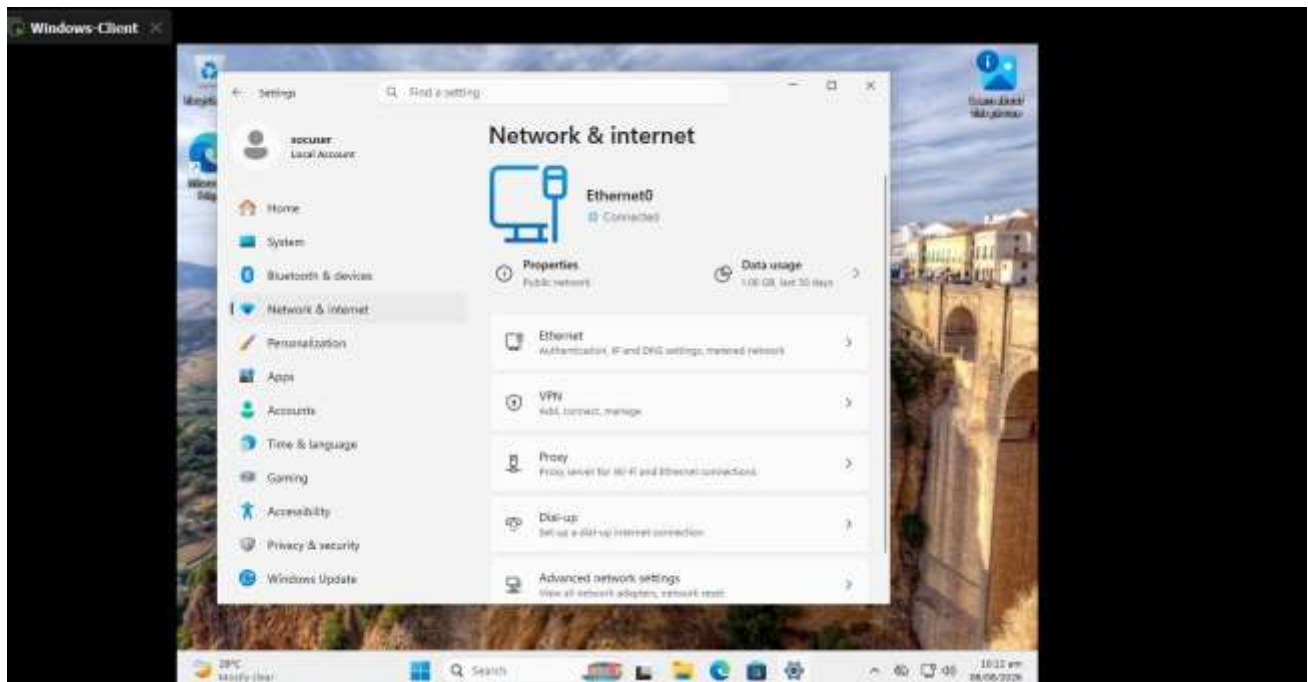






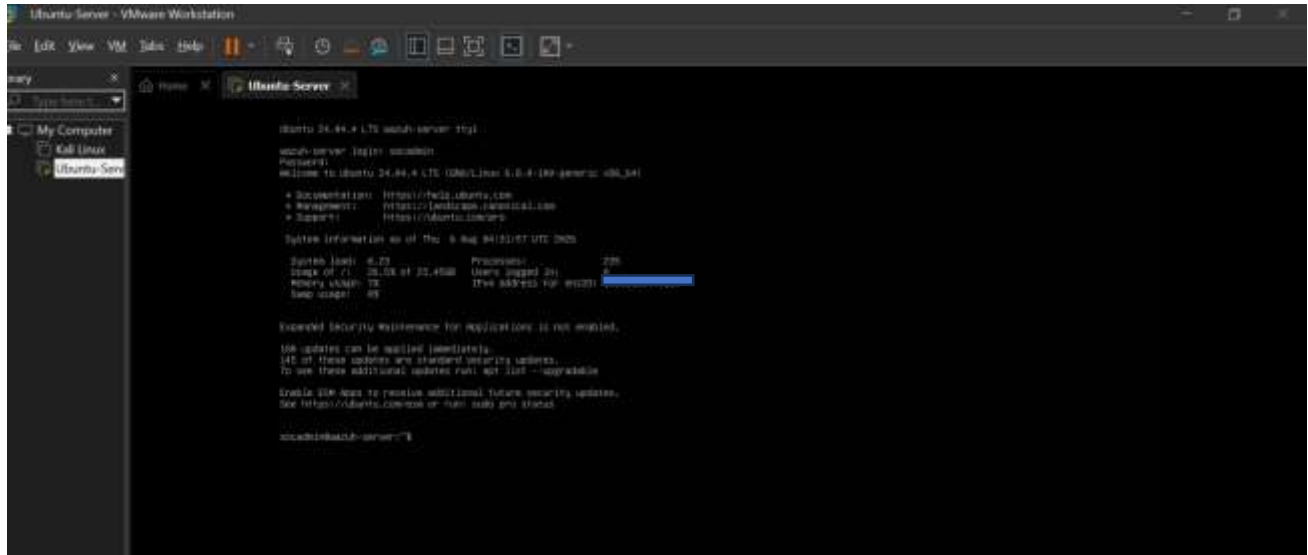






17 Appendix E: Wazuh Evidences

17.1 Wazuh Server:



```
Ubuntu Server - VMware Workstation
File Edit View VM Help
Library
My Computer
Kali Linux
Ubuntu-Server

Ubuntu 24.04.4 LTS wazuh-server tty1
wazuh-server login: wazuh-server
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic amd64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/support

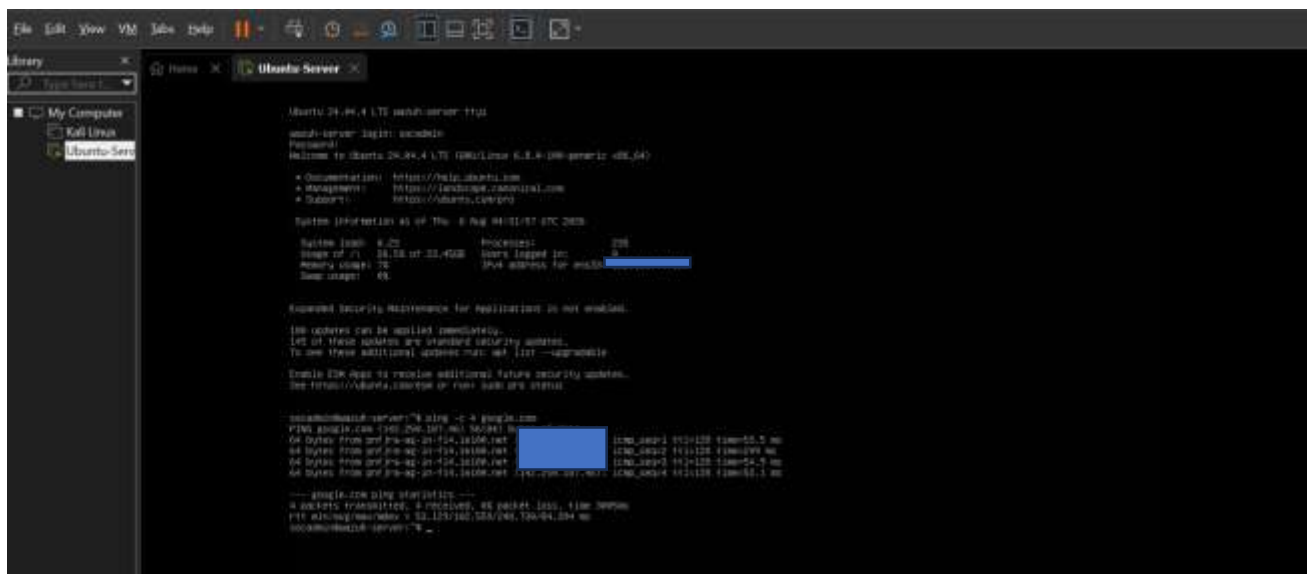
System information as of the 6 Aug 04:01:07 UTC 2025:

System load:  6.73          Processes:    226
Usage of /:   25.2% of 32.4GB    Users logged in:  0
Memory usage: 7%             IPv6 address for eno1: 
Swap usage:   0%

Expanded security maintenance for applications is not enabled.
IMM updates can be applied immediately.
145 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: apt-get install esm

wazuh@wazuh-server:~$
```



```
Ubuntu 24.04.4 LTS wazuh-server tty1
wazuh-server login: wazuh-server
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic amd64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/support

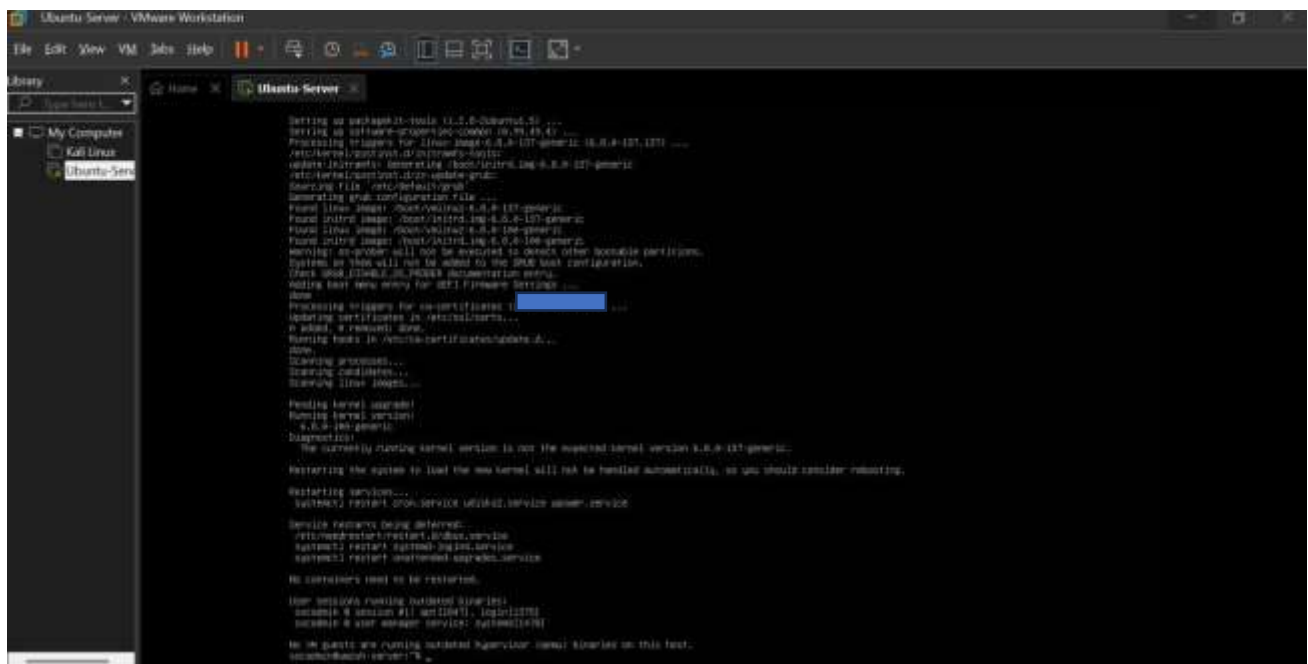
System information as of the 6 Aug 04:01:07 UTC 2025:

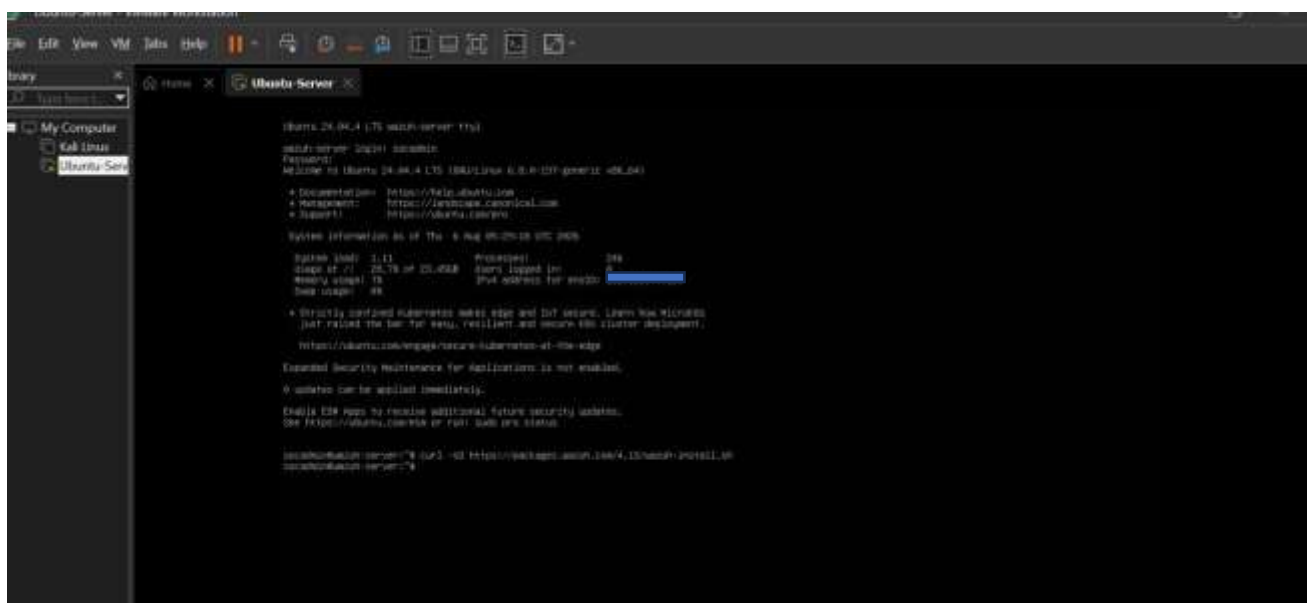
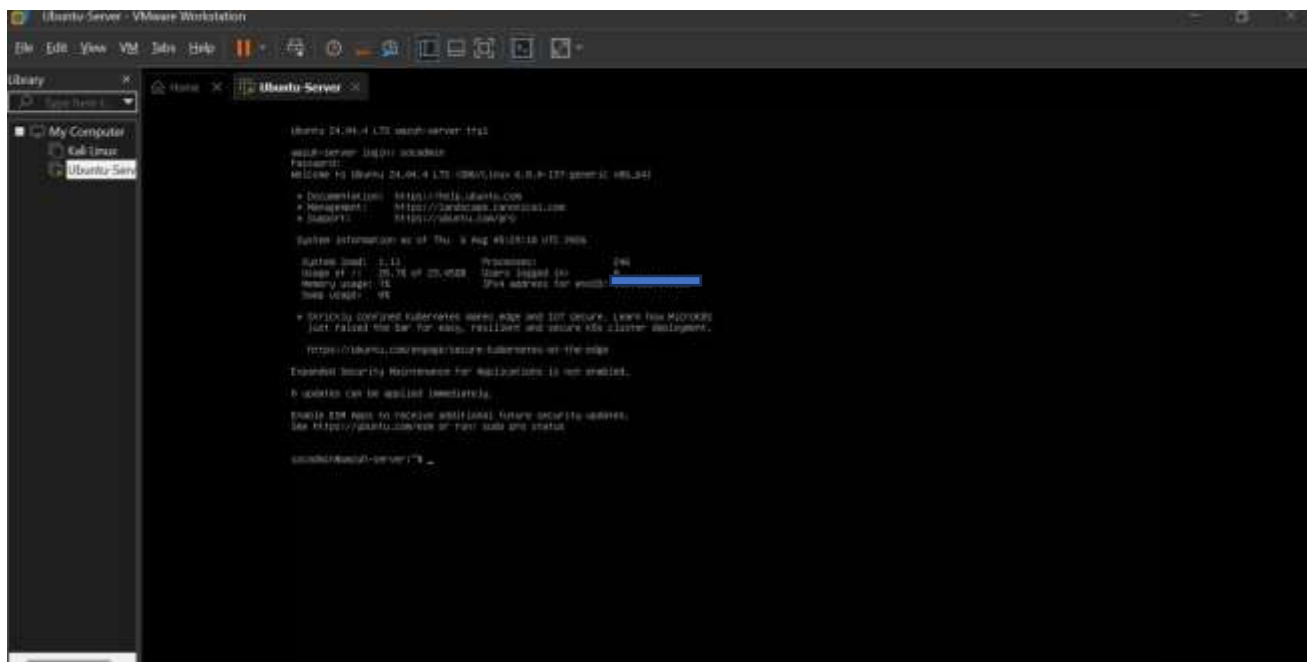
System load:  6.73          Processes:    226
Usage of /:   25.2% of 32.4GB    Users logged in:  0
Memory usage: 7%             IPv6 address for eno1: 
Swap usage:   0%

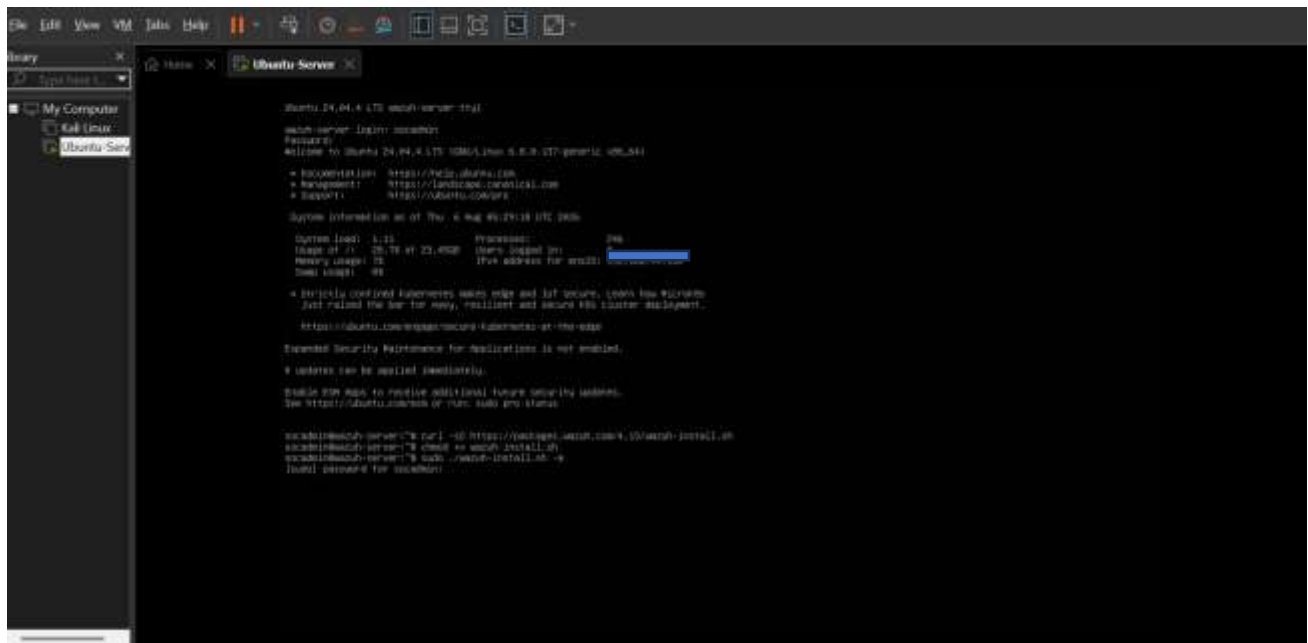
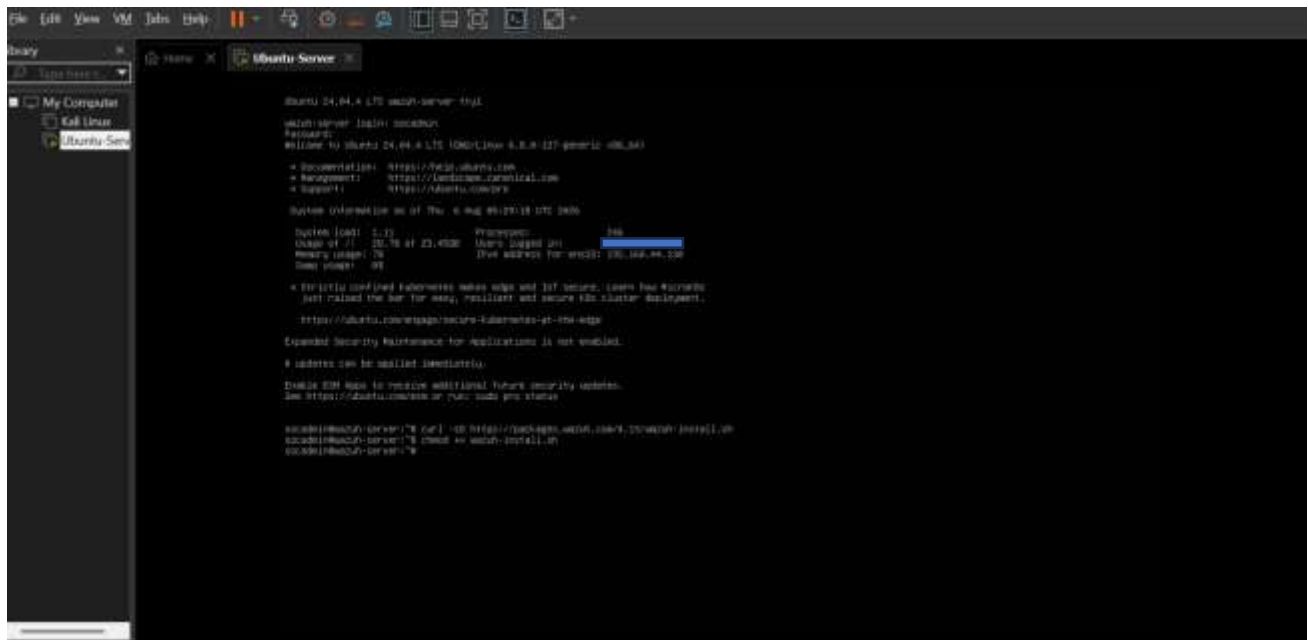
Expanded security maintenance for applications is not enabled.
IMM updates can be applied immediately.
145 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: apt-get install esm

wazuh@wazuh-server:~$ ping -c 4 google.com
PING google.com (142.250.190.160): 64 bytes of data: 64 bytes from google.com: icmp_seq=1 ttl=64 time=55.5 ms
64 bytes from google.com: icmp_seq=2 ttl=64 time=55.9 ms
64 bytes from google.com: icmp_seq=3 ttl=64 time=54.5 ms
64 bytes from google.com: icmp_seq=4 ttl=64 time=51.1 ms
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 200ms
rtt min/avg/max/mdev = 52.123/52.524/56.726/4.104 ms
wazuh@wazuh-server:~$
```







```
System: Ubuntu 18.04 LTS
Kernel: 4.15.0-47-generic
Architecture: x86_64
CPU: Intel Core i7-8700K
Memory: 32GB
Swap: 2GB
Disk: 1TB
Network: Ethernet
IP Address: 192.168.1.100
Gateway: 192.168.1.1
DNS: 8.8.8.8, 8.8.4.4
Hostname: ubuntu-server

# Verify if correct subnets are used and not secure. Learn how Mikrotik
# just raised the bar for easy, resilient and secure VPN cluster deployment.
https://ubuntus.com/vpnpage/secure-subnetted-at-the-edge

Expanded Security Maintenance for Applications is not enabled.
# updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run sudo apt update.

$ sudo apt-get update
$ sudo apt-get install -y mikrotik-vpn

$ sudo systemctl status mikrotik-vpn
● mikrotik-vpn.service
   Loaded: loaded (/etc/systemd/system/mikrotik-vpn.service; vendor preset: enabled)
   Active: active (running) since 2020-06-04 13:00:00; 1h ago
     Main PID: 1000 (mikrotik-vpn)
       CGroup: /system.slice/mikrotik-vpn.service
               └─ 1000 /usr/bin/mikrotik-vpn

$ sudo journalctl -u mikrotik-vpn
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting mikrotik-vpn installation assistant. Mikrotik version: 4.30.1
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik logging redirected to /var/log/mikrotik-install.log
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Verifying that your system meets the recommended Mikrotik hardware requirements.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik web interface port will be 441.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: --- Proceeding ---
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Installing apt repository script.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Installing debhelper.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik repository added.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: --- Configuration files ---
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generating configuration files.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generating the root certificate.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generating Mikrotik certificates.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generating Mikrotik cluster certificates.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generating Mikrotik dashboard certificates.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generated mikrotik-install-files.tar. It contains the Mikrotik cluster key, certificates, and passwords necessary for installation.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: --- Mikrotik cluster ---
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting Mikrotik cluster installation.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik cluster installation finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik cluster post-install configuration finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting service mikrotik-vpn.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik cluster service started.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Initializing Mikrotik cluster security settings.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik cluster security configuration initialized.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik cluster installation finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: --- Mikrotik cluster ---
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting the Mikrotik manager installation.
```

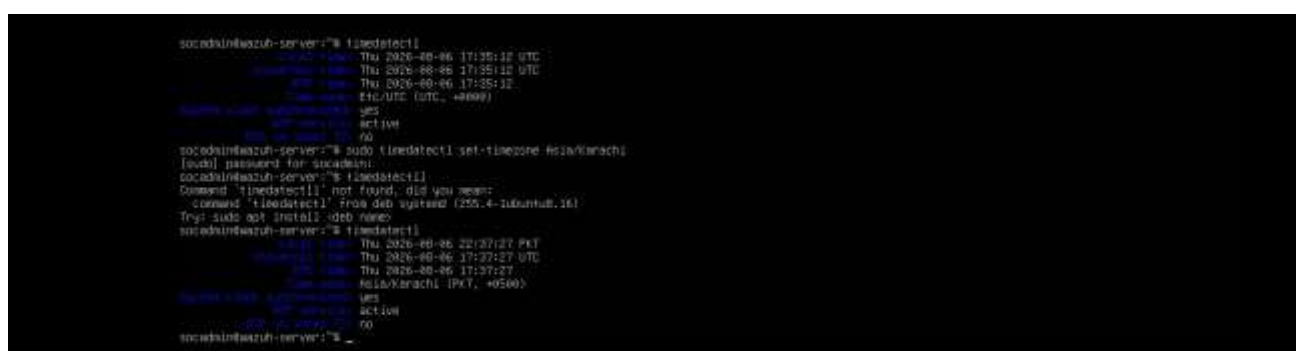
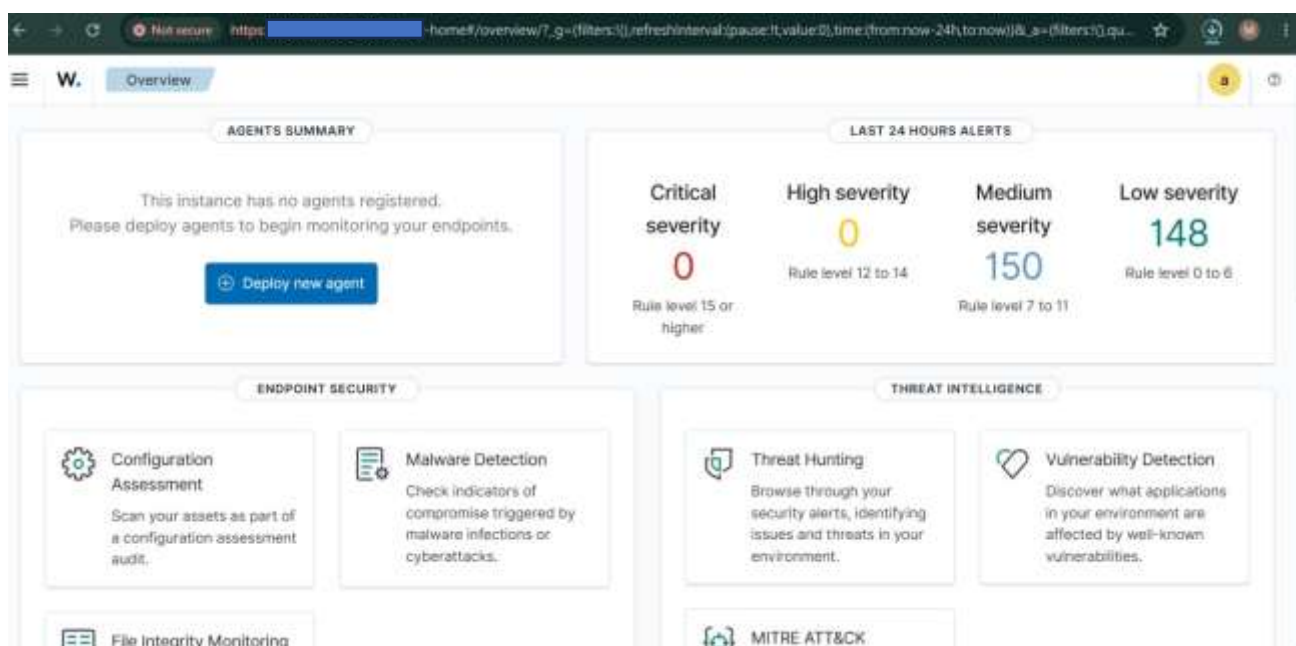
```
$ updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run sudo apt update.

$ sudo apt-get update
$ sudo apt-get install -y mikrotik-vpn

$ sudo systemctl status mikrotik-vpn
● mikrotik-vpn.service
   Loaded: loaded (/etc/systemd/system/mikrotik-vpn.service; vendor preset: enabled)
   Active: active (running) since 2020-06-04 13:00:00; 1h ago
     Main PID: 1000 (mikrotik-vpn)
       CGroup: /system.slice/mikrotik-vpn.service
               └─ 1000 /usr/bin/mikrotik-vpn

$ sudo journalctl -u mikrotik-vpn
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting mikrotik-vpn installation assistant. Mikrotik version: 4.30.1
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik logging redirected to /var/log/mikrotik-install.log
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Verifying that your system meets the recommended Mikrotik hardware requirements.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik web interface port will be 441.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: --- Proceeding ---
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Installing apt repository script.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Installing debhelper.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik repository added.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: --- Configuration files ---
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generating configuration files.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generating the root certificate.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generating Mikrotik certificates.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generating Mikrotik cluster certificates.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generating Mikrotik dashboard certificates.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Generated mikrotik-install-files.tar. It contains the Mikrotik cluster key, certificates, and passwords necessary for installation.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: --- Mikrotik cluster ---
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting Mikrotik cluster installation.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik cluster installation finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik cluster post-install configuration finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting service mikrotik-vpn.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik cluster service started.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Initializing Mikrotik cluster security settings.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik cluster security configuration initialized.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik cluster installation finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: --- Mikrotik cluster ---
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting the Mikrotik manager installation.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik manager installation finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik manager post-install configuration finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting service mikrotik-vpn.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik manager service started.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik manager installation finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik manager post-install configuration finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting service mikrotik-vpn.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik manager service started.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik manager installation finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik manager post-install configuration finished.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Starting service mikrotik-vpn.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik manager service started.
Jun 04 13:00:00 ubuntu-server: mikrotik-vpn: Mikrotik manager installation finished.
```

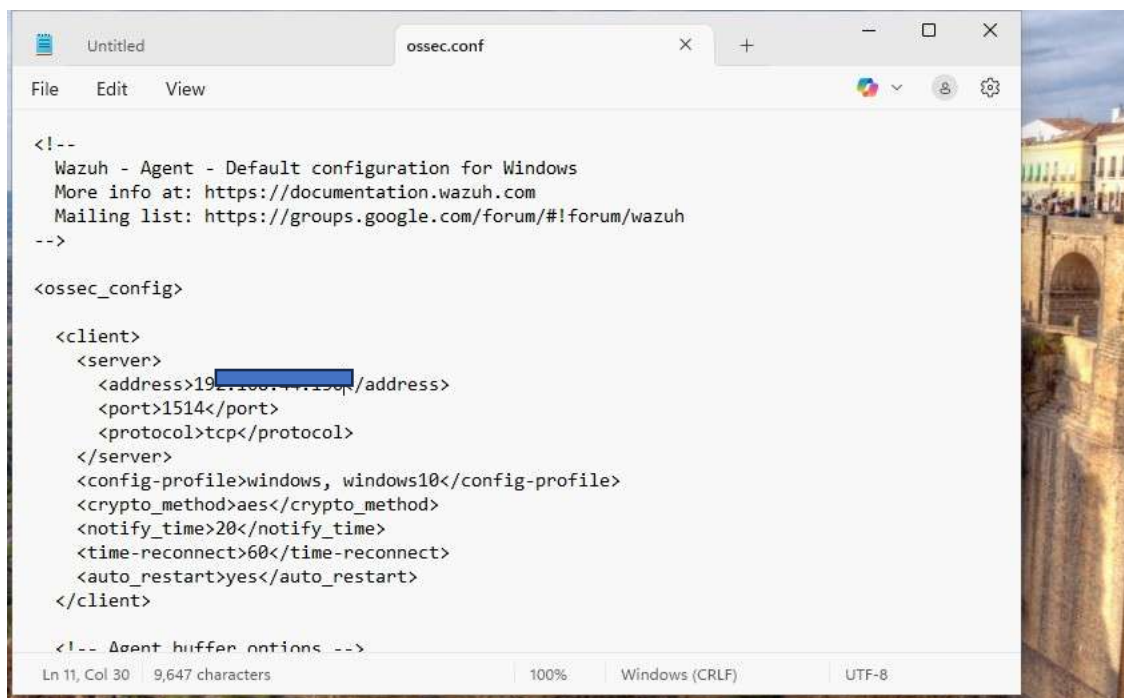



17.2 Wazuh Agent:

```
3 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo apt status  
  
socat@wazuh-server:~$ hostname -i  
socat@wazuh-server:~$ _
```

```
C:\Users\socuser>ping 192.168.0.136  
  
Pinging 192.168.0.136 [192.168.0.136]: 32 bytes of data:  
Reply from 192.168.0.136: bytes=32 time=1ms TTL=64  
Reply from 192.168.0.136: bytes=32 time=1ms TTL=64  
Reply from 192.168.0.136: bytes=32 time=1ms TTL=64  
Reply from 192.168.0.136: bytes=32 time=1ms TTL=64  
  
Ping statistics for 192.168.0.136:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milliseconds:  
        Minimum = 1ms, Maximum = 1ms, Average = 1ms  
C:\Users\socuser>
```

The screenshot shows a web browser displaying the Wazuh documentation page for deploying agents on Windows. The page title is "wazuh." and the version is "Version 4.13". The breadcrumb navigation is "Installation guide / Wazuh agent / Deploying Wazuh agents on Windows endpoints". An attention box states: "Attention This documentation does not apply to the most recent version of Wazuh. Check out the docs for the latest version." The main heading is "Deploying Wazuh agents on Windows endpoints". The text describes the Wazuh agent's role and deployment range. A note box states: "Note You must have administrator privileges to perform the installation." The browser's address bar shows the URL: "https://documentation.wazuh.com/4.13/installation-guide/wazuh-agent/wazuh-agent-package-windo...". The Windows taskbar at the bottom shows the date and time as "1:26 pm 10/08/2026".



```
<!--
Wazuh - Agent - Default configuration for Windows
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>

  <client>
    <server>
      <address>192.168.1.100</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>windows, windows10</config-profile>
    <crypto_method>aes</crypto_method>
    <notify_time>20</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
  </client>

  <!-- Agent buffer options -->

Ln 11, Col 30 | 9,647 characters | 100% | Windows (CRLF) | UTF-8
```

```
PS C:\WINDOWS\system32> Select-String -Path "C:\Program Files (x86)\ossec-agent\ossec.conf" -Pattern "<address>"
C:\Program Files (x86)\ossec-agent\ossec.conf:11:      <address>192.168.1.100</address>

PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> & "C:\Program Files (x86)\ossec-agent\agent-auth.exe" -n [redacted] -A Windows-Client
2026/08/10 14:06:28 agent-auth: INFO: Started (pid: [redacted])
2026/08/10 14:06:28 agent-auth: INFO: Requesting a key from server: [redacted]
2026/08/10 14:06:29 agent-auth: INFO: No authentication password provided
2026/08/10 14:06:29 agent-auth: INFO: Using agent name as: Windows-Client
2026/08/10 14:06:29 agent-auth: INFO: Waiting for server reply
2026/08/10 14:06:29 agent-auth: INFO: Valid key received
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> & "C:\Program Files (x86)\ossec-agent\agent-auth.exe" -m [REDACTED] -A Windows-Client
2026/08/10 14:06:28 agent-auth: INFO: Started (pid: [REDACTED]).
2026/08/10 14:06:28 agent-auth: INFO: Requesting a key from server: [REDACTED]
2026/08/10 14:06:29 agent-auth: INFO: No authentication password provided
2026/08/10 14:06:29 agent-auth: INFO: Using agent name as: Windows-Client
2026/08/10 14:06:29 agent-auth: INFO: Waiting for server reply
2026/08/10 14:06:29 agent-auth: INFO: Valid key received
PS C:\WINDOWS\system32> Start-Service -Name wazuh
PS C:\WINDOWS\system32> Get-Service -Name wazuh
```

Status	Name	DisplayName
Running	WazuhSvc	wazuh

Activate Windows

